Flex GUI

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CHAPTER

ONE

FLEX GUI

Flex GUI is a flexible GUI that can be customized to suit your needs.

1.1 Description

Creating a GUI with Flex requires NO programming or coding experience. All you need to do is drag and drop widgets into a Qt Main Window and change the object name to match the function. It's just that simple.

- Uses stock Qt Designer 5 or 6 so no special libraries are needed
- Widget names are used to connect controls to the correct code
- Widget functions are auto-discovered at startup
- Special widgets only need Dynamic Properties to be discovered and created
- Your GUI can have exactly the controls and labels you want
- You can create and use your own style sheet, changing fonts, colors, etc.
- All Flex GUI configuration is done in the .ini file
- Flex GUI remembers the size and position of your GUI
- Flex GUI has built in Touch Screen support for all data entry widgets
- Flex GUI works on both PC and Raspberry Pi
- Flex GUI includes NC Code Generators and Calculators

For more information about Dynamic Properties see *Dynamic Properties*

1.1.1 Features

- Status Labels
- Menu Actions
- Button Actions
- Button Controls
- Plotter
- MDI Input and Buttons
- Spindle Controls
- Probing with Spindle Safety

- Tool Change Controls
- Coordinate System Controls
- HAL Buttons, Spinboxes, Sliders, Labels and LCDs
- HAL I/O Controls
- Touch Screen Controls and Popups
- Multiple Spindle Support

1.1.2 Acronyms

The following acronyms will be found in this document

- NC = *Numerical Control* code includes G, M and O codes
- G = *Geometric* code is used for geometric movements
- M = *Miscellaneous* code is used for non-movement functions
- O = *Organizing* code is used in CNC programs to control flow

CHAPTER

TWO

GETTING STARTED

1. Install options for Flex GUI
    - Install using Debian Apt - See *Install with Apt*
    - Install the prebuilt .deb file - See *Install a Deb*
    - Clone and build a .deb file and install - See *Build and Install*
2. Install Qt Designer - See *Installing the Qt Designer*
3. Copy Flex GUI Examples - See *Copy Example Files*
4. Building Your GUI the way *you* want it the Flexible Way
    - In the *linuxcnc/configs/flex_examples/starters* copy one of the examples
    - Rename the directory, the ini and the ui files to suit.
    - In the ini file change the name of the file in GUI to the name you gave it
5. Running Your GUI
    - From the CNC menu run Linuxcnc and select your configuration
    - Check *Create Desktop Shortcut* for a fast way to run your configuration

CHAPTER

THREE

INSTALLING

3.1 Install with Apt

The advantage of using apt to install Flex GUI is when a new version of Flex GUI is released apt will know a new version is available when you run *sudo apt update*. This will allow you to install the new version of Flex GUI along with other Debian software.

Note: on a brand-new Rpi5 using the linuxcnc.iso, use apt *–fix-broken install* to install several qt6 sub-dependencies

The first command will ask for your password. Neither command will print anything in the terminal.

For a PC to create an apt sources file for Flex GUI copy and paste this command in a terminal

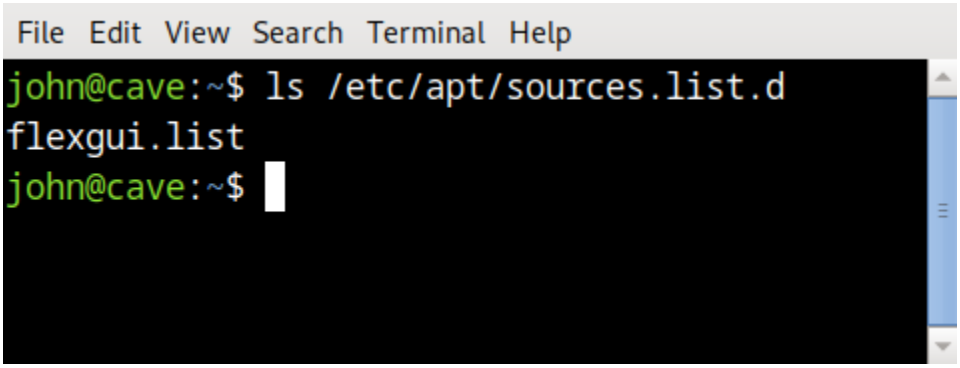
```
echo 'deb [arch=amd64] https://gnipsel.com/flexgui/apt-repo stable main' | sudo tee /etc/
↪apt/sources.list.d/flexgui.list
```

For a Raspberry Pi create an apt sources file for Flex GUI copy and paste this command in a terminal

```
echo 'deb [arch=arm64] https://gnipsel.com/flexgui/apt-repo stable main' | sudo tee /etc/
↪apt/sources.list.d/flexgui.list
```

To check the above command worked you can list the file with this command

```
ls /etc/apt/sources.list.d
```



Next get the public key for Flex GUI and copy it to trusted.gpg.d

```
sudo curl --silent --show-error https://gnipsel.com/flexgui/apt-repo/pgp-key.public -o /
↪etc/apt/trusted.gpg.d/flexgui.asc
```

If curl is not installed you can install it with the following command

```
sudo apt install curl
```

Next update apt

```
sudo apt update
```

If you have Flex GUI installed you can see what packages can be upgraded with the following command

```
apt list --upgradable
```

If Flex GUI is not installed you can install it with the following command

```
sudo apt install flexgui
```

3.2 Install a Deb

You can download the deb from github and install with gdebi if that works better for you. If you don't have an internet connection this is the best way to install Flex GUI.

Note: Between releases the deb will have the latest bug fixes

Installing Flex GUI Tutorial

Download the latest deb file from >HERE<. Right click on the link and select open link in new tab.

If the link is not clickable, copy and paste the following URL into your browser

```
https://github.com/jethornton/flexgui/releases
```

~amd64.deb is for PC's and ~arm64.deb is for Raspberry Pi.

Select the latest release and click on the .deb to start a download.

Right click on the deb file and select *Open with GDebi Package Installer*. If that option is not there then GDebi is not installed, open a terminal and run this command to install it:

```
sudo apt install gdebi
```

An alternative is to install from the terminal outright using *dpkg*. Make sure the version number is correct for the deb you have the following command may be an older version.

```
sudo dpkg -i flexgui_1.4.1_amd64.deb
```

3.3 Build and Install

If you plan on changing code in Flex GUI you can clone the repository and build the deb after making changes. The target directory is optional.

```
git clone https://github.com/jethornton/flexgui.git (target/directory)
```

Before building the deb you will need to install some programs that do the building. Open a terminal and run the following to install devscripts

```
sudo apt install devscripts
```

Open a terminal in the top most flexgui directory and use this command to build a deb file.

```
debuild -us -uc
```

3.4 Copy Example Files

After installing Flex GUI, a menu item *Copy Flex Examples* is added to the *CNC* menu. This will copy the Flex GUI example files to ~/linuxcnc/configs/*flex_examples*.

Note: After updating the Flex GUI some examples may have changed. To get a fresh copy of the examples delete the *linuxcnc/configs/flex_examples* or rename it.

CHAPTER

FOUR

QT DESIGNER

4.1 Installing the Qt Designer

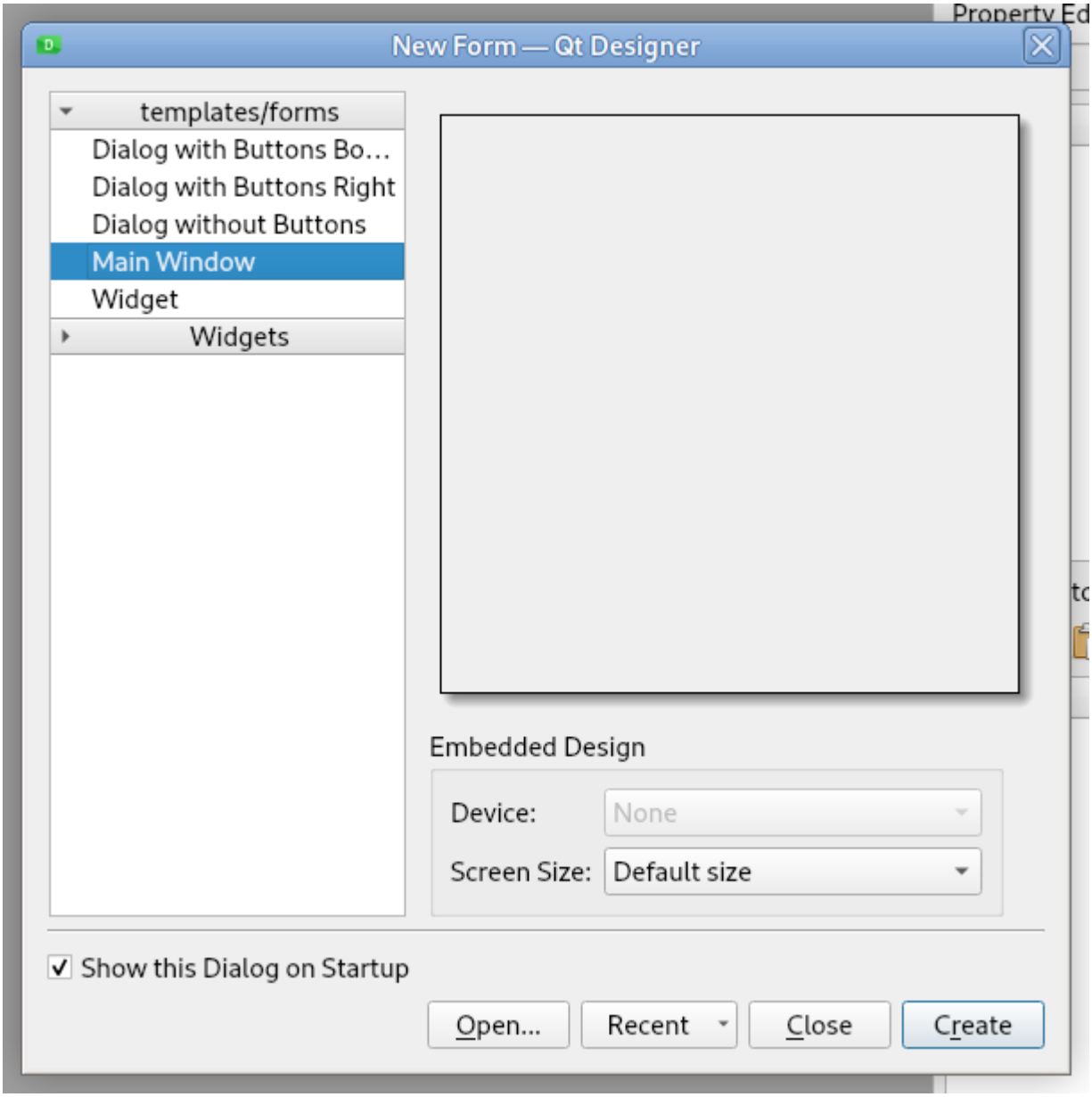
In a terminal, install Qt Designer 5 with

```
sudo apt install qttools5-dev-tools
```

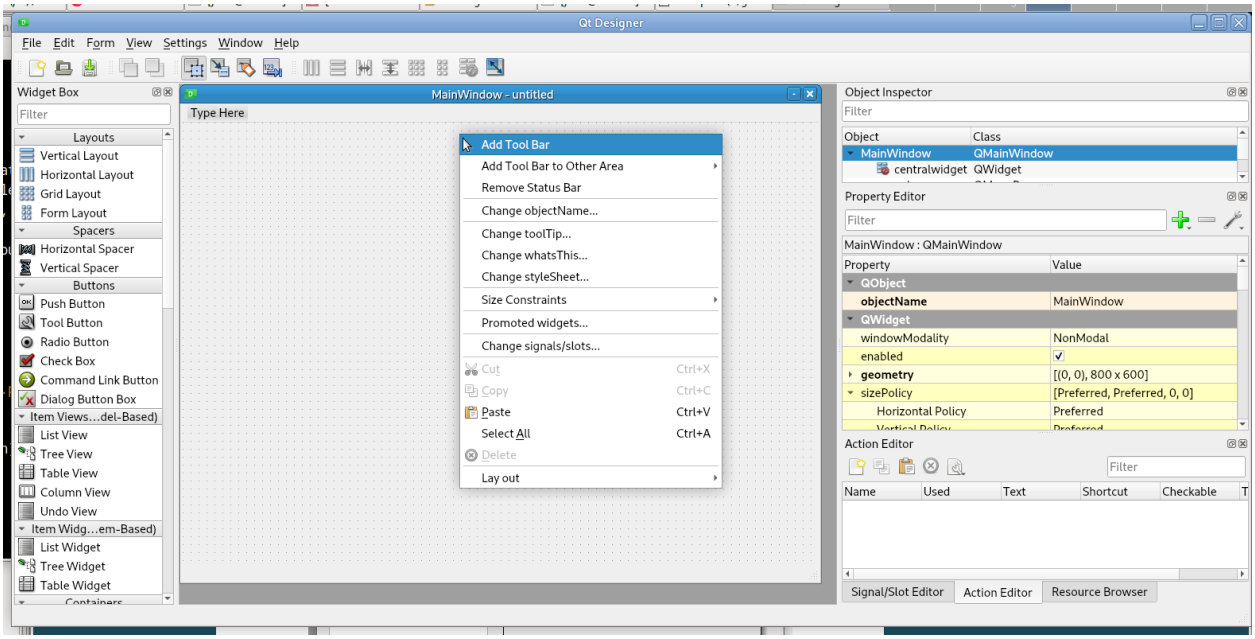
Note: The Qt6 Designer is not required nor better; Qt5 Designer is fine.

4.2 Building a GUI

Run the Qt Designer from the Applications > Programming menu and create a new *Main Window*

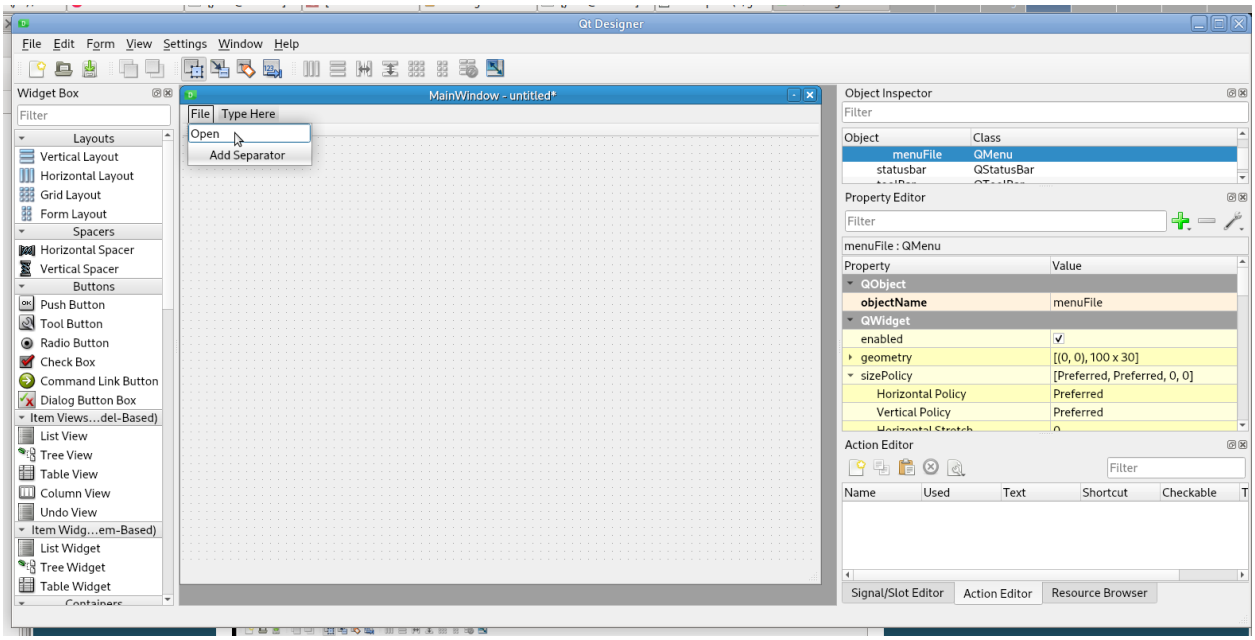


To add a Tool Bar, right click on the main window and select *Add Tool Bar*

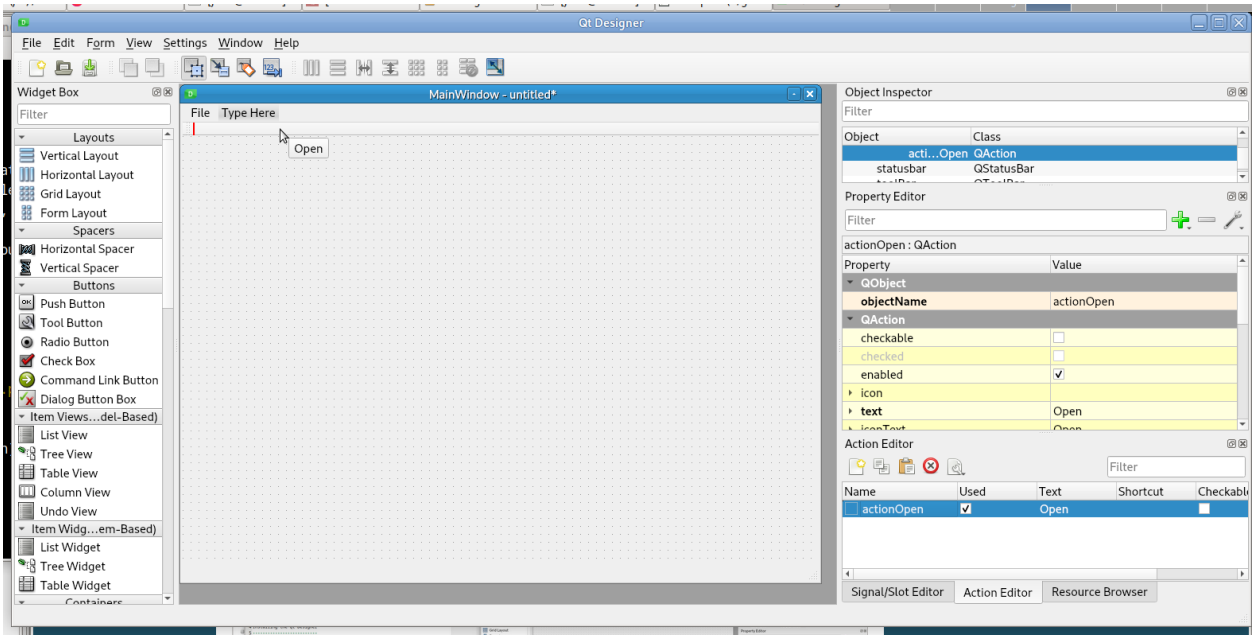


To add a Menu, type in the menu area and press enter.

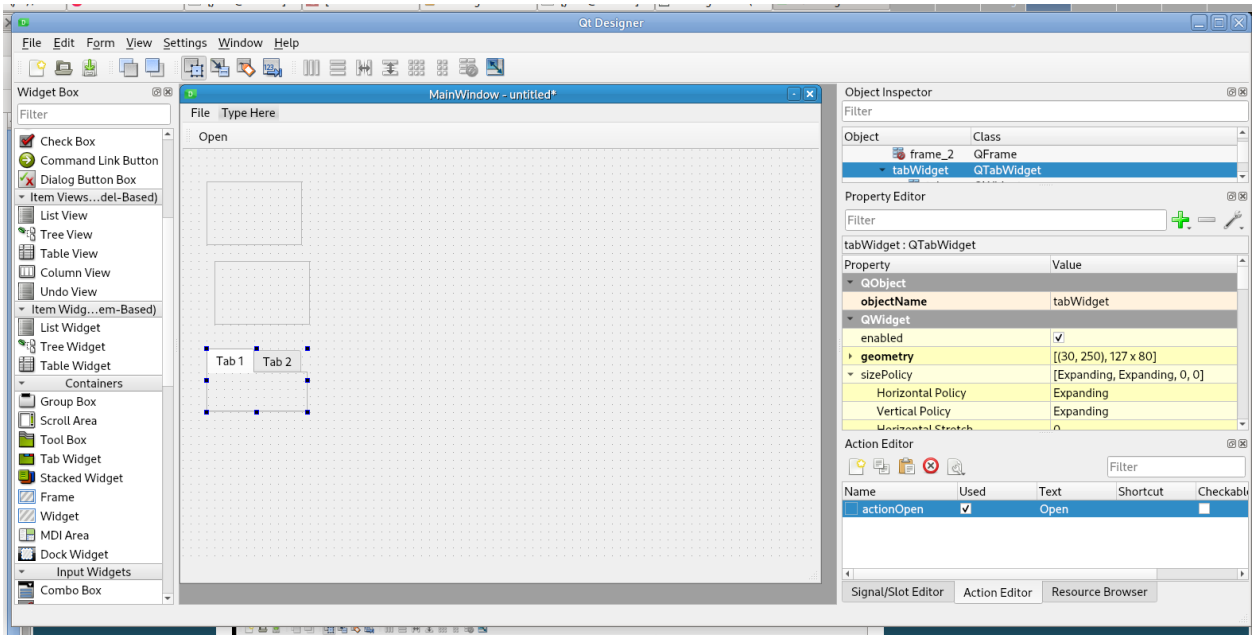
Note: You must press the enter key to add the menu item to the GUI.



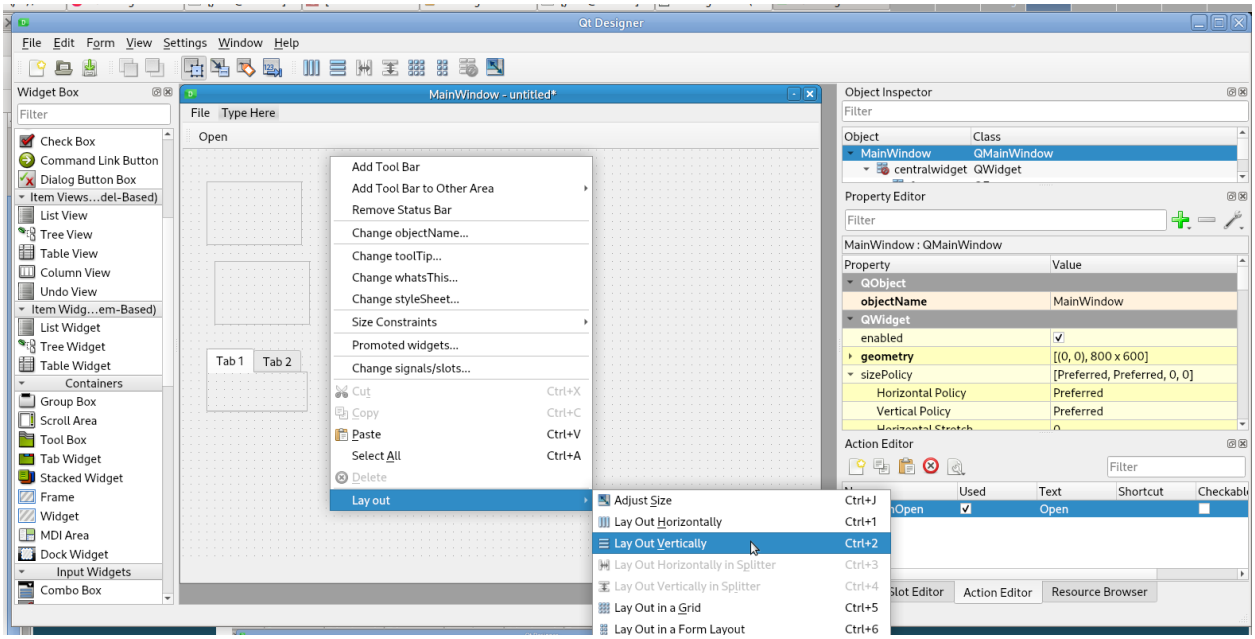
When you create a Menu item it creates an action; this action can be dragged to the Tool Bar to create a tool bar button



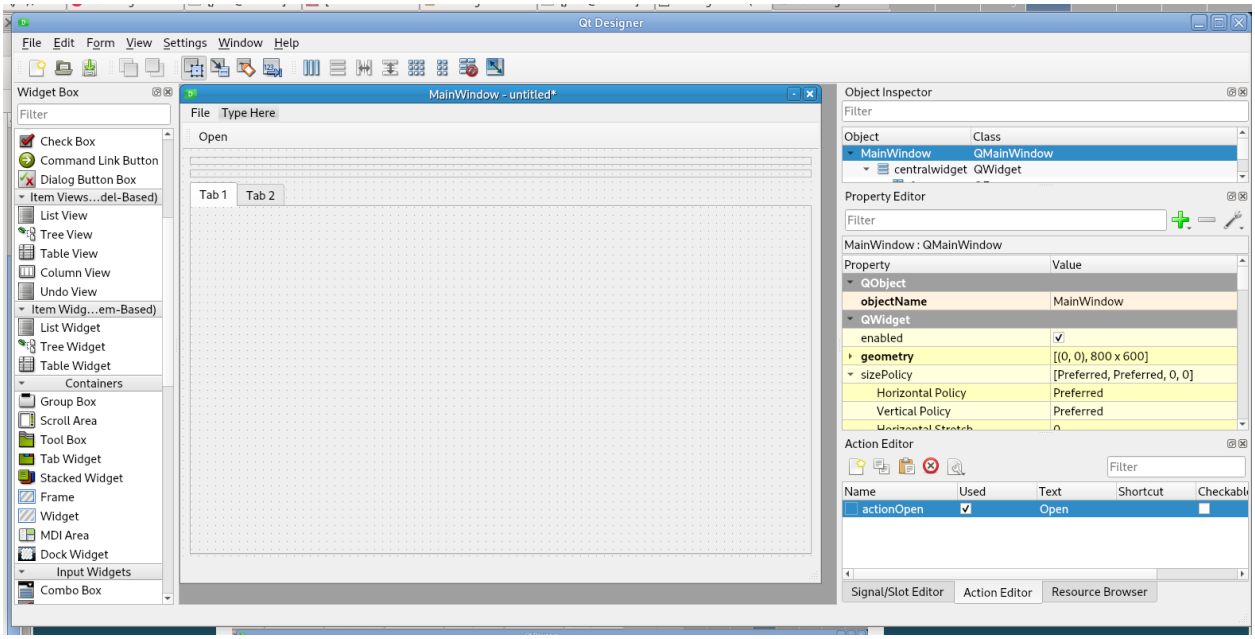
Adding items from the Widget Box is drag-and-drop. To create a basic layout from Containers, add two Frames and a Tab Widget



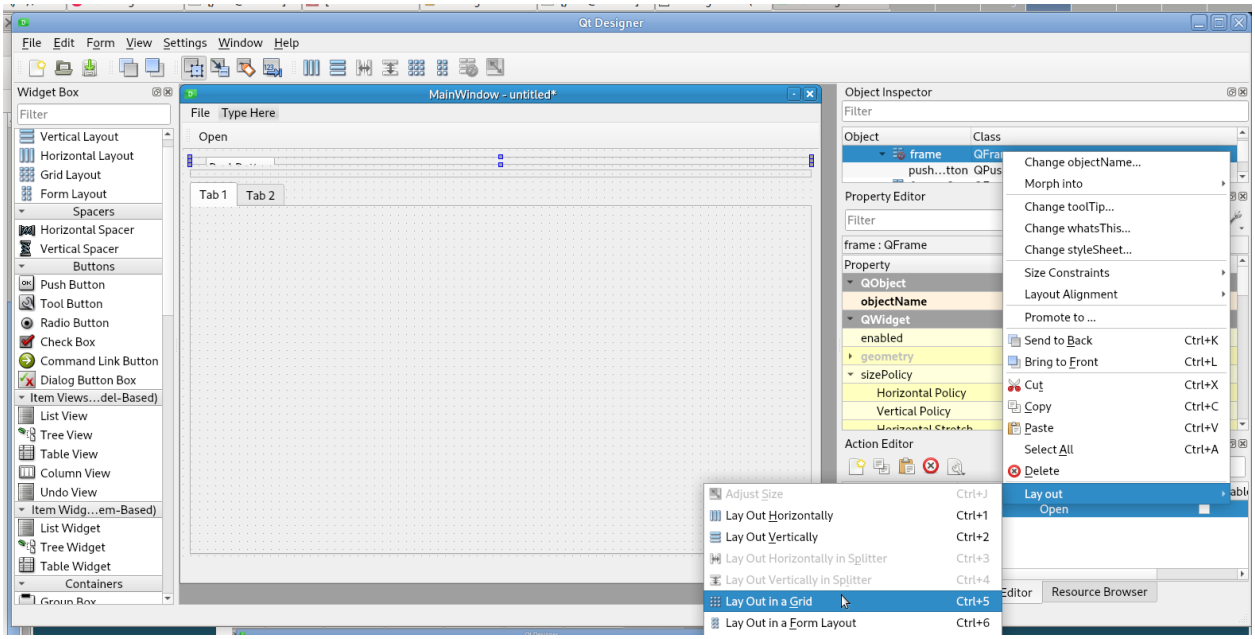
Right click in the QMainWindow and select Lay out –> Lay out Vertically



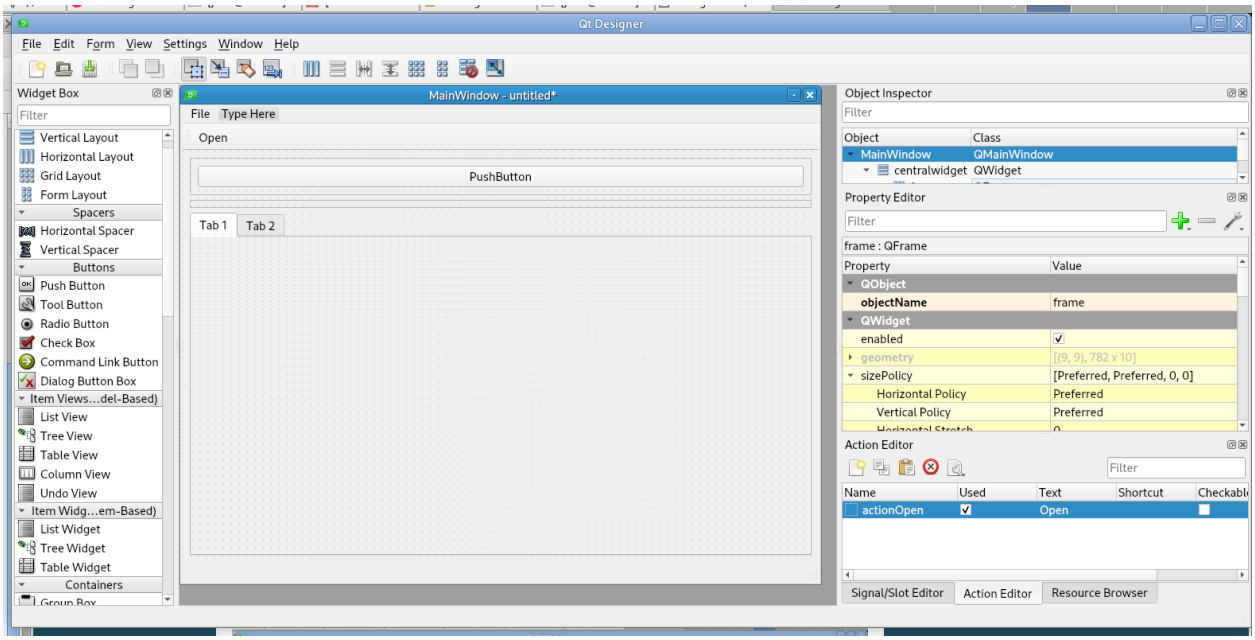
Once you add a layout to the main window the widgets will fill the main window.



Add a Push Button to the QFrame, then right click on the frame or the QFrame in the Object Inspector and set the lay out to *grid*



Once you add a layout to a container the container will resize to match the objects in the containerl.



After dragging a widget into the window, make sure you use the correct objectName for that widget. For example the E-Stop button is called estop_pb.

Note: Each object name must be unique; designer will not allow duplicate names.

Save the GUI in the configuration directory alongside your .ini file.

You can start Qt5 Designer from a terminal with *designer &* which spawns a new process (gives you back the terminal prompt.)

Note: There is an documented issue with Qt5 Designer and bold fonts not appearing properly. Set Font size and weight in the QSS file.

4.3 Qt6 Designer

Qt6 Designer can be installed from a terminal with

```
sudo apt install designer-qt6
```

To run Qt6 Designer you must use the full path to the executable

```
/usr/lib/qt6/bin/designer
```

CHAPTER

FIVE

BUILDING A GUI

After installing Flex GUI if you have not copied the examples from the CNC menu select Copy Flex Examples. This will put the Flex examples in the directory linuxcnc/configs/flex_examples.

The starters have all the files needed to run a simulation without any coding needed. The starters have a very simple GUI to start, just enough to show you that they work.

5.1 Copy a Starter

From the linuxcnc/configs/flex_examples/starters copy the directory and all the contents of one of the starter types to the linuxcnc/configs directory. This will give you a fully working simulator without any coding.

- Rename the directory to the name of your choice.
- Rename the .ui and .ini files to the name of your choice.
- Edit the .ini file and change the GUI to the name of your .ui file.
- Edit the MIN_LIMIT and MAX_LIMIT for each axis and joint to match your machine
- From the CNC menu select LinuxCNC and pick your configuration, check Create Desktop Shortcut then click OK to run your configuration.

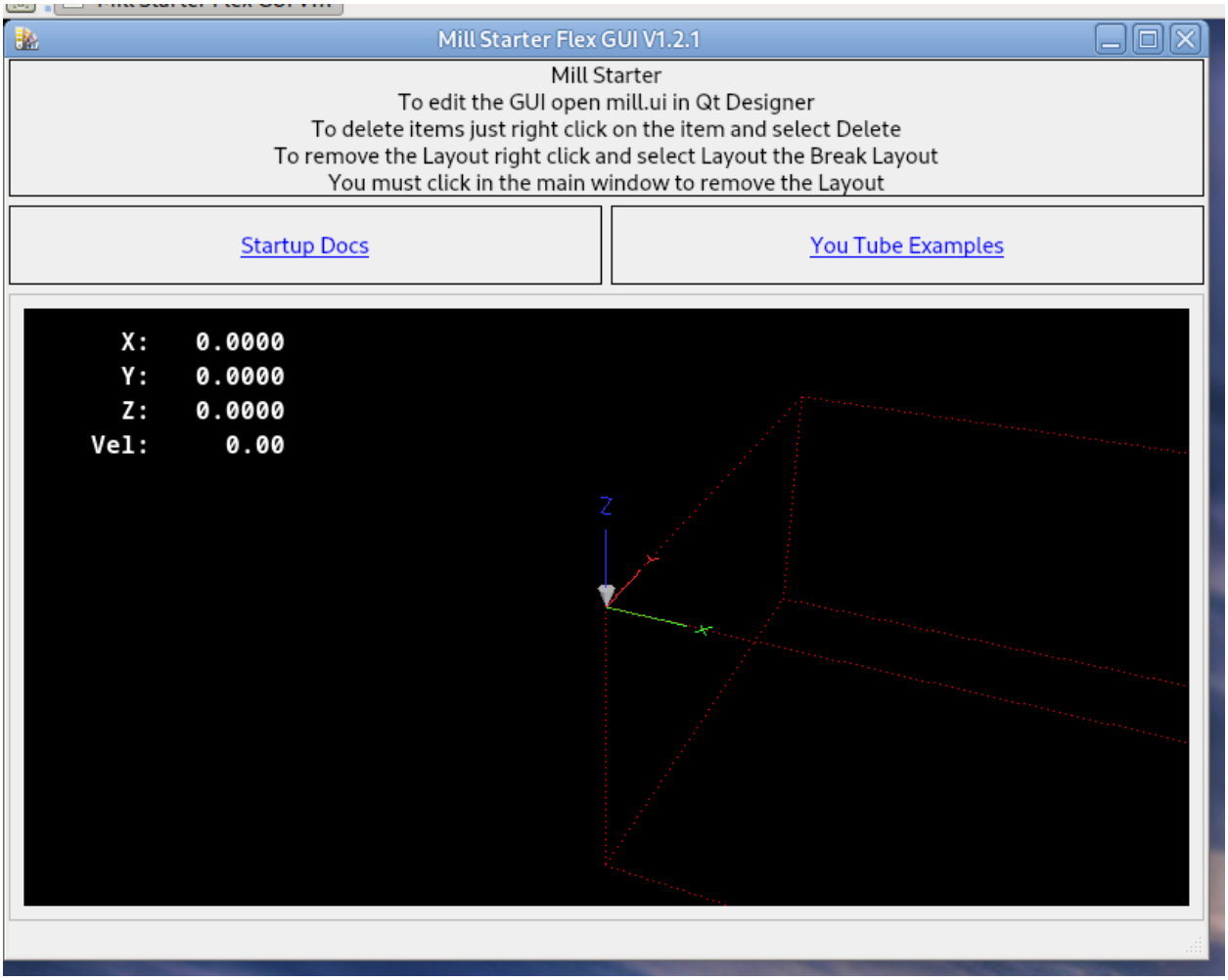


Fig. 1: Mill Starter Example

CHAPTER

SIX

INI SETTINGS

Video Tutorial

6.1 [EMC]

The optional DEBUG key in the [EMC] section can be used to show various levels of debug information when running in a terminal. The DEBUG key is not required.

```
DEBUG = option number
```

Debug Options

```
EMC_DEBUG_CONFIG               2
EMC_DEBUG_VERSIONS             8
EMC_DEBUG_TASK_ISSUE          10
EMC_DEBUG_NML                 40
EMC_DEBUG_MOTION_TIME         80
EMC_DEBUG_INTERP             100
EMC_DEBUG_RCS                200
EMC_DEBUG_INTERP_LIST        800
EMC_DEBUG_IOCONTROL         1000
EMC_DEBUG_OWORD             2000
EMC_DEBUG_REMAP             4000
EMC_DEBUG_PYTHON            8000
EMC_DEBUG_NAMEDPARAM       10000
EMC_DEBUG_GDBONSIGNAL      20000
EMC_DEBUG_STATE_TAGS       80000
EMC_DEBUG_UNCONDITIONAL 40000000
EMC_DEBUG_ALL           7FFFFFFF
```

6.2 [DISPLAY]

In the [DISPLAY] section of the .ini file you can tell LinuxCNC to use Flex GUI and which .ui file to load.

6.2.1 Flex GUI

To use the Flex GUI (as opposed to Axis or others), change the DISPLAY value to

```
DISPLAY = flexgui
```

If no GUI is specified then the default GUI will be used.

Note: Any Flex GUI .ui and .qss files must be in the same LinuxCNC configuration directory as the .ini file.

To use your .ui file (created with Qt Designer), add a GUI key to the .ini with its *filename*

```
GUI = my-file-name.ui
```

6.2.2 Startup File

To automatically open a NC file on startup, add the OPEN_FILE key with any valid path. Use ~/ as a shortcut to the users home directory. Use ./ to indicate that the file is in the configuration directory

```
Full Path to the file
OPEN_FILE = /home/john/linuxcnc/configs/myconfig/welcome.ngc
or use the ~ for the users home directory
OPEN_FILE = ~/linuxcnc/configs/flex_examples/probe_sim/square.ngc
or use the ./ to use the current configuration directory
OPEN_FILE = ./welcome.ngc
or use the ../ to use the parent directory of the configuration
OPEN_FILE = ../welcome.ngc
```

6.2.3 File Location

Likewise, to specify a default location for NC files, add the PROGRAM_PREFIX item.

```
Full Path to the file
PROGRAM_PREFIX = /home/john/linuxcnc/my_files
or use the ~ for the users home directory
PROGRAM_PREFIX = ~/linuxcnc/my_files
or use the ./ to use the current configuration directory
PROGRAM_PREFIX = ./my_local_files
or use the ../ to use the parent directory of the configuration
PROGRAM_PREFIX = ../my_files
```

6.2.4 Tool Table Editor

To specify a different tool table editor add an entry to the [DISPLAY] section. If no entry is found then the default tool editor is used

```
TOOL_EDITOR = tooledit
```

To control the columns displayed by the default tool editor add any of the valid column specifiers separated by a space.

```
TOOL_EDITOR = tooledit x y z a b c u v w diam front back orien
```

If no entry is found then the axes in the configuration and diameter are shown. Tool, Pocket and Comment are always shown.

6.2.5 File Extensions

The keyboard file dialog defaults to **.ngc* and this ignores case. To specify the file extensions you want the file dialog to show, add an EXTENSIONS key with the desired extensions separated by a comma. The extensions must be in the format **.ext* with the asterisk and dot

```
EXTENSIONS = `*.nc`, `*.G-code`, `*.ngc`, `*.txt`
```

6.3 [FLEXGUI]

6.3.1 Window Title

You can have a custom window title by adding WINDOW_TITLE to the [FLEXGUI] section.

```
WINDOW_TITLE = Mill #3
```

6.3.2 Cycle Time

The update cycle time can be set to 50-200 ms using the CYCLE_TIME key. If the CYCLE_TIME key is not found in the [FLEXGUI] section the cycle time will be set to 100 ms. A slower computer may benifit from a longer cycle time.

```
CYCLE_TIME = 100
```

6.3.3 Splash Screen

You can add a splash screen with custom title and time to flex. The image file name is the only required item. If title and time are not specified then the default *Flex Dragon Slayer* and 5 seconds are used. The image file must be in the configuration directory. If the image is a jpg you might have to open it in GIMP and save it as png if it does not show up. If the font is not an exact match the nearest font will be used. Some fixed width fonts are Monospace and Courier. Some variable width fonts are Cantarell and Serif. Font colors are simple base colors only.

```
SPLASH_IMAGE = name.png
SPLASH_TITLE = Flex
SPLASH_TIME = 10
```

(continues on next page)

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```
SPLASH_FONT = Name of Font
SPLASH_FONT_SIZE = 12
SPLASH_FONT_BOLD = True or False
SPLASH_FONT_COLOR = black
```

6.3.4 Flash Timer

The Flash Timer default is 1000ms to change it add the following to the [FLEXGUI] section. The value is in ms so 500 is 0.5 seconds.

```
FLASH_TIME = 500
```

6.3.5 Themes

Themes are just style sheets that get applied to the widgets. The theme files are in the themes directory of the example files if you want to copy and customize one of the themes.

```
blue.qss
blue-touch.qss
dark.qss
dark-touch.qss
keyboard.qss
touch.qss
```

To use a built-in theme with no color changes add one of the following to the [FLEXGUI] section of the ini file.

```
THEME = touch
THEME = keyboard
```

To use a built in theme with coloring add one of the following to the [FLEXGUI] section of the ini file.

```
THEME = blue
THEME = blue-touch
THEME = dark
THEME = dark-touch
```

Note: Touch themes use tabs set to South for rounding and non touch use tabs set to North.

Note: THEME is checked first then QSS so the first entry found is used.

Jog velocity settings are in [TRAJ] LINEAR_UNITS per second. These settings will setup the jog velocity slider minimum, maximum and setting at startup.

```
MIN_JOG_VELOCITY = 0.1
DEFAULT_JOG_VELOCITY = 0.2
MAX_JOG_VELOCITY = 1.0
```

6.3.6 Keyboard Jog

Keyboard Jog options can be added by adding the following to the INI file in the [FLEXGUI] section.

See *Keyboard Jog* for more information.

```
KEYBOARD_JOG = True
KB_JOG_INCREMENT = True
KB_JOG_FOCUS = True
```

6.3.7 Style Sheet

To use a custom .qss style sheet you created add the name of the stylesheet to the QSS option in [FLEXGUI] section of the ini file.

```
QSS = name_of_stylesheet.qss
```

For more information on style sheets see *StyleSheet*

6.3.8 Resource File

To use a .py resource file (to add images to buttons with your qss stylesheet) place the .py resource file in the configuration directory and add the following line to the .ini file

```
RESOURCES = resources.py
```

See the *Resources* section for more info.

6.3.9 Screen Size

To control the initial size of the screen, add of the following values.

```
SIZE = minimized
SIZE = normal
SIZE = maximized
SIZE = full
SIZE = rpi_full
```

Note: the values are case sensitive

Note: The Raspberry Pi 5 with Wayland needs to use the rpi_full value

Warning: Full size screen does not have any window controls. Make sure there is a way to close the GUI like an Exit button or you may not be able to close the application. As a last-resort, pressing ALT-F4 will close it.

6.3.10 Plotter

The plotter background color can be set in the [FLEXGUI] section of the ini. The value is the Red,Green,Blue color numbers from 0 to 1 with no space. So an entry of 0.0,0.0,0.0 is black and 1.0,1.0,1.0 is white. Use a RGB 0-1 Color Picker to select the RGB values.

```
[FLEXGUI]
PLOT_BACKGROUND_COLOR = 0.0,0.0,0.0
```

The plotter orientation can be set to one of the following x, x2, y, y2, z, or p.

```
[FLEXGUI]
PLOT_VIEW = x
```

The font size for the plotter can be set in the ini by adding the following to the FLEXGUI section. The font size must be an integer.

```
[FLEXGUI]
DRO_FONT_SIZE = 12
```

The plotter DRO can be set to automatically change units when the program units G20/G21 change. Setting to *True* disables the manual unit change controls.

```
[FLEXGUI]
PLOT_UNITS = True
```

The plot grid sizes are set in the ini file. Fraction and decimal numbers are valid. Fractions are always in Inch units. Valid suffixes are *mm*, *cm*, *um*, *in*, *inch*, *mil*. If fraction does not have a suffix then *inch* will be added as the suffix. If no suffix is found the the machine units suffix will be added. If the size does not have a valid suffix or is not a number it will be skipped and a warning will appear.

```
[FLEXGUI]
PLOT_GRID = 1/2, 1 in, 1 1/2, 2 inch
or for metric
PLOT_GRID = 10mm, 20mm, 50mm, 100mm
```

The first item in the list will be considered the default and set when FlexGUI starts. If you wish disable grid by default, add a *0* item to the beginning of the list.

To toggle the plotter grid on and off with a checkbox add the following to the ini file in the [FLEXGUI] section. Units are [TRAJ] LINEAR_UNITS from the ini file. The value must be either an integer or a float.

```
[FLEXGUI]
DEFAULT_GRID = 0.5
```

6.3.11 DRO

To set the DRO labels to follow the program units add the following

```
[FLEXGUI]
DRO_UNITS = True
```

6.3.12 Jog Increments

The Jog Increments Combobox is populated from the *JOG_INCREMENTS* entry. Units can be mm, cm, um, in, inch, mil or left out. A space can be between the distance and the units for better readability. Fractions are are in inch units and can be a whole number with a space then the fraction.

```
[FLEXGUI]
JOG_INCREMENTS = 0.250, 0.100, 0.010, 0.001
or
JOG_INCREMENTS = 1 1/2, 1 inch, 0.5 in, 1 cm, 1 mm
```

Warning: [FLEXGUI] JOG_INCREMENTS must be a comma seperated list or it will be ignored.

Note: Jog incremnts can have unit labels, the following are valid unit labels cm, mm, um, inch, in or mil. If no unit labels are found the the configuration units are used.

6.3.13 LED Defaults

LED buttons can have defaults set in the ini file. This makes it easier to have consistent LED size, position and colors. These options go in the [FLEXGUI] section.

The color options can be specified using HEX, RGB or RGBA.

Valid RGB(A) Red, Green, Blue (Alpha) values are 0 to 255.

Valid HEX values are #000000 to #ffffff

In PyQt6 the Alpha channel is 0 to 255. 0 represents a fully transparent color, while 255 represents a fully opaque color. If Alpha is ommitted then it's set to fully opaque or 255.

The Diameter and Offset values are whole numbers only.

```
[FLEXGUI]
LED_DIAMETER = 15
LED_RIGHT_OFFSET = 5
LED_TOP_OFFSET = 5
LED_ON_COLOR = 0, 255, 0
LED_OFF_COLOR= 125, 0, 0, 255
```

For more information see *LED Buttons*

NEW 2.0

HAL LED Text labels can have background and text colors based on the state of the HAL pin.

HAL LED Text Labels can have default colors for the background and text set in the INI file. As above the colors can be HEX, RGB or RGBA.

```
[FLEXGUI]
LED_ON_BG_COLOR = 0, 255, 0
LED_OFF_BG_COLOR= 125, 0, 0, 255
LED_ON_TEXT_COLOR = 0, 255, 0
LED_OFF_TEXT_COLOR= #000000
```

6.3.14 Touch Screens

Options for touch screen users.

To set the style sheet used by the popup number pad, NC code pad and popup keyboard add the following to the FLEXGUI section with the name of the .qss stylesheet to use. If *POPUP_QSS* is not found then the default .qss file will be used.

```
[FLEXGUI]
POPUP_QSS = somefile.qss
```

Set the touch screen file chooser to automatically adjust the width by adding the following to the FLEXGUI section.

```
[FLEXGUI]
TOUCH_FILE_WIDTH = True
```

Add popup keypad to all spin boxes.

```
[FLEXGUI]
TOUCH_SPINBOX = True
```

To add a manual tool change popup to add the following to the ini file in the [FLEXGUI] section.

```
[FLEXGUI]
MANUAL_TOOL_CHANGE = True
```

6.3.15 Keyboard Shortcuts

To add keyboard jog increment to Flex GUI add the following to the FLEXGUI section. The keys 0 through the maximum index position will set the index position in the Jog Modes Combobox. A maximum of 10 positions can be used 0-9. The Jog Modes Combobox must have jog increments set in the INI section [FLEXGUI] JOG_INCREMENTS.

To change the jog increment the control must be out of E-Stop and in manual mode and the Ctrl key must be pressed and held as you press a number key.

```
[FLEXGUI]
KB_JOG_INCREMENT = True
```

6.3.16 I/O

LinuxCNC defaults to 4 each of Digital I/O and Analog I/O. If you configured to have more than 4 Digital I/O or Analog I/Os and you want to track them in Flex GUI you must tell Flex how many you have configured.

```
[FLEXGUI]
DIO = Number of Digital I/O
AIO = Number of Analog I/O
```

6.4 [SPINDLE_n]

The spindles are numbered 0-7 so replace the n with the spindle number. The default spindle is spindle 0. If any of these are not found the default values show here will be used.

```
MIN_RPM = 0
MAX_RPM = 1000
INCREMENT = 100
MIN_OVERRIDE = 0
MAX_OVERRIDE = 100
DEFAULT_RPM = 0
```

The keys *MIN_RPM* and *MAX_RPM* set the minimum and maximum RPM for the spindle. The *INCREMENT* key sets the increment for the faster, slower controls and the speed spinbox up/down arrow key increment.

The *MIN_OVERRIDE* and *MAX_OVERRIDE* set the limits for the spindle speed override slider. The value is the percent of override to allow. So if you set *MIN_OVERRIDE* to 50 the minimum speed override is 50% of the current programmed or manual setting of the spindle. The key *MAX_OVERRIDE* must be set at 100 or higher.

The *DEFAULT_RPM* will be the RPM setting at startup.

6.5 [HAL]

To make HAL connections to Flex GUI HAL objects it must be done in the post gui HAL file and the file must be added to the [HAL] section of the ini file.

```
POSTGUI_HALFILE = postgui.hal
```

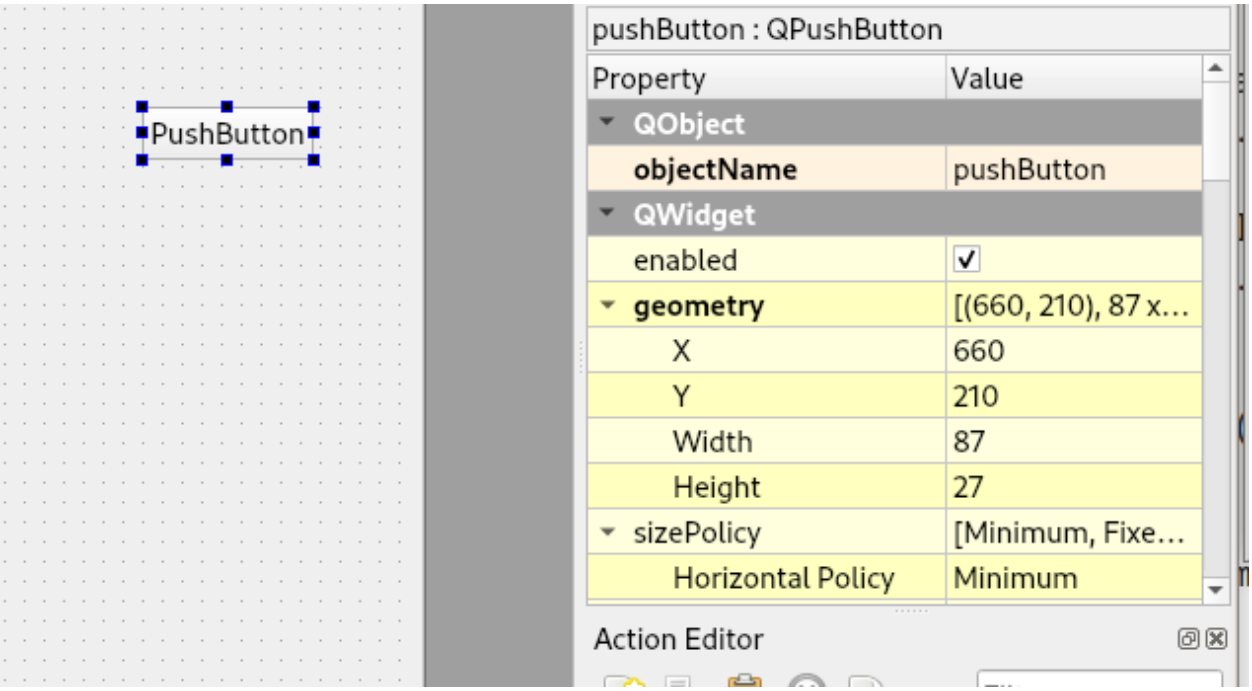
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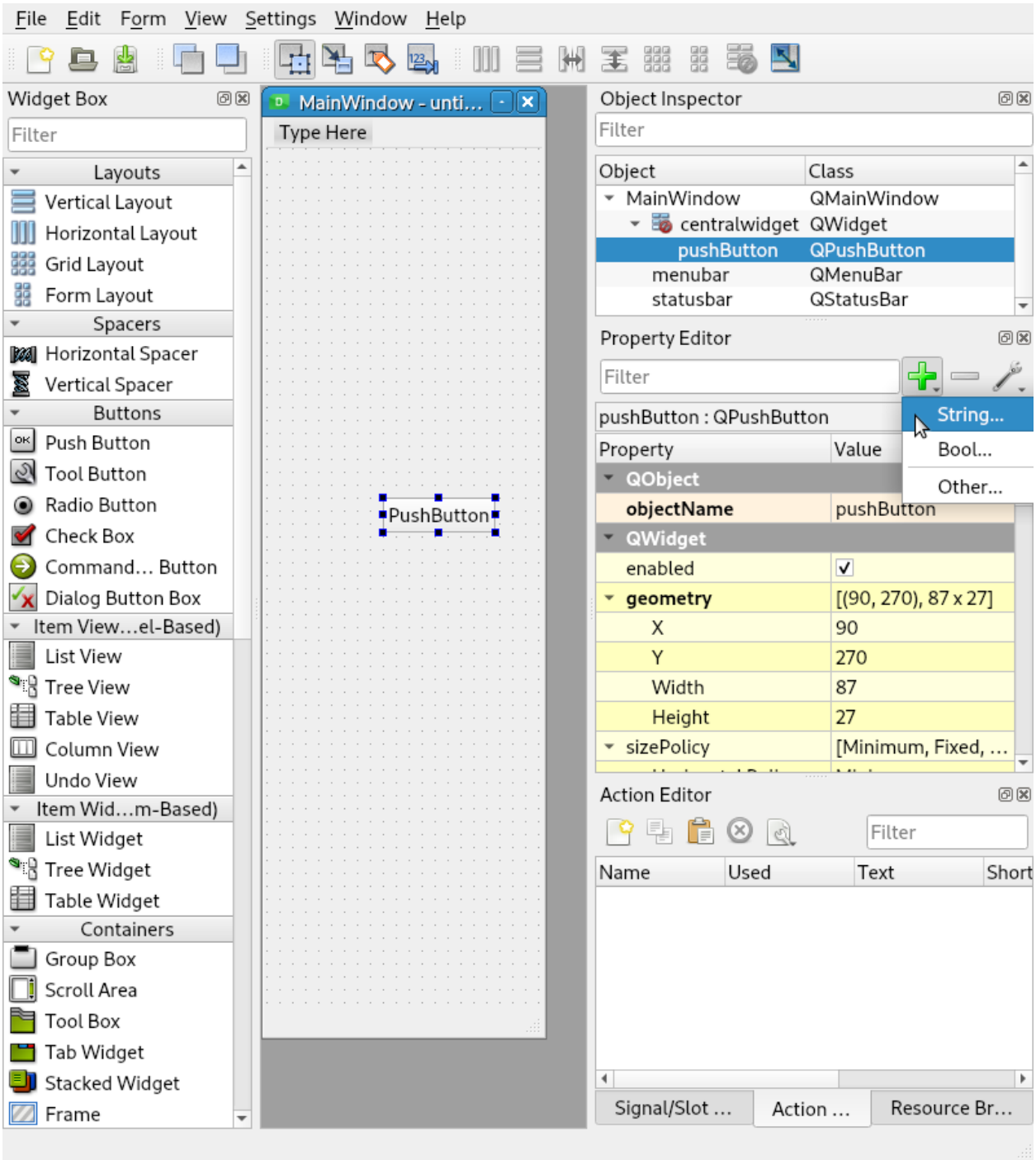
DYNAMIC PROPERTIES

Flex GUI uses a lot Dynamic Properties to customize widget behavior and add more functionality to a widget.

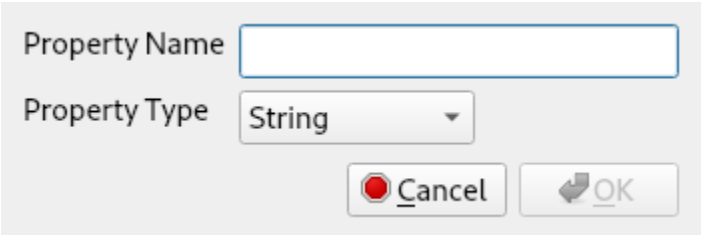
To Create a Dynamic Property the first thing you do is select the widget.



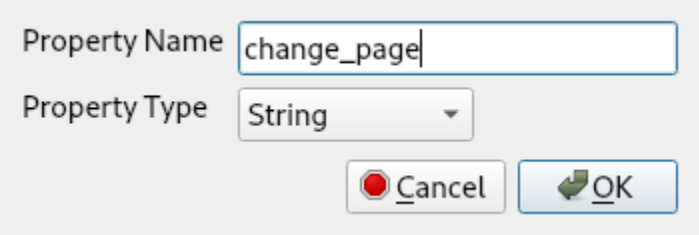
Next left click on the green plus sign in the Property Editor. Select the type of property, select Other… if the property is not a String or Bool.



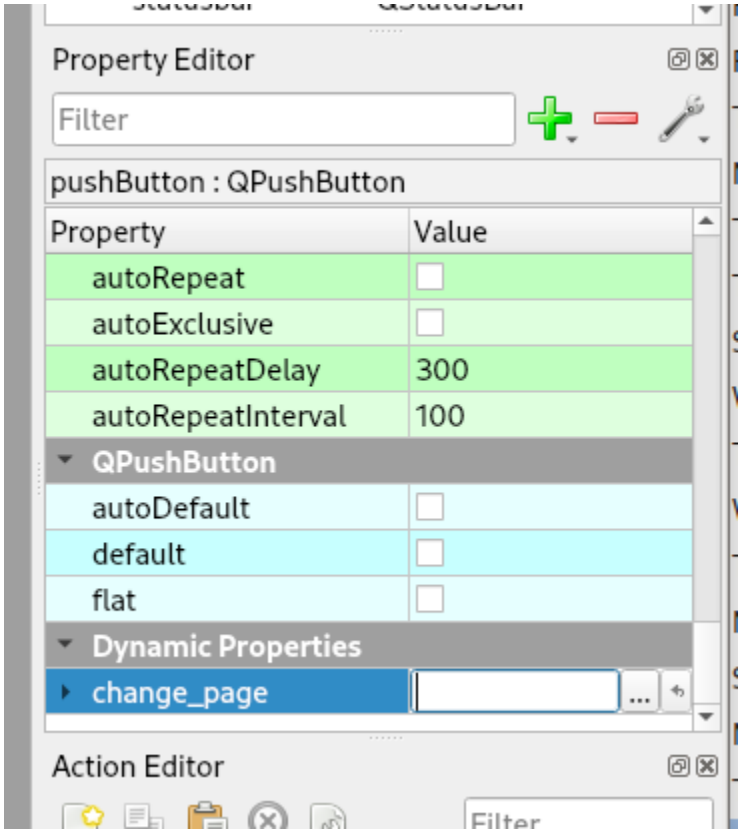
Typically Flex GUI uses a string type Dynamic Property with some exceptions.



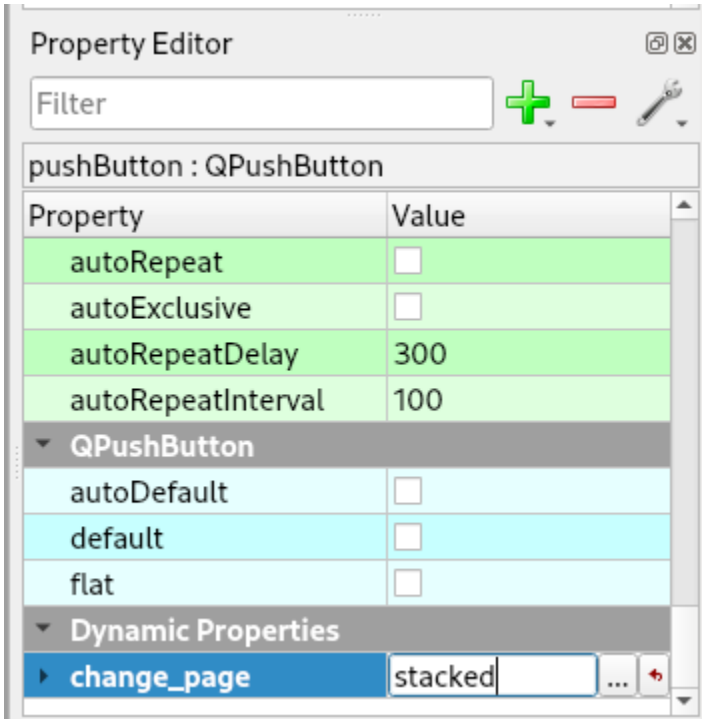
Next enter the Dynamic Property Name, which must be exactly as shown in the documents and Dynamic Property Names are case sensitive.



Press OK then the Dynamic Property will show up in the Property Editor.



Now you can enter the Value for that Dynamic Property.



Note: The Dynamic Properties section does not show up in the Property Editor until you create a Dynamic Property.

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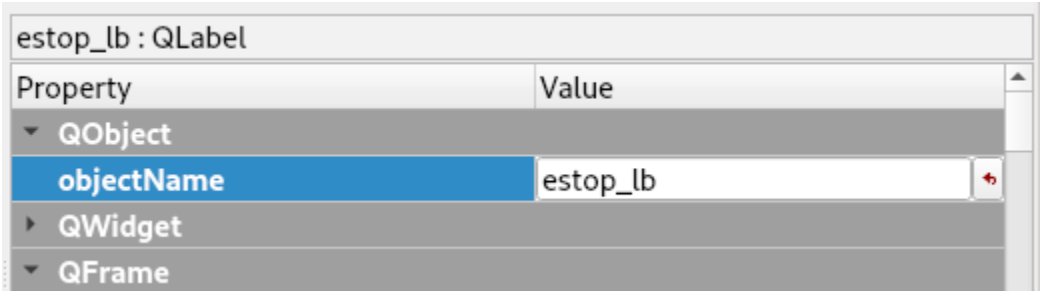
STATUS LABELS

Status Labels Tutorial

Status labels are created using a QLabel and setting the *Object Name* to the name for that status. Status labels come in two forms. A single-status-label like *flood_lb* only contains one piece of information, such as *OFF* or *ON*.

Multiple status labels like the *axis* or *joint* dictionaries have multiple items and displays for all joints. Multiple-status-labels use a number identifier to select the axis, joint, or spindle information wanted.

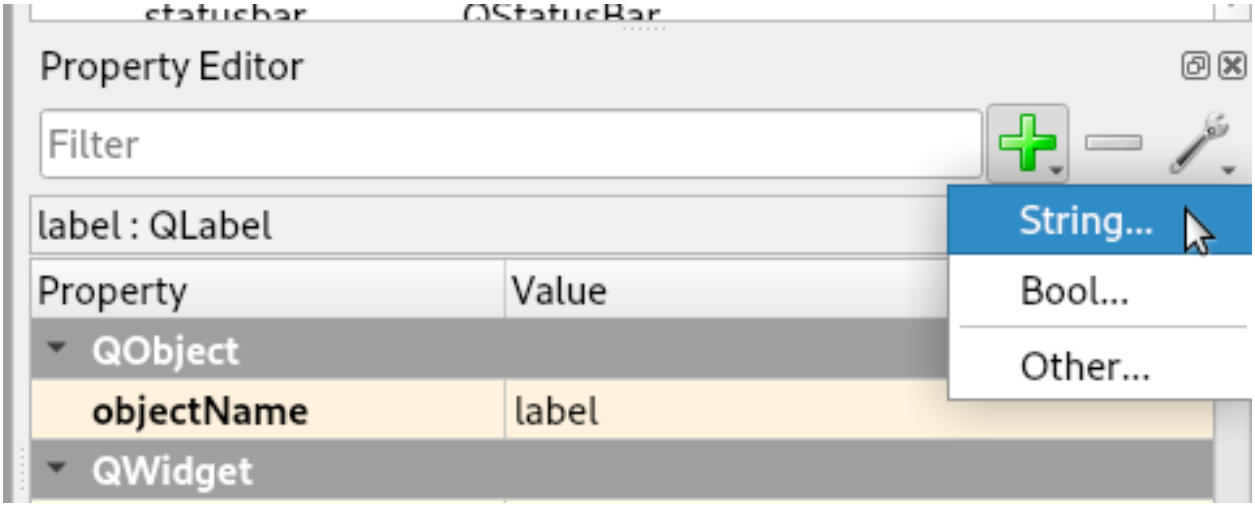
When creating a status label, set the *objectName* to the status you want to display



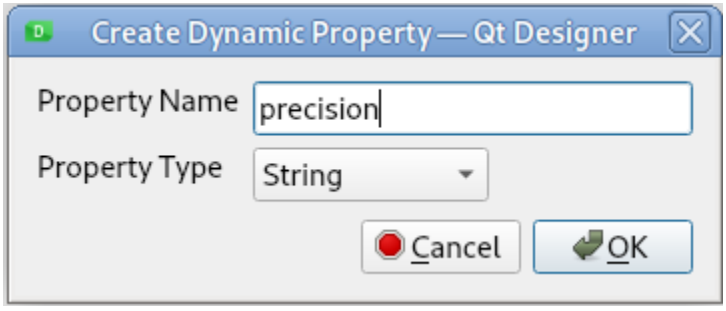
8.1 Precision

Labels that return float values default to 3 decimal places for metric and 4 for inch.

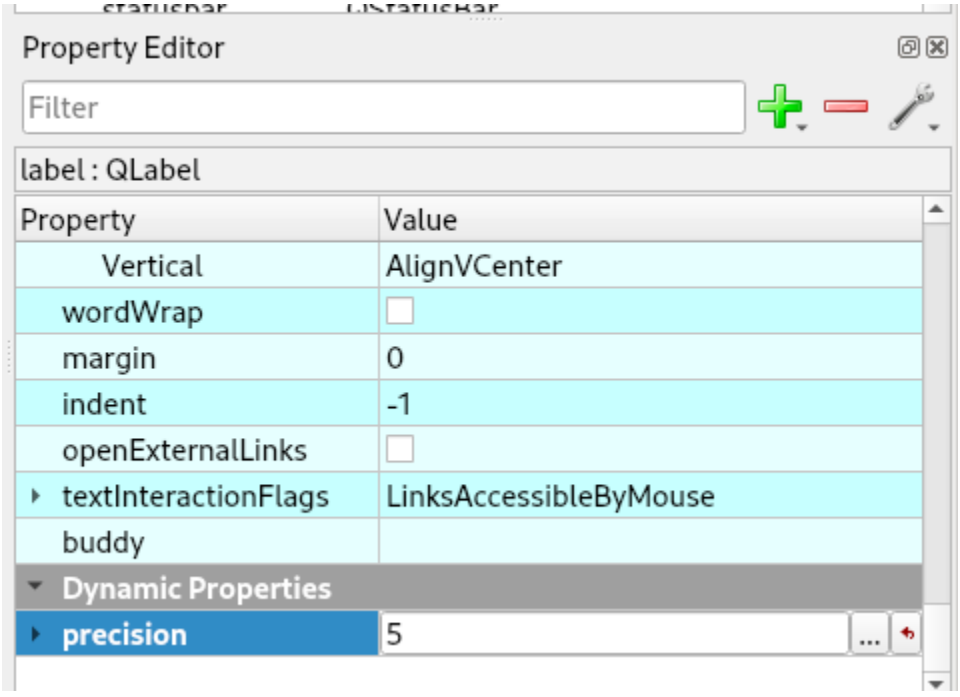
To override the default, select the label then click on the Green Plus sign in the Property Editor to add a Dynamic Property and select String. See *Dynamic Properties*



Set the Property Name to *precision*:



Set the Value to how many decimal places you want for that status label



For more information about status labels read the LinuxCNC Python Interface Status Attributes

8.2 Status Labels

Status Labels are created by adding a QLabel and changing the Object Name to one of the following. Some status labels use a dictionary to look up the name instead of displaying the integer the name is displayed.

```
acceleration_lb - returns float
reflects the INI entry [TRAJ]DEFAULT_LINEAR_ACCELERATION, if that entry is not
found it returns 1e+99

active_queue_lb - returns integer
number of motions blending

actual_position_lb - returns tuple of floats
current trajectory position, (x y z a b c u v w) in machine units
```

See the *Axis Position* labels for individual axis positions.

```
adaptive_feed_enabled_lb - returns boolean
status of adaptive feedrate override

ain_lb - returns tuple of floats
current value of the analog input pins

angular_units_lb - returns float - precision can be set
machine angular units per deg, reflects [TRAJ]ANGULAR_UNITS

aout_lb - returns tuple of floats
current value of the analog output pins

axis_lb - returns tuple of dicts
reflecting current axis values

axis_mask_lb - returns integer
sum of the axes by [TRAJ]COORDINATES X=1, Y=2, Z=4, A=8, B=16, C=32, U=64, V=128, W=256

block_delete_lb - returns boolean
block delete current status

call_level_lb - returns integer
current subroutine depth. - 0 If not in a subroutine

command_lb - returns string
currently executing command

current_line_lb - returns integer
currently executing line

current_vel_lb - returns float - precision can be set
current velocity in user units per second

cycle_time_lb - returns float - precision can be set
thread period
```

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```
debug_lb - returns integer
debug flag from the INI file

delay_left_lb - returns float - precision can be set
remaining time on G4 dwell command in seconds

din_lb - returns tuple of integers
current value of the digital input pins

distance_to_go_lb - returns float - precision can be set
remaining distance of current move as reported by trajectory planner

echo_serial_number_lb - returns integer
The serial number of the last completed command sent by a UI to task.
All commands carry a serial number. Once the command has been executed,
its serial number is reflected in echo_serial_number

enabled_lb - returns boolean
trajectory planner enabled flag

estop_lb - returns integer
Returns either STATE_ESTOP = 1) or not = 0)

exec_state_lb - returns integer that is used to lookup the state name.
task execution state. One of EXEC_ERROR = 1, EXEC_DONE = 2,
EXEC_WAITING_FOR_MOTION = 3, EXEC_WAITING_FOR_MOTION_QUEUE = 4,
EXEC_WAITING_FOR_IO = 5, EXEC_WAITING_FOR_MOTION_AND_IO = 7,
EXEC_WAITING_FOR_DELAY = 8, EXEC_WAITING_FOR_SYSTEM_CMD = 9,
EXEC_WAITING_FOR_SPINDLE_ORIENTED = 10).

feed_hold_enabled_lb - returns boolean
enable flag for feed hold

feed_override_lb - returns boolean
 enable flag for feed override

file_lb - returns string
currently loaded filename

flood_lb - returns integer that is used to lookup the state of OFF or ON
Flood status, either FLOOD_OFF = 0) or FLOOD_ON = 1)

g5x_index_lb - returns integer that is used to lookup the coordinate system name
currently active coordinate system, G54=1, G55=2 etc

g5x_offset_lb - returns tuple of floats
offsets of the currently active coordinate system X, Y, Z, U, V, W, A, B, C

gcodes_lb - returns tuple of integers
Active G-codes for each modal group.
The integer values reflect the nominal G-code numbers multiplied by 10.
(Examples: 10 = G1, 430 = G43, 923 = G92.3)
```

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```
homed_lb - returns tuple of integers
currently homed joints, 0 = not homed, 1 = homed
```

See the *Axis Homed* labels for individual axis home status.

```
ini_filename_lb - returns string
path to the INI file passed to linuxcnc

inpos_lb - returns boolean
machine-in-position flag

input_timeout_lb - returns boolean
flag for M66 timer in progress

interp_state_lb - returns integer that is used to lookup the state name
current state of RS274NGC interpreter. One of INTERP_IDLE = 1,
INTERP_READING = 2, INTERP_PAUSED = 3, INTERP_WAITING = 4

interpreter_errcode_lb - returns integer that is used to lookup the error name
current RS274NGC interpreter return code
        INTERP_OK = 0,
        INTERP_EXIT = 1,
        INTERP_EXECUTE_FINISH = 2,
        INTERP_ENDFILE = 3,
        INTERP_FILE_NOT_OPEN = 4,
        INTERP_ERROR = 5

joint_lb - returns tuple of dicts
reflecting current joint values
```

See the *Joint Status* labels for individual joint status items.

```
joint_actual_position_lb - returns tuple of floats
actual joint positions

joint_position_lb - returns tuple of floats
desired joint positions

joints_lb - returns integer
number of joints. Reflects [KINS]JOINTS INI value

kinematics_type_lb - returns integer that is used to lookup the kinematics name
The type of kinematics
        KINEMATICS_IDENTITY = 1
        KINEMATICS_FORWARD_ONLY = 2
        KINEMATICS_INVERSE_ONLY = 3
        KINEMATICS_BOTH = 4

limit_lb - returns tuple of integers
axis limit masks. minHardLimit=1, maxHardLimit=2, minSoftLimit=4, maxSoftLimit=8
```

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```
linear_units_lb - returns float - precision can be set
machine linear units per mm, reflects [TRAJ]LINEAR_UNITS INI value

lube_lb - returns integer
lube on flag

lube_level_lb - returns integer
reflects iocontrol.0.lube_level

machine_units_lb - return string - "MM" or "INCH"
The machine's native units, reflects [TRAJ]LINEAR_UNITS INI value and is
updated once on start-up.

All labels with an object name that starts with `machine_units_lb` will be
updated with this value for example `machine_units_lb_1`. The label can be
used multiple times in the UI but the trailing number must be unique for each
label. Add a dynamic property `suffix` (string) to add a suffix, for example
/min to display <machine units>/min

max_acceleration_lb - returns float - precision can be set
maximum acceleration. Reflects [TRAJ]MAX_ACCELERATION

max_velocity_lb - returns float - precision can be set
maximum velocity. Reflects the current maximum velocity. If not modified by
halui.max-velocity or similar it should reflect [TRAJ]MAX_VELOCITY

min_jog_vel_lb - returns int
minimum jog velocity slider setting. Reflects the [DISPLAY] MIN_LINEAR_VELOCITY
setting in user units per minute.

max_jog_vel_lb - returns int
maximum jog velocity slider setting. Reflects the [DISPLAY] MAX_LINEAR_VELOCITY
setting in user units per minute.

mcodes_lb - returns tuple of 10 integers
currently active M-codes

mist_lb - returns integer
Mist status, either MIST_OFF = 0 or MIST_ON = 1

motion_line_lb - returns integer
source line number motion is currently executing

motion_mode_lb - returns integer that is used to lookup the motion mode name
This is the mode of the Motion controller.
        TRAJ_MODE_FREE = 1
        TRAJ_MODE_COORD = 2
        TRAJ_MODE_TELEOP = 3

motion_type_lb - returns integer that is used to lookup the motion type name
The type of the currently executing motion. One of:
        MOTION_TYPE_TRAVERSE = 1
```

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```
        MOTION_TYPE_FEED = 2
        MOTION_TYPE_ARC = 3
        MOTION_TYPE_TOOLCHANGE = 4
        MOTION_TYPE_PROBING = 5
        MOTION_TYPE_INDEXROTARY = 6
        Or 0 if no motion is currently taking place.

optional_stop_lb - returns integer
option stop flag

paused_lb - returns boolean
motion paused flag

pocket_prepped_lb - returns integer
A Tx command completed, and this pocket is prepared. -1 if no prepared pocket

position_lb - returns tuple of floats
trajectory position

probe_tripped_lb - returns boolean
True if probe has tripped

probe_val_lb - returns integer
reflects value of the motion.probe-input pin

probed_position_lb - returns tuple of floats
position where probe tripped

probing_lb - returns boolean
True if a probe operation is in progress

program_units_lb - return string - "MM" or "INCH"
G20 and G21 program mode.
All labels with an object name that starts with `program_units_lb` will be
updated with this value for example `program_units_lb_1`.The label can be
used multiple times in the UI but the trailing number must be unique.

queue_lb - returns integer
current size of the trajectory planner queue

queue_full_lb - returns boolean
the trajectory planner queue is full

rapid_override_lb - returns percent
rapid override percent

rapidrate_lb - returns float - precision can be set
rapid override scale, 1.0 = 100%

read_line_lb - returns integer
line the RS274NGC interpreter is currently reading
```

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```
rotation_xy_lb - returns float - precision can be set
current XY rotation angle around Z axis

spindle_lb - returns tuple of dicts
returns the current spindle status
```

See the *Spindle* section for individual spindle status items

```
spindles_lb - returns integer
returns number of spindles in [TRAJ] SPINDLES

state_lb - returns integer that is used to lookup the state name
current command execution status
One of RCS_DONE = 1, RCS_EXEC = 2, RCS_ERROR = 3

task_mode_lb - returns integer that is used to lookup the task mode name
current task mode
One of  MODE_MANUAL = 1, MODE_AUTO = 2, MODE_MDI = 3

task_paused_lb - returns integer
task paused flag, not paused = 0, paused = 1

task_state_lb - returns integer that is used to lookup the task state name
current task state
One of STATE_ESTOP = 1, STATE_ESTOP_RESET = 2, STATE_OFF = 3 STATE_ON = 4
STATE_OFF is never seen

tool_in_spindle_lb - returns integer
current tool number in spindle (0 if no tool loaded)

tool_from_pocket_lb - returns integer
pocket number for the currently loaded tool (0 if no tool loaded)

tool_offset_lb - returns tuple of floats
offset values of the current tool

tool_table_lb - returns tuple of tool_results
list of tool entries. Each entry is a sequence of the following fields: id,
xoffset, yoffset, zoffset, aoffset, boffset, coffset, uoffset, voffset,
woffset, diameter, frontangle, backangle, orientation. The id and orientation
are integers and the rest are floats.
If id = -1 no tools are in the tool table.
```

Note: You don't have to use all the labels; only use the ones you need.

8.3 Axis Status

The Axis status contains status items for all 9 axes. Replace the *n* with the number of the axis. Axis numbers start at 0 and go through 8. Returns a float

Table 1: **Axis Status Labels**

axis_n_min_position_limit_lb	axis_n_max_position_limit_lb
axis_n_velocity_lb	axis_n_vel_per_min_lb

Note: The Axis velocity label only reports back *jogging* speed; use the joint velocity label for *linear* speed.

8.4 Joint Status Labels

The Joint status contains status items for 16 joints. Replace the *n* with the number of the joint. Joint numbers start at 0 and go through 15

Table 2: **Joint Status Labels**

joint_n_backlash_lb	joint_n_enabled_lb
joint_n_fault_lb	joint_n_ferror_current_lb
joint_n_ferror_highmark_lb	joint_n_homed_lb
joint_n_homing_lb	joint_n_min_soft_limit_lb
joint_n_inpos_lb	joint_n_input_lb
joint_n_jointType_lb	joint_n_max_ferror_lb
joint_n_max_hard_limit_lb	joint_n_max_position_limit_lb
joint_n_max_soft_limit_lb	joint_n_min_ferror_lb
joint_n_min_hard_limit_lb	joint_n_min_position_limit_lb
joint_n_output_lb	joint_n_override_limits_lb
joint_n_units_lb	joint_n_vel_min_lb
joint_n_vel_sec_lb	

8.5 Joint Descriptions

backlash (returns float) - Backlash in machine units. configuration parameter, reflects [JOINT_n]BACKLASH.

enabled (returns integer) - non-zero means enabled

fault (returns integer) - non-zero means axis amp fault.

ferror_current (returns float) - current following error.

ferror_highmark (returns float) - magnitude of max following error.

homed (returns integer) - non-zero means joint is been homed.

homing (returns integer) - non-zero means homing in progress.

inpos(returns integer) - non-zero means in position.

input (returns float) - current input position.

jointType (returns integer) - type of axis configuration parameter, reflects [JOINT_n]TYPE. LINEAR=1, ANGULAR=2. See Joint INI configuration for details.

max_ferror (returns float) - maximum following error. configuration parameter, reflects [JOINT_n]FERROR.

max_hard_limit (returns integer) - non-zero means max hard limit exceeded.

max_position_limit (returns float) - maximum limit (soft limit) for joint motion, in machine units. configuration parameter, reflects [JOINT_n]MAX_LIMIT.

max_soft_limit non-zero means max_position_limit was exceeded, int

min_ferror (returns float) - configuration parameter, reflects [JOINT_n]MIN_FERROR.

min_hard_limit (returns integer) - non-zero means min hard limit exceeded.

min_position_limit (returns float) - minimum limit (soft limit) for joint motion, in machine units. configuration parameter, reflects [JOINT_n]MIN_LIMIT.

min_soft_limit (returns integer) - non-zero means min_position_limit was exceeded.

output (returns float) - commanded output position.

override_limits (returns integer) - non-zero means limits are overridden.

units (returns float) - joint units per mm, or per degree for angular joints. (joint units are the same as machine units, unless set otherwise by the configuration parameter [JOINT_n]UNITS)

velocity (returns float) - current velocity.

8.6 Special Labels

Run from line label *start_line_lb* displays the current selected line in the NC code file.

8.6.1 Axis Position Labels

Axis machine position labels (no offsets.) Returns a float

Table 3: **Machine Absolute Position Status Labels**

actual_lb_x	actual_lb_y	actual_lb_z
actual_lb_a	actual_lb_b	actual_lb_c
actual_lb_u	actual_lb_v	actual_lb_w

Axis position labels *including* all offsets. Returns a float

Table 4: **DRO Relative Status Labels**

dro_lb_x	dro_lb_y	dro_lb_z
dro_lb_a	dro_lb_b	dro_lb_c
dro_lb_u	dro_lb_v	dro_lb_w

8.6.2 Axis Distance to Go labels

Table 5: **Distance to Go Labels**

dtg_lb_x	dtg_lb_y	dtg_lb_z
dtg_lb_a	dtg_lb_b	dtg_lb_c
dtg_lb_u	dtg_lb_v	dtg_lb_w

8.6.3 Axis Homed Labels

Table 6: **Axis Homed Labels**

home_lb_0	home_lb_1	home_lb_2
home_lb_3	home_lb_4	home_lb_5
home_lb_6	home_lb_7	home_lb_8

8.6.4 Offset Labels

Offsets for the currently active G5x coordinate system. Returns a float

Table 7: **G5x Status Labels**

g5x_lb_x	g5x_lb_y	g5x_lb_z
g5x_lb_a	g5x_lb_b	g5x_lb_c
g5x_lb_u	g5x_lb_v	g5x_lb_w

Offsets for G92. Returns a float

Table 8: **G92 Status Labels**

g92_lb_x	g92_lb_y	g92_lb_z
g92_lb_a	g92_lb_b	g92_lb_c
g92_lb_u	g92_lb_v	g92_lb_w

8.6.5 Velocity Labels

NEW

Version 2.0.0 allows multiple Velocity Labels

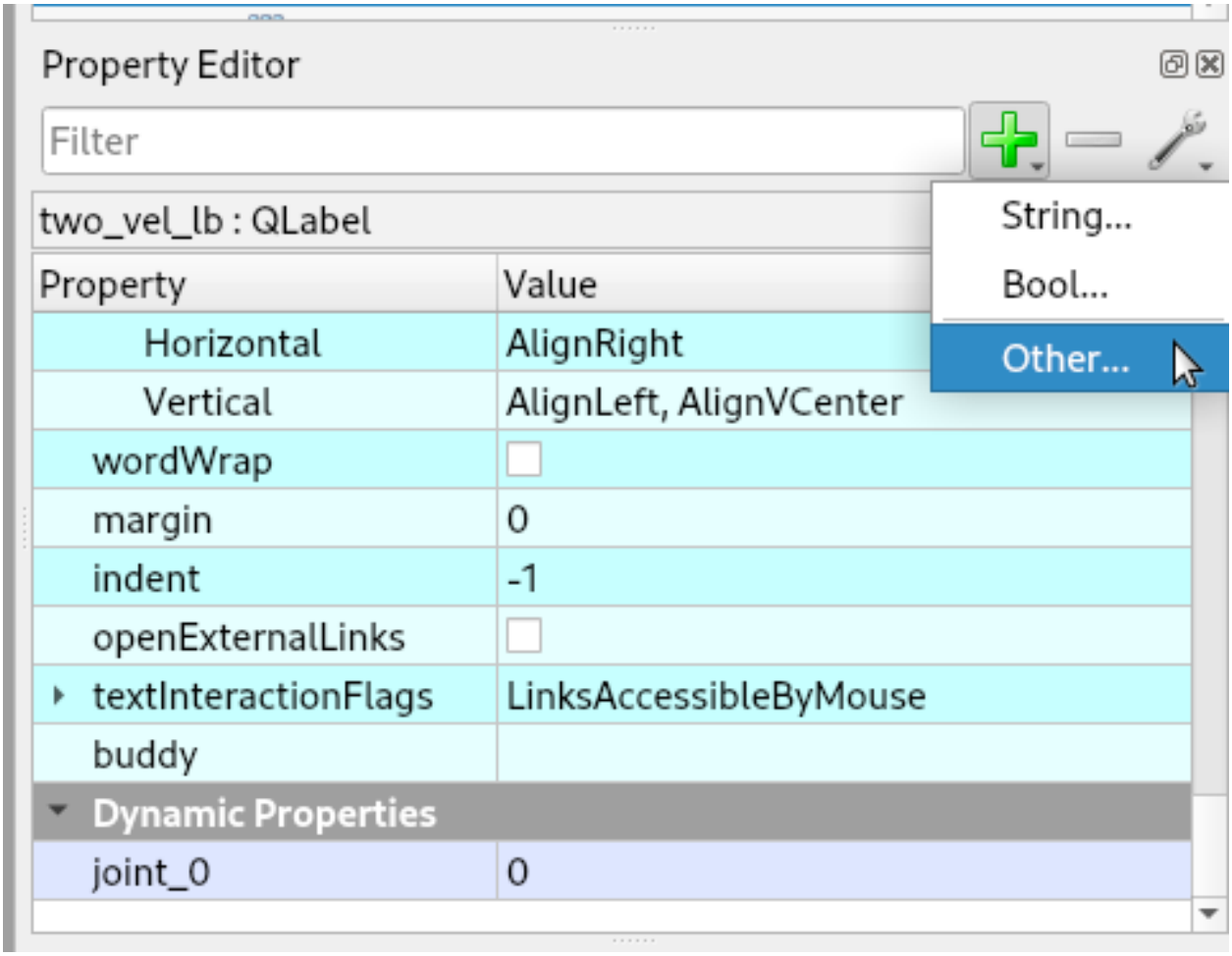
Tool velocity using two perpendicular joint velocities.

To create label(s) that show the combined velocity of two joints add the following Dynamic Properties. You can have as many two joint velocity labels as you want.

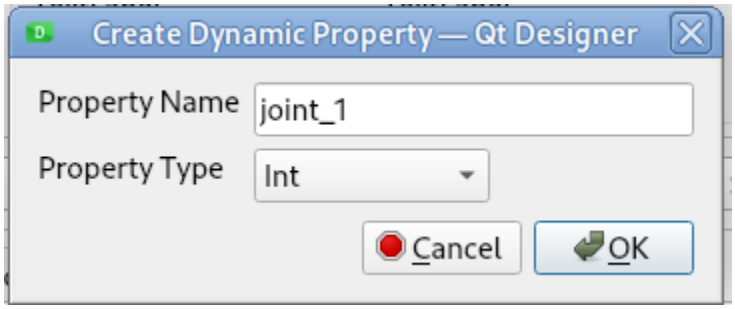
Table 9: **Two Joint Velocicy**

Property Type	**Property Name**	**Pin Value**
String	function	two_joint_velocity
Int	joint_0	the first joint number
Int	joint_1	the second joint number

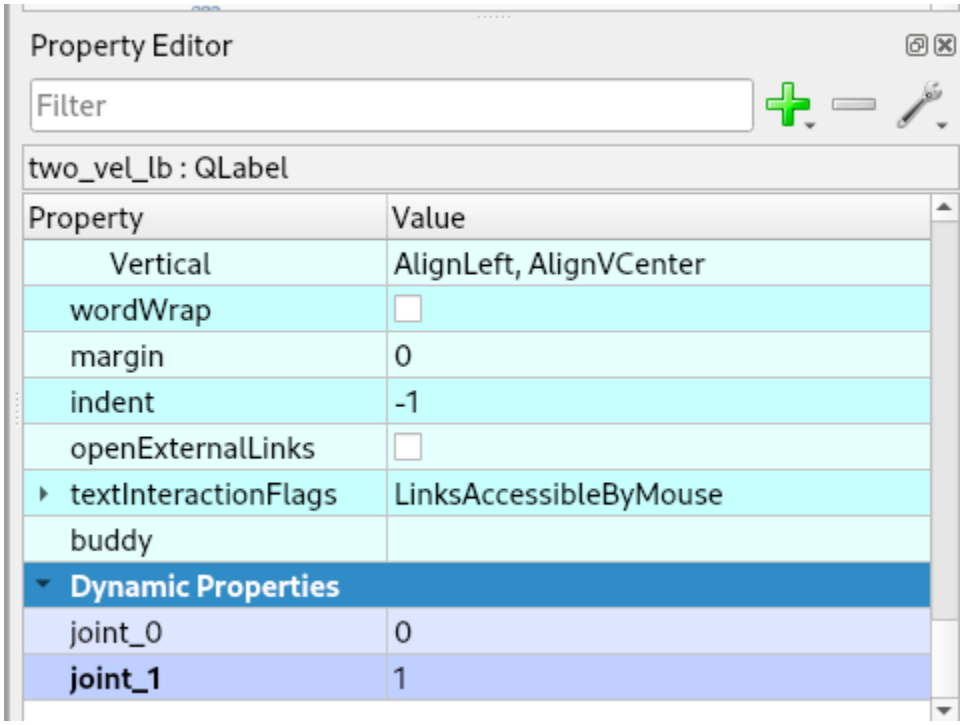
To select an int type of Dynamic Property, select *Other* after clicking on the green plus sign



Then select the Property Type of *int*



The two Dynamic Properties should look like this



Tool velocity using *three* perpendicular joint velocities.

To create label(s) that show the combined velocity of three joints add the following Dynamic Properties. You can have as many three joint velocity labels as you want.

Table 10: **Three Joint Velocicy**

Property Type	**Property Name**	**Pin Value**
String	function	three_joint_velocity
Int	joint_0	the first joint number
Int	joint_1	the second joint number
Int	joint_2	the third joint number

8.7 I/O Status

The I/O status contains status items for up to 64 I/O's. The default is 4 I/O pins for each type. If you have configured more than 4 pins you will need to add an .ini entry to show the number of pins.

For more information see see *I/O Settings* in the .ini file.

Analog I/O are a float. Digital I/O are a bool.

For example a QLabel with an object name of din_0_lb will show the status of the *motion.digital-in-00* HAL pin.

Table 11: **I/O Status Labels**

HAL Pin	Label Name
motion.analog-in-nn	ain_n_lb
motion.analog-out-nn	aout_n_lb
motion.digital-in-nn	din_n_lb
motion.digital-out-nn	dout_n_lb

Replace the *n* with the number of the I/O. I/O numbers start at 0 and go through 63 if you have that many configured.

CHAPTER

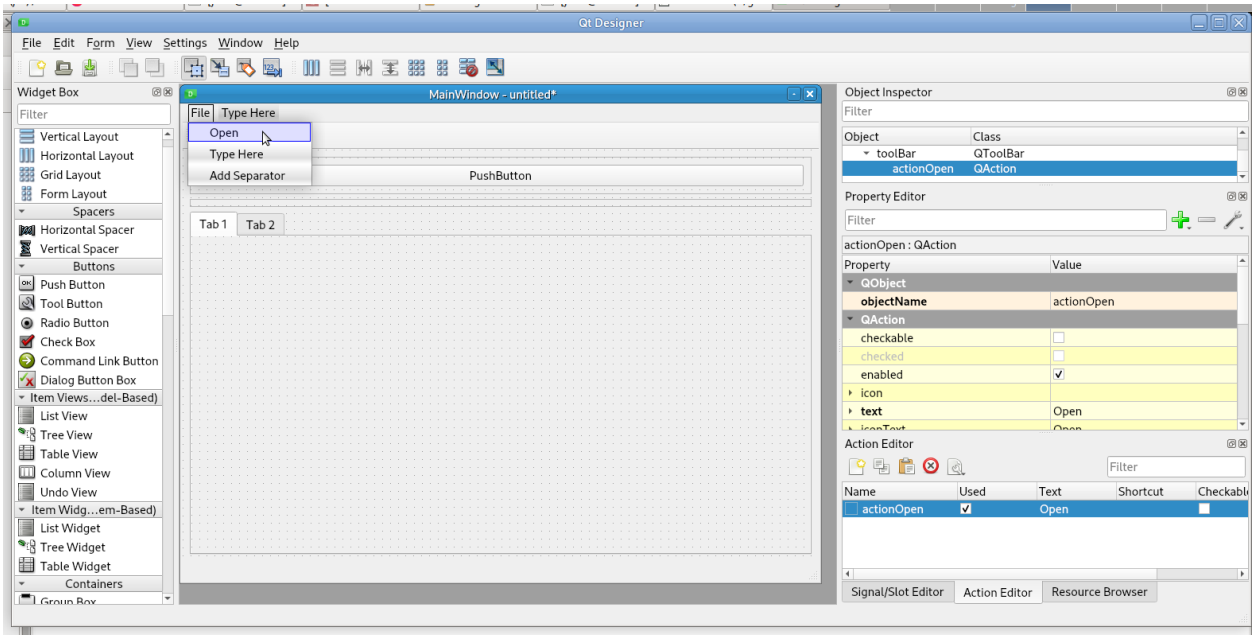
NINE

MENU

Adding Menu Items Tutorial Tool Bar Buttons Tutorial

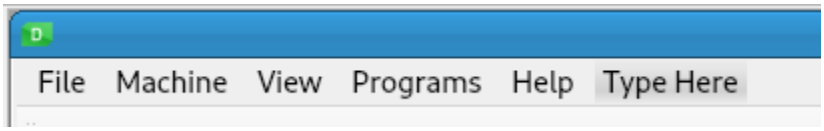
Note: Every menu item has a command button, so you don't need to use any menu items if you don't want to.

Adding a menu item creates an action. When you create File > Open menu, the *actionOpen* action is created.



Warning: If you use the full-screen option, you will not be able to exit the application if you don't have the Exit action or an Exit Push Button or Press ALT-F4 to close the GUI.

This shows the typical menu categories which are the first items in each menu. The image is from the Qt Designer.



The following table shows the menu name you type into Qt Designer and the action name that is created by the Qt Designer. Menu categories like *File* don't create an action name.

Table 1: **Menu Items**

File	**Action Name**
Open	actionOpen
Edit	actionEdit
Reload	actionReload
Save	actionSave
Save As	actionSave_As
Edit Tool Table	actionEdit_Tool_Table
Reload Tool Table	actionReload_Tool_Table
Ladder Editor	actionLadder_Editor
Quit	actionQuit
Machine	**Action Name**
E Stop	actionE_Stop
Power	actionPower
Run	actionRun
Run From Line	actionRun_From_Line
Step	actionStep
Pause	actionPause
Resume	actionResume
Stop	actionStop
Clear MDI History	actionClear_MDI_History
Copy MDI History	actionCopy_MDI_History
Save MDI	actionSave_MDI
Homing	this creates a home menu item for each axis
Unhoming	this creates a unhome menu item for each axis
Clear Offsets	this creates a clear offsets for each coordinate system
Programs	**Action Name**
Show HAL	actionShow_HAL
HAL Meter	actionHAL_Meter
HAL Scope	actionHAL_Scope
View	**Action Name**
DRO	actionDRO
Limits	actionLimits
Extents Option	actionExtents_Option
Live Plot	actionLive_Plot
Velocity	actionVelocity
Metric Units	actionMetric_Units
Program	actionProgram
Rapids	actionRapids
Tool	actionTool
Lathe Radius	actionLathe_Radius
DTG	actionDTG
Offsets	actionOffsets
Overlay	actionOverlay
Clear Live Plot	actionClear_Live_Plot
Grids	actionGrids
Help	**Action Name**

continues on next page

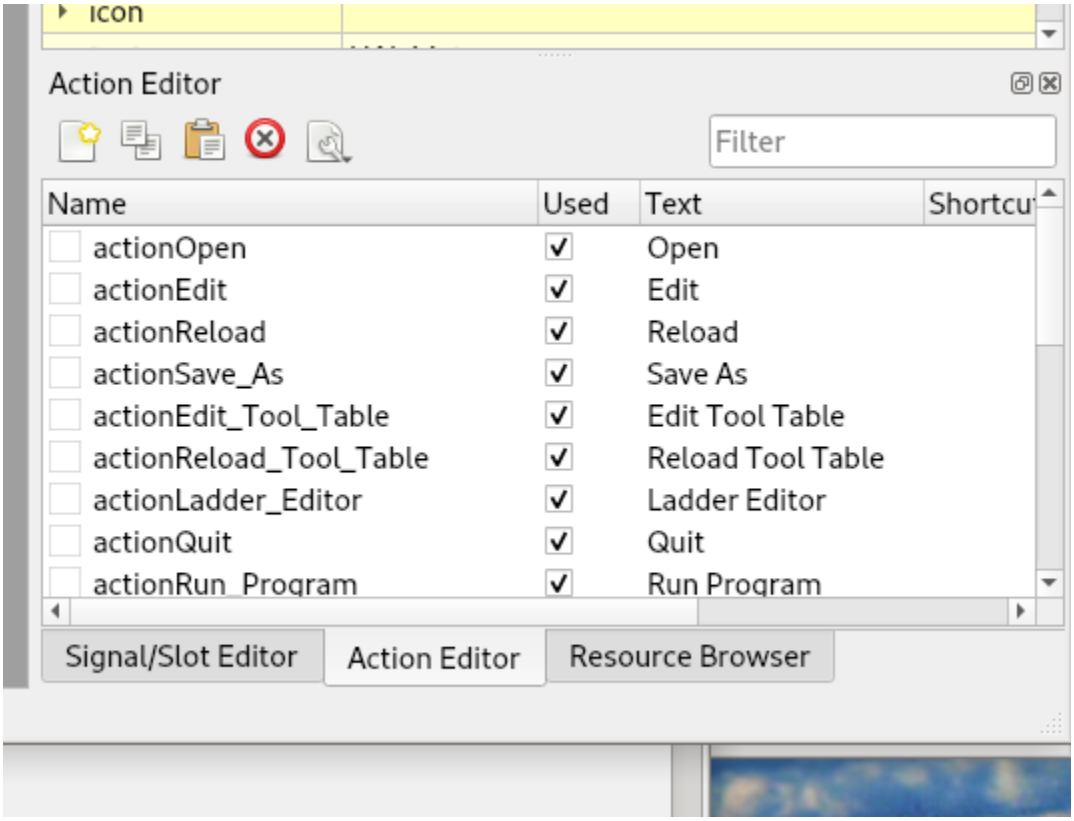
Table 1 – continued from previous page

About	actionAbout
Quick Reference	actionQuick_Reference

9.1 Action Names

When you add a menu item, it creates an action and the Object Name is created from the menu name automatically.

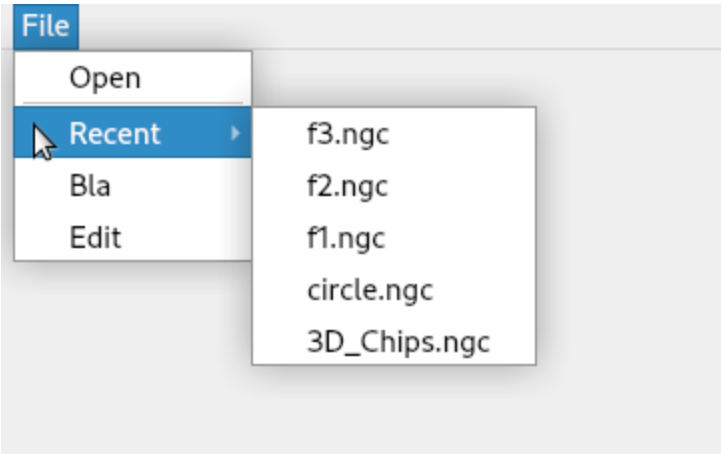
The Object Name must match the above items *exactly* in order to be discovered by Flex GUI:



9.2 Recent Files

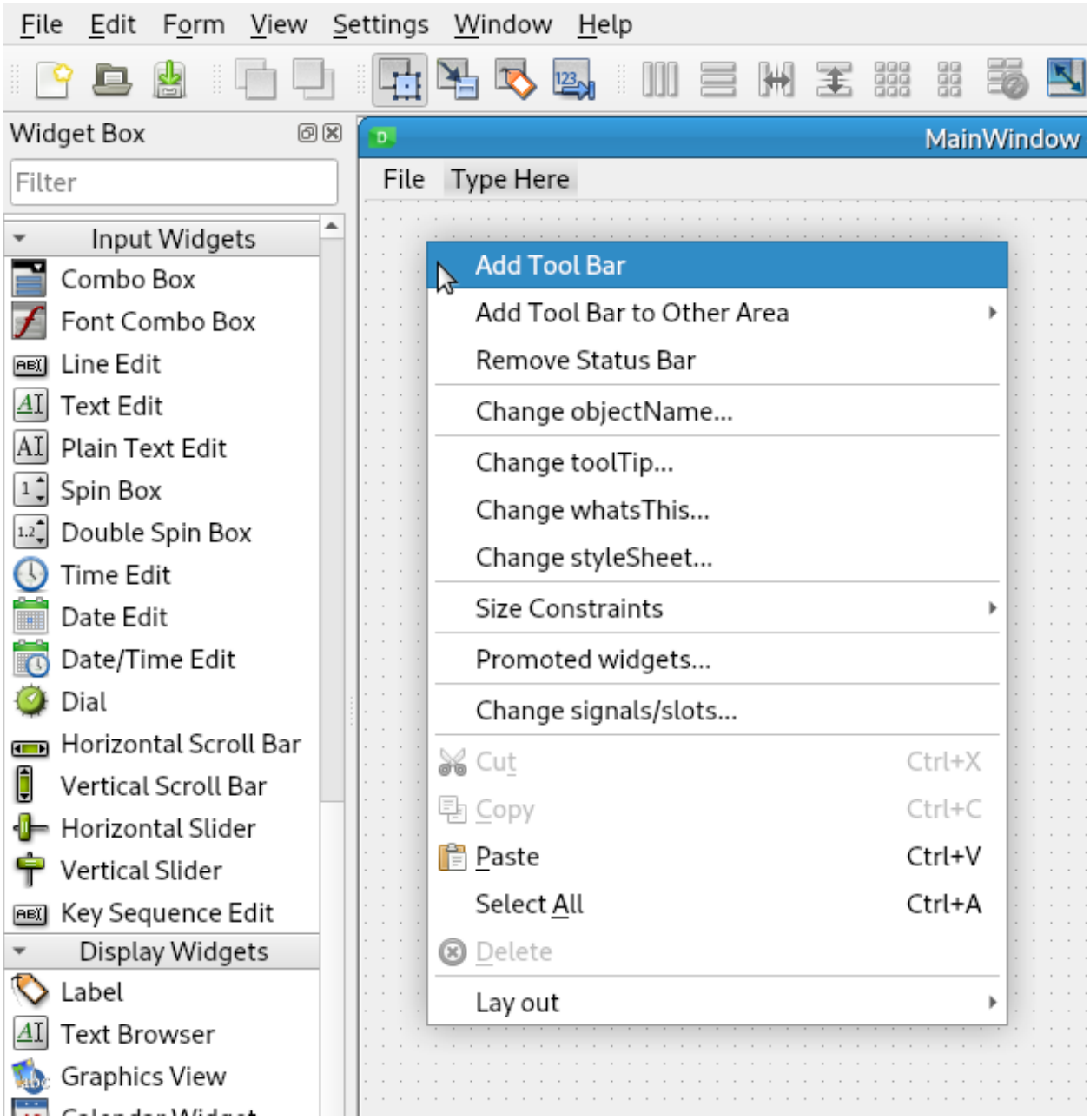
Note: The Recent menu item is added after the Open menu. There must be at least one menu item after Open for the Recent menu to be added.

Location of the Recent menu after the Open menu:

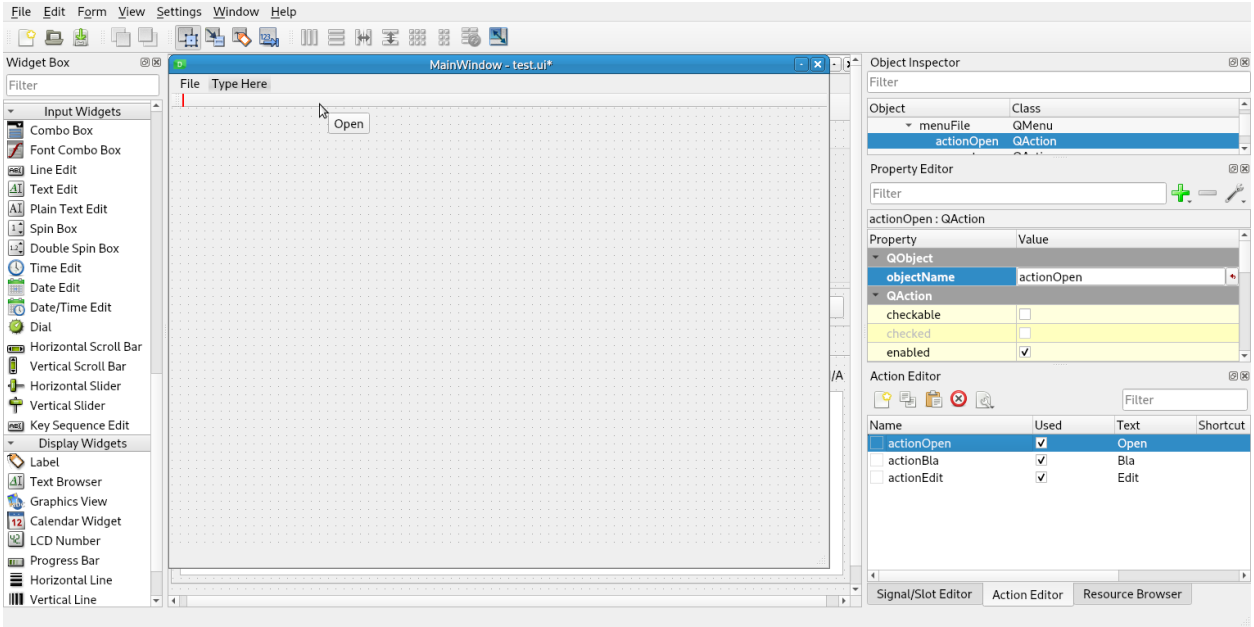


9.3 Tool Bars

If you right-click on the main window, you can add a Tool Bar:



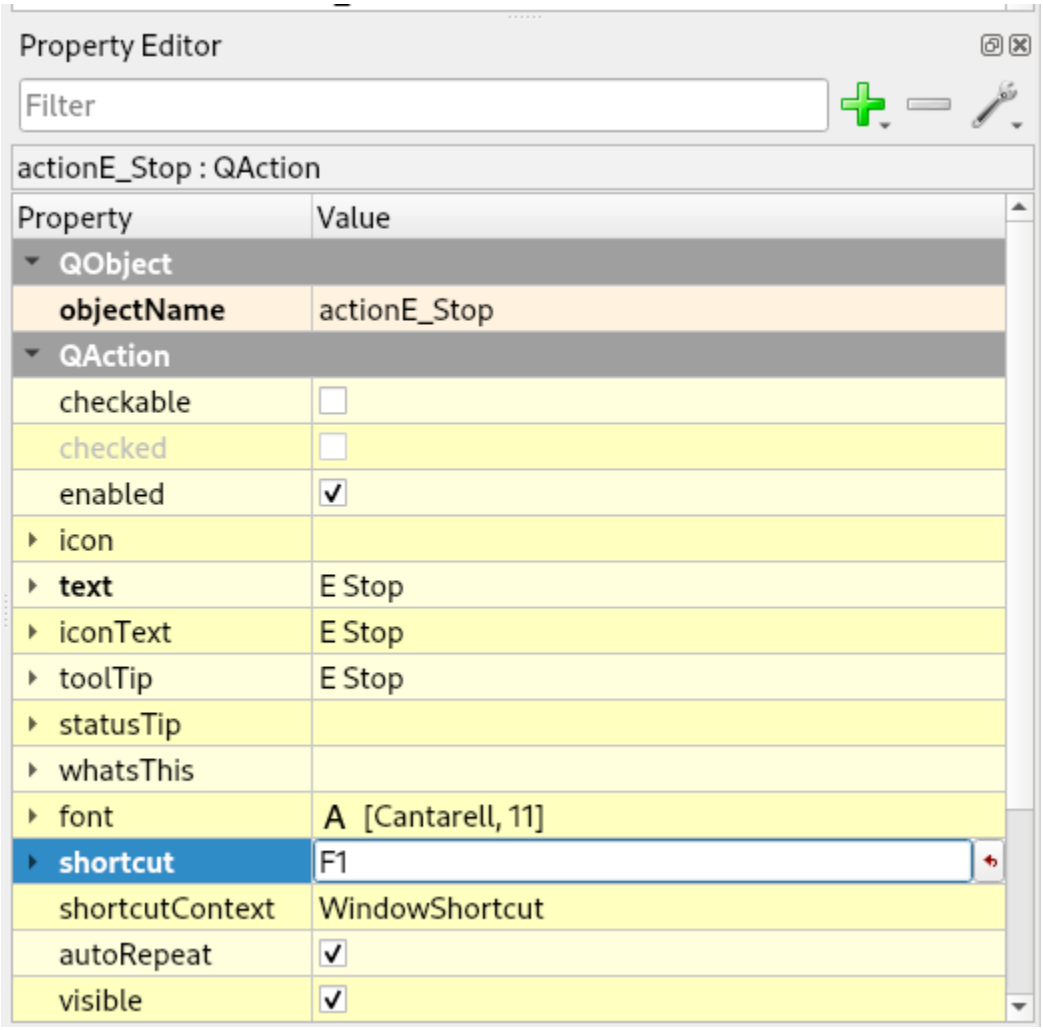
To add actions to the Tool Bar, drag them from the Action Editor and drop them in the Tool Bar:



To set the style of a Tool Bar Button, use the action name and replace action with *flex_* for example the actionQuit would be *flex_Quit*. See *Tool Bar Buttons* in the stylesheet examples.

9.4 Shortcut Keys

Shortcut keys can be added in the Property Editor by clicking in the shortcut Value box and pressing the key or key combination you want to use. You can change text, icon Text, or tool Tip also.



Note: A toolTip can be handy, however they might not work on touchscreens.

CHAPTER

TEN

CONTROLS

Command Buttons Tutorial Home Controls Tutorial

10.1 Push Buttons

Controls are QPushButtons that can be placed anywhere you like. Use the Name from the list below for each control widget objectName. Replace the *(0-8)* with the joint number or *Axis Index*. More controls are in *Tools*.

Table 1: **Control Push Buttons**

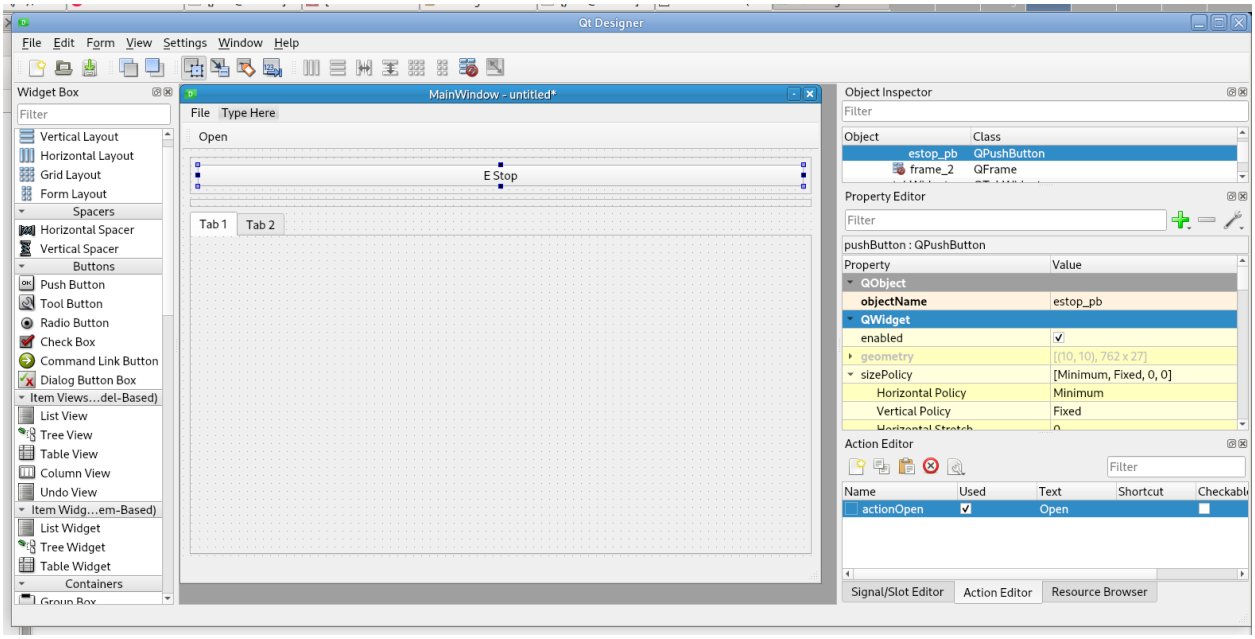
Control Name	**Object Name**	**Function**
Open	open_pb	Open a G-code File
Edit	edit_pb	Edit a G-code File
Reload	reload_pb	Reload a G-code File
Edit Tool Table	edit_tool_table_pb	Edit the tool table
Edit Ladder	edit_ladder_pb	Open the Ladder Editor
Reload Tool Table	reload_tool_table_pb	Reload the tool table
Save	save_pb	Save the current file
Save As	save_as_pb	Save the current file as new name
Quit the Program	quit_pb	Exit the Program
E-Stop	estop_pb	Toggle the E-Stop state
Power Toggle	power_pb	Toggle the Power state
Run	run_pb	Run a Loaded G-code File
Run From Line	run_from_line_pb	Run from the current selected line
Step	step_pb	Run one Logical Line
Pause	pause_pb	Pause a Running Program
Resume	resume_pb	Resume a Paused Program
Stop	stop_pb	Stop a Running Program
Home All	home_all_pb	Home All Joints if possible
Home	home_pb_(0-8)	Home a Joint (0-8)
Unhome All	unhome_all_pb	Unhome all Joints
Unhome	unhome_pb_(0-8)	Unhome a Joint (0-8)
Clear an Axis Offset	clear_(axis letter)_pb	Clear axis offset in the current
Manual Mode	manual_mode_pb	Change to Manual mode
Probe Enable	probing_enable_pb	Enables Probing and disables other controls
Flood	flood_pb	Toggle the flood output
Mist	mist_pb	Toggle the mist output
Clear Error History	clear_errors_pb	Clears the Errors text box
Copy Error History	copy_errors_pb	Copy the Errors to the clipboard

continues on next page

Table 1 – continued from previous page

Clear Information History	clear_info_pb	Clears the Information text box
Show HAL	show_hal_pb	Runs the Show HAL program
HAL Meter	hal_meter_pb	Runs the HAL Meter program
HAL Scope	hal_scope_pb	Runs the HAL Scope program
Help About	about_pb	Opens the About dialog
Quick Reference	quick_reference_pb	Opens the Quick Reference Dialog

Note: You don't have to use any of these controls; Flex GUI is flexible.



Note: Touch-Off buttons require a Double Spin Box named *touchoff_dsb*

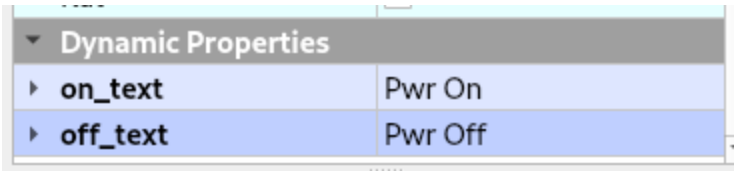
Note: Tool-Touch-Off buttons require a Double Spin Box named *tool_touchoff_dsb*

10.2 E Stop and Power

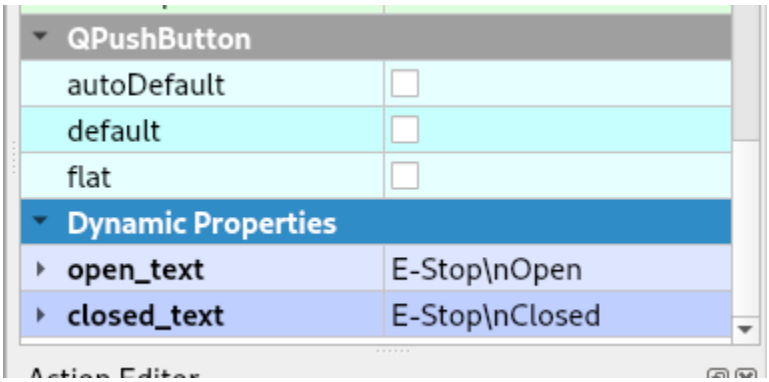
The E Stop push button Open/Closed state text can be set by adding two string type Dynamic Properties *off_text* and *on_text*. The text in those two properties will be used if found. See *Dynamic Properties*



The Power push button On/Off state text can be set by adding two string type Dynamic Properties *on_text* and *off_text*. The text in those two properties will be used if found.



Note: To have two words be above and below insert a backslash and n between the words.



This is how the above looks in the GUI



10.3 External E Stop

To add an external E Stop you need to change the HAL code. First you need to find and remove the following code

```
net estop-out <= iocontrol.0.user-enable-out
net estop-out => iocontrol.0.emc-enable-in
```

Next you need to add the estop_latch component and make the connections to iocontrol.

```
loadrt estop_latch
addf estop-latch.0 servo-thread
net estop-loopout iocontrol.0.emc-enable-in <= estop-latch.0.ok-out
net estop-loopin iocontrol.0.user-enable-out => estop-latch.0.ok-in
net estop-reset iocontrol.0.user-request-enable => estop-latch.0.reset
net remote-estop estop-latch.0.fault-in <= your.physical.input.pin.name
```

When the physical E Stop is on the estop-latch.0.fault-in is True that prevents the iocontrol from changing from ESTOP state to ESTOP_RESET state. If you have an E Stop button on the GUI the checked state is kept in sync with the state. If your physical E Stop is normally closed you need to reverse the logic with a not HAL component or change it to normally open.

10.4 LED Buttons

Some push buttons can have a LED to indicate on and off states. The LED is added to the push button with a Bool type Dynamic Property called *led_indicator*. All other properties are optional.

Table 2: **LED Buttons**

Control Name	Object Name	Function
E Stop	estop_pb	E Stop Toggle Button
Power	power_pb	Power Toggle Button
Run	run_pb	Runs a loaded NC file
Flood	flood_pb	Turns on the flood coolant
Mist	mist_pb	Turns on the mist coolant
Probe Enable	probing_enable_pb	Enables Probing and disables other controls
Home All	home_all_pb	Homes all the joints
Save	save_pb	Saves the NC Code to the same file
Save As	save_as_pb	Saves the NC Code to a new file

Adding the Bool type Dynamic Property *led_indicator* to the above control buttons will add the default round LED to the upper right corner of the button. Each control button can have different options. The default colors are Red when Off and Green when On.

Table 3: **LED Button Dynamic Properties**

Property Type	Property Name	Function
Bool	led_indicator	Creates a LED
Optional		
Int	led_diameter	Sets the Diameter of the LED in pixels
Int	led_right_offset	Sets the offset from the right edge in pixels
Int	led_top_offset	Sets the offset from the top edge in pixels
Color	led_on_color	Sets the color of the LED when on
Color	led_off_color	Sets the color of the LED when off

To change the LED default options they can be set in the INI file.

See *LED Settings* for more information.

Tip: A space after the button text gives more room for the LED

10.5 Coordinate System Controls

A QPushButton can be used to clear the current coordinate system by using 0 as the index or any one of the 9 coordinate systems with (1-9).

To clear the G92 coordinate system use 10 as the index.

To clear the current G54x coordinate system rotation use 11 as the index.

Table 4: **Clear Coordinate System Offsets**

Function	Object Name
Clear Current G5x	clear_coord_0
Clear G5x Coordinate System	clear_coord_(1-9)
Clear G92 Coordinate System	clear_coord_10
Clear Current G5x Rotation	clear_coord_11

To clear an axis offset in the current G5x coordinate system use the following.

Table 5: **Clear Axis Offsets**

Function	Object Name
Clear Current G5x X axis	clear_(axis letter)_pb

10.6 Options

The QPushButton options are toggle-type buttons; press to turn on, press again to turn off. They are normal push buttons but Flex automatically makes them *checkable*.

Table 6: **Options**

Function	Widget	Name
Flood Toggle	QPushButton	flood_pb
Mist Toggle	QPushButton	mist_pb
Optional Stop at M1	QPushButton	optional_stop_pb
Block Delete line that starts with /	QPushButton	block_delete_pb
Feed Override Enable/Disable	QPushButton	feed_override_pb

10.7 Axis Index

```
X 0
Y 1
Z 2
A 3
B 4
C 5
U 6
V 7
W 8
```

10.8 Overrides

Overrides Tutorial

A QSlider is used to control the following functions and the corresponding label shows the value of the slider:

Table 7: **Overrides**

Function	Widget	Object Name
Feed Override Slider	QSlider	feed_override_sl
Feed Override Percent	QLabel	feed_override_lb
Rapid Override Slider	QSlider	rapid_override_sl
Rapid Override Percent	QLabel	rapid_override_lb
Spindle Override Slider	QSlider	spindle_override_n_sl
Spindle Override Percent	QLabel	spindle_override_n_lb
Maximum Velocity	QSlider	max_vel_sl
Override Limits	QCheckBox	override_limits_cb

For more information on the Spindle Overrides see *Spindle* Documents

The following settings can be used in the DISPLAY section of the ini file:

```
Feed Override maximum            MAX_FEED_OVERRIDE
Spindle Override maximum         MAX_SPINDLE_OVERRIDE
```

10.9 Override Presets

Feed, Rapid and Spindle overrides can have a preset button(s) for different preset amounts. Replace the nnn with the percent of override you want that button to use. For Spindle Override presets see *Spindle Override Presets*

Table 8: **Override Presets**

Function	Widget	Object Name
Feed Override Preset	QPushButton	feed_percent_nnn
Rapid Override Preset	QPushButton	rapid_percent_nnn

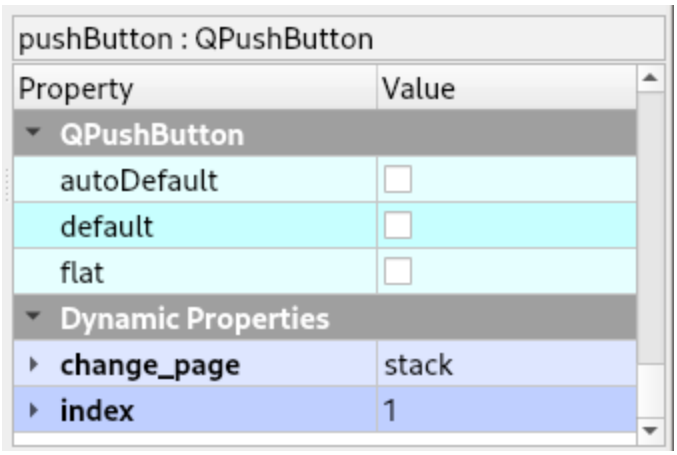
Note: The maximum override for Rapid is 100

10.10 Stacked Widget

To change to a specific page on a QStackedWidget add a QPushButton on each page and set a couple of Dynamic Properties. See *Dynamic Properties*

Table 9: **Stacked Widget Change Page**

Dynamic Property Name	Value
change_page	QStackedWidget Object Name
index	index of page to change to



To create a Next Page and Previous Page buttons for a QStackedWidget add two QPushButtons with the following Dynamic Properties. See *Dynamic Properties*

Table 10: **Stacked Widget Next/Previous Page**

Button Function	Dynamic Property Name	Value
Next Page	next_page	QStackedWidget Object Name
Previous Page	previous_page	QStackedWidget Object Name

Note: The Forward and Backward Buttons should not be in the QStackedWidget

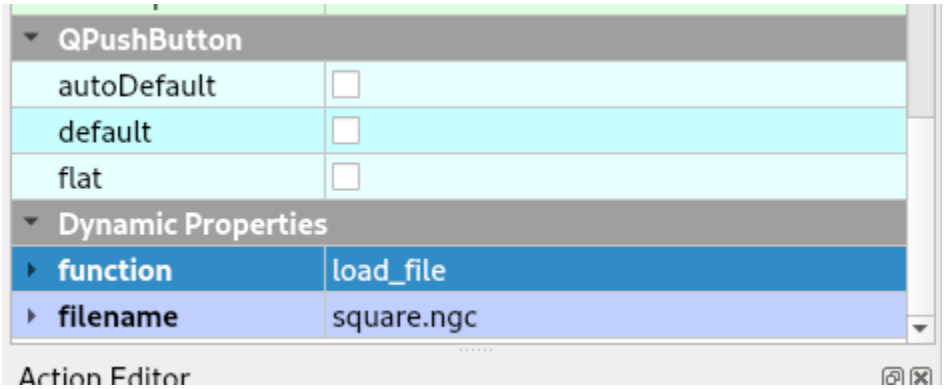
10.11 File Load Buttons

To create a QPushButton to load a specific file add two Dynamic Properties.

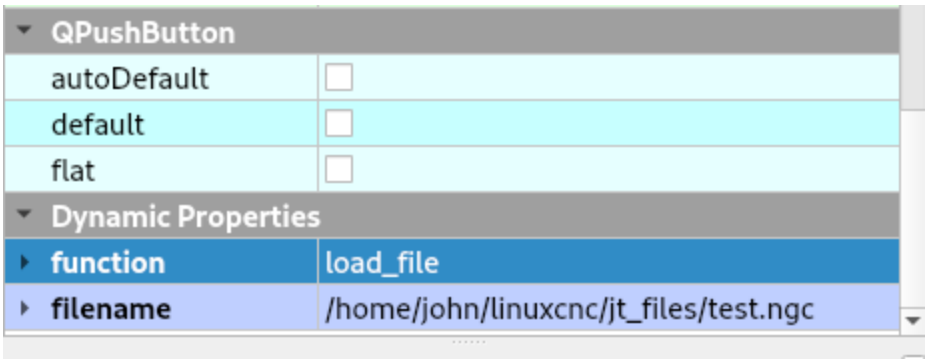
Table 11: **File Load Buttons**

Button Function	Dynamic Property Name	Value
Load File Button	function	load_file
Path to file	filename	file to load

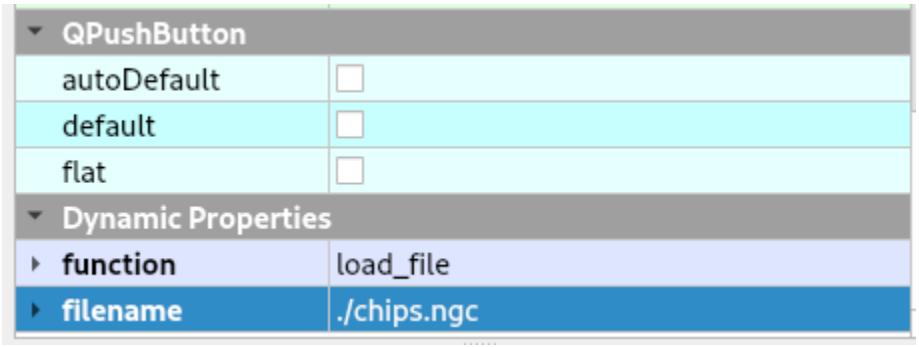
File Name in the PROGRAM_PREFIX path.



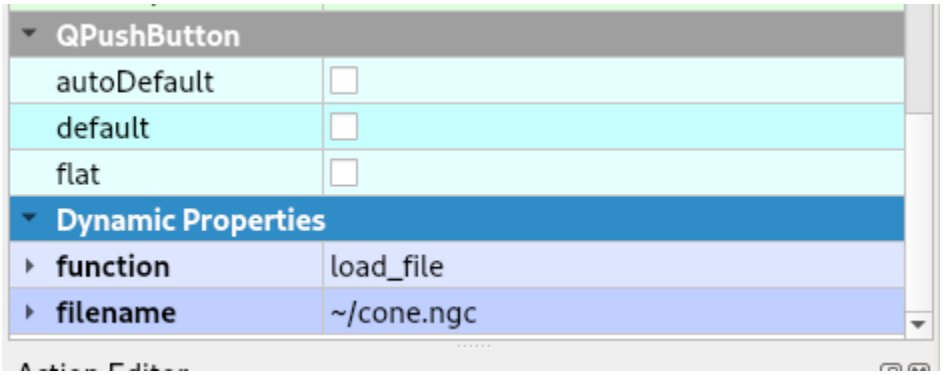
File Name with Full Path



File Name in the Configuration Directory



File Name relative to the Users Home Directory



The file name can be just the name and extension if it's in the PROGRAM_PREFIX path. Or it can be any valid path and file name.

File Name Examples

```
A file in the PROGRAM_PREFIX path
somefile.ngc

A file in the configuration directory
./anotherfile.ngc

A file up one directory from the configuration directory
../up_one.ngc

A file relative to the users home directory
```

(continues on next page)

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```
/home/fred/linuxcnc/my_files/afile.ngc
could be
~/linuxcnc/my_files/afile.ngc
```

Warning: The file must be in the directory specified by the INI entry PROGRAM_PREFIX in the [DISPLAY] section or have a valid path.

This is useful for probe routine buttons to load the nc code so the path can be viewed in the plotter and for programs that are ran frequently.

Note: The file is not added to the Recent Files list.

CHAPTER

ELEVEN

JOGGING

Jogging requires a *Jog Velocity Slider* and *Jog Mode Selector*. If either is not found, Jogging will be disabled.

Jogging increments are set the ini entry [FLEXGUI] JOG_INCREMENTS section. See *Jog Settings* for more information.

Table 1: **Required Jog Widgets**

Function	Widget	Name
Jog Velocity Slider	QSlider	jog_vel_sl
Jog Mode Selector	QComboBox	jog_modes_cb

The Jog Velocity Label shows the current jog velocity setting from the Jog Velocity Slider

Table 2: **Optional Jog Widgets**

Function	Widget	Name
Jog Velocity Label	QLabel	jog_vel_lb

11.1 Keyboard Jogging

Keyboard jogging uses the following keys to jog the X, Y, Z, A and B or C axes.

- Right Arrow jogs the X axis in the positive direction
- Left Arrow jogs the X axis in the negative direction
- Up Arrow jogs the Y axis in the positive direction
- Down Arrow jogs the Y axis in the negative direction
- Page Up jogs the Z axis in the positive direction
- Page Down jogs the Z axis in the negative direction
- Insert jogs the A axis in the positive direction
- Delete jogs the A axis in the negative direction
- Home jogs the B or C axis in the positive direction
- End jogs the B or C axis in the negative direction

Keyboard jogging can jog more than one axis at the same time.

There are two ways to enable keyboard jogging, the recommended way is to add the following to the ini file.

```
[FLEXGUI]
KEYBOARD_JOG = True
```

When keyboard jogging is enabled in this manner either Ctrl key must be pressed before pressing one of the jog keys. If you release the Ctrl key all axes will stop jogging.

The second way to enable keyboard jogging is to add a QCheckBox to turn keyboard jog on and off. Currently the jog keys work all the time when the QCheckBox is checked.

Table 3: **Keyboard Jogging**

Function	Widget	Name
Jog Enable	QCheckBox	keyboard_jog_cb

11.2 Keyboard Turbo Jog

The *Shift* key can be used to jog at turbo speed. The turbo speed is set in the INI file in Section FLEXGUI. The speed is in user units per second and it applies to all axes. If your configuration is in inches a value of 1.0 would be 60 inches per minute.

Releasing *Shift* key while jogging in Turbo mode will stop the jog.

```
[FLEXGUI]
KEYBOARD_SHIFT_VEL = 1.0
```

11.3 Keyboard Jog Increment

To use the keyboard to select the jog increment in the *jog_modes_cb* combobox add the following to the [FLEXGUI] section of the INI file. When keyboard jogging is enabled you can press and hold the Ctrl key while pressing a number key to select jog incrments in the *jog_modes_cb* combobox.

```
[FLEXGUI]
KB_JOG_INCREMENT = True
```

11.4 Auto Keyboard Jog Focus

When using the *keybaord_jog_cb* QCheckbox to control jogging, the arrow keys may conflict with some controls (eg. edit lines, text boxes, spin boxes, and other controls that use the arrow keys)

FlexGUI can disable keyboard jogging when the keyboard focus is in one of these controls, and re-enable jogging when focus leaves that control.

To enable this mode, add the following configuration to the ini file:

```
[FLEXGUI]
KB_JOG_FOCUS = True
```

Note: This mode cannot be used with the *KEYBOARD_JOG* mode, and requires the *keyboard_jog_cb* QCheckBox to function.

Note: Qt Designer has various "focus policies" and options for navigating the GUI using the keyboard. Be sure that you understand how these options interact with one another and with keyboard jogging. Test your GUI's keyboard jog and keyboard focus configuration thoroughly.

11.5 Jog Button Controls

Jog Controls Tutorial

This type of jog controls provides a button for each axis and jog direction.

Table 4: **Jog Button Widgets**

Function	Widget	Name
Jog Plus Axis (0-8)	QPushButton	jog_plus_pb_(0-8)
Jog Minus Axis (0-8)	QPushButton	jog_minus_pb_(0-8)

Note: Jog Plus/Minus buttons use the *Axis Index*. So *Jog Y Plus* is *jog_plus_pb_1*.

Note: *Jog Mode Selector* reads the ini entry [FLEXGUI] JOG_INCREMENTS and if not found, only *Continuous* will be an option.

11.6 Jog Selected Axis Controls

To add Axis style jog controls where you select an axis then the plus/minus buttons jog the selected axis add a QRadioButton for each axis and a QPushButton for Plus and Minus. Axes are 0-8 for X, Y, Z, A, B, C, U, V, W.

Table 5: **Jog Selected Widgets**

Function	Widget	Name
Axis Select (0-8)	QRadioButton	axis_select_(0-8)
Jog Plus	QPushButton	jog_selected_plus
Jog Minus	QPushButton	jog_selected_minus

11.7 Axis Index

```
X 0
Y 1
Z 2
A 3
B 4
C 5
U 6
V 7
W 8
```

CHAPTER

TWELVE

PLOTTER

Plotter Tutorial

To add a live G-code plotter, add a QWidget or QFrame and name it *plot_widget*.

12.1 Controls

If you're using a touch-screen, add pan, zoom, and rotate controls for the plotter

Table 1: **Display Controls**

Control	**Widget**	**Object Name**
Rotate View Up	QPushButton	view_rotate_up_pb
Rotate View Down	QPushButton	view_rotate_down_pb
Rotate View Left	QPushButton	view_rotate_left_pb
Rotate View Right	QPushButton	view_rotate_right_pb
Pan View Up	QPushButton	view_pan_up_pb
Pan View Down	QPushButton	view_pan_down_pb
Pan View Left	QPushButton	view_pan_left_pb
Pan View Right	QPushButton	view_pan_right_pb
Zoom View In	QPushButton	view_zoom_in_pb
Zoom View Out	QPushButton	view_zoom_out_pb

The following controls set-predefined views

Table 2: **Display Views**

Control	**Widget**	**Object Name**
View Perspective	QPushButton	view_p_pb
View X	QPushButton	view_x_pb
View Y	QPushButton	view_y_pb
View Y2	QPushButton	view_y2_pb
View Z	QPushButton	view_z_pb
View Z2	QPushButton	view_z2_pb

To clear the Live plot

Table 3: **Display Functions**

Control	Widget	Name
Clear Live Plot	QPushButton	view_clear_pb

12.2 Display

The DRO overlaid onto the plotter can be customized by turning on or off various features. Use either a QCheckbox or a QPushButton to toggle these

Table 4: **Display Checkbox Options**

Function	Widget	Object Name
View DRO	QCheckBox	view_dro_cb
View Machine Limits	QCheckBox	view_limits_cb
View Extents Option	QCheckBox	view_extents_option_cb
View Live Plot	QCheckBox	view_live_plot_cb
View Velocity	QCheckBox	view_velocity_cb
Use Metric Units	QCheckBox	view_metric_units_cb
View Program	QCheckBox	view_program_cb
View Rapids	QCheckBox	view_rapids_cb
View Tool	QCheckBox	view_tool_cb
View Lathe Radius	QCheckBox	view_lathe_radius_cb
View Distance to Go	QCheckBox	view_dtg_cb
View Offsets	QCheckBox	view_offsets_cb
View Overlay	QCheckBox	view_overlay_cb

Table 5: **Display PushButton Options**

Function	Widget	Object Name
View DRO	QPushButton	view_dro_pb
View Machine Limits	QPushButton	view_limits_pb
View Extents Option	QPushButton	view_extents_option_pb
View Live Plot	QPushButton	view_live_plot_pb
View Velocity	QPushButton	view_velocity_pb
Use Metric Units	QPushButton	view_metric_units_pb
View Program	QPushButton	view_program_pb
View Rapids	QPushButton	view_rapids_pb
View Tool	QPushButton	view_tool_pb
View Lathe Radius	QPushButton	view_lathe_radius_pb
View Distance to Go	QPushButton	view_dtg_pb
View Offsets	QPushButton	view_offsets_pb
View Overlay	QPushButton	view_overlay_pb

Note: Don't set the checked property to checked in Qt Designer as this is already handled in the code. Once you check an option it is remembered.

12.3 Menu

The following menu items can set display options. *Menu Name* is what you type when creating the Menu, then press enter. All the items are checkbox type menu items that stay coordinated with the checkbox of the same option.

Table 6: **Plot Menu Options**

Function	Menu Name	Object Name
View DRO	DRO	actionDRO
View Machine Limits	Limits	actionLimits
View Extents Option	Extents Option	actionExtents_Option
View Live Plot	Live Plot	actionLive_Plot
View Velocity	Velocity	actionVelocity
Use Metric Units	Metric Units	actionMetric_Units
View Program	Program	actionProgram
View Rapids	Rapids	actionRapids
View Tool	Tool	actionTool
View Lathe Radius	Lathe Radius	actionLathe_Radius
View Distance to Go	DTG	actionDTG
View Offsets	Offsets	actionOffsets
View Overlay	Overlay	actionOverlay

Note: Once a view selection has been set, Flex GUI remembers it.

The live plot can be cleared from the menu with this menu item.

Table 7: **Clear Live Plot Menu**

Function	Menu Name	Object Name
Clear Live Plot	Clear Live Plot	actionClear_Live_Plot

12.4 Grids

Note: Grids are not visible in any perspective view.

The plotter can optionally display a grid when in orthogonal views. To add the option of grids add a menu item named *Grids*. The grid size selection will be added to the *Grids* menu item. The default grid sizes are based on the machine units. The *Grids* menu can be part of the main menu bar, or added to a submenu.

Table 8: **Plot Menu Grids**

Function	Menu Name	Object Name
Plotter Grid	Grids	actionGrids

To configure the size options of the grid see the *Plotter Section* of the INI configuration documents.

The grid can be toggled on and off using a default grid size from the ini file.

```
[FLEXGUI]
DEFAULT_GRID = 0.5
```

Table 9: **Plot Grid Toggle**

Function	Widget	Object Name
Toggle Grid	QCheckBox	view_grid_cb

12.5 DRO

The font size can be set in the ini file by adding in the [FLEXGUI] section DRO_FONT_SIZE = n where n is an integer. The default size is 12.

CHAPTER

THIRTEEN

MANUAL DATA INPUT (MDI)

MDI Tutorial

13.1 MDI Interface

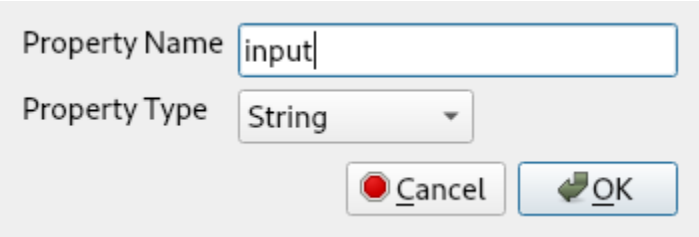
The MDI Interface uses a QLineEdit named *mdi_command_le* to enter commands.

For touch screens there are two options a *NC Popup* is a touch screen that has G and M words and a number keypad or a *Keyboard Popup* has a full keyboard.

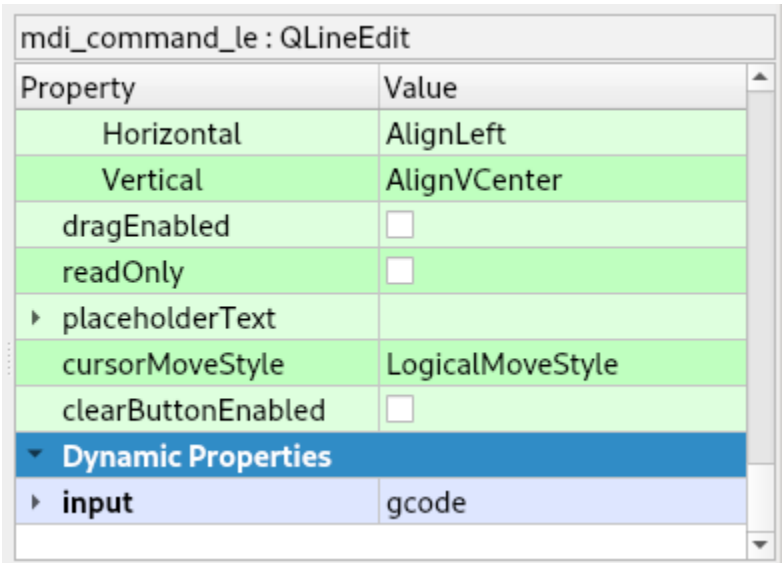
To enable a popup add a Dynamic string type Property to the *mdi_command_le* QLineEdit and name it *input* and set the value to either *nccode* or *keyboard*.

Note, QLineEdit widgets with object names that begin with 'mdi' do not respond to the *return_button* dynamic property as the keyboard and mouse behavior of these objects are handled internally by FlexGUI.

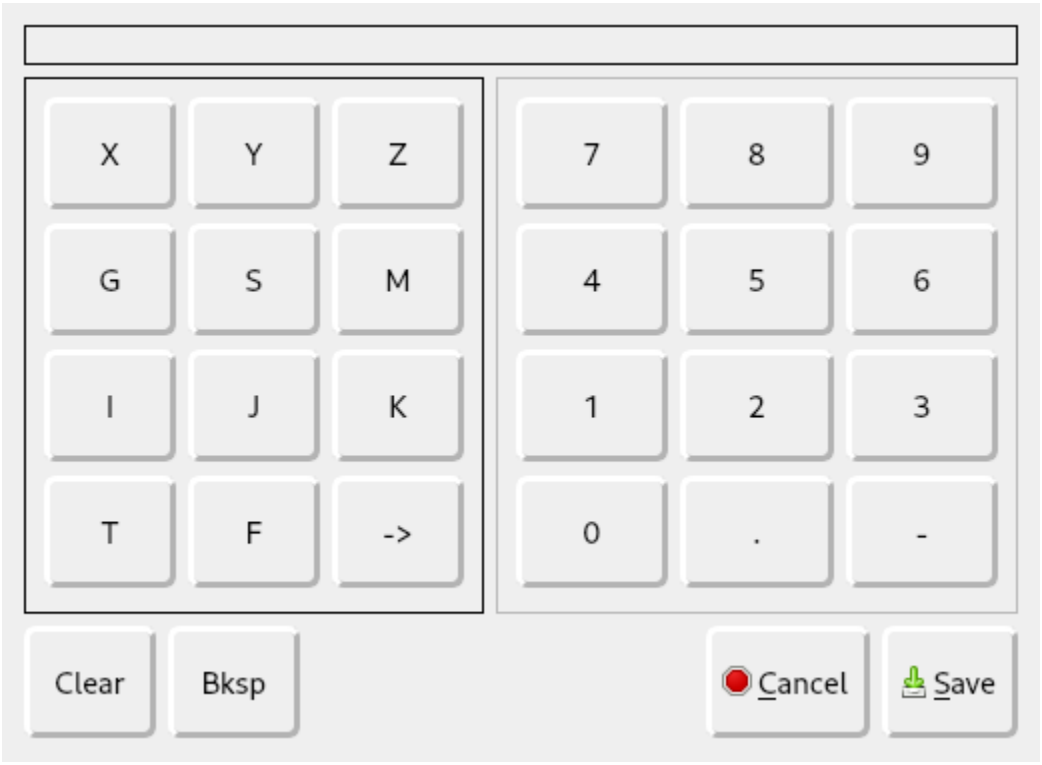
Dynamic Property



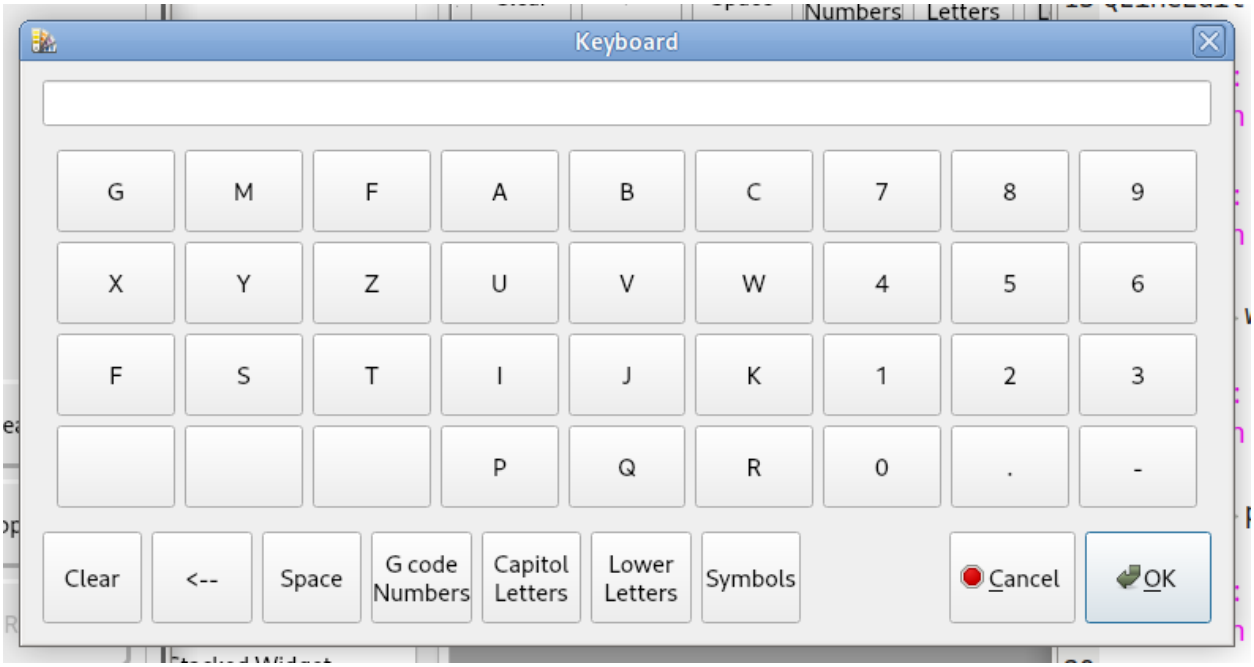
Setting the value



NC code popup window



Keyboard popup window



13.2 MDI History

MDI history uses a QListWidget named *mdi_history_lw* to display the MDI history. You can click on a line in the history display to copy the command to the MDI Interface, ready for running.



The MDI history is kept in a file named *mdi_history.txt* in the configuration directory.

13.3 MDI Controls

The following QPushButtons can be used to execute, copy, and clear MDI command history

Table 1: **MDI Push Buttons**

Function	Widget	Object Name
Run MDI Command	QPushButton	run_mdi_pb
Copy the MDI History to the Clipboard	QPushButton	copy_mdi_history_pb
Save the MDI History to a file	QPushButton	save_mdi_history_pb
Clear the MDI History	QPushButton	clear_mdi_history_pb

13.4 MDI Button

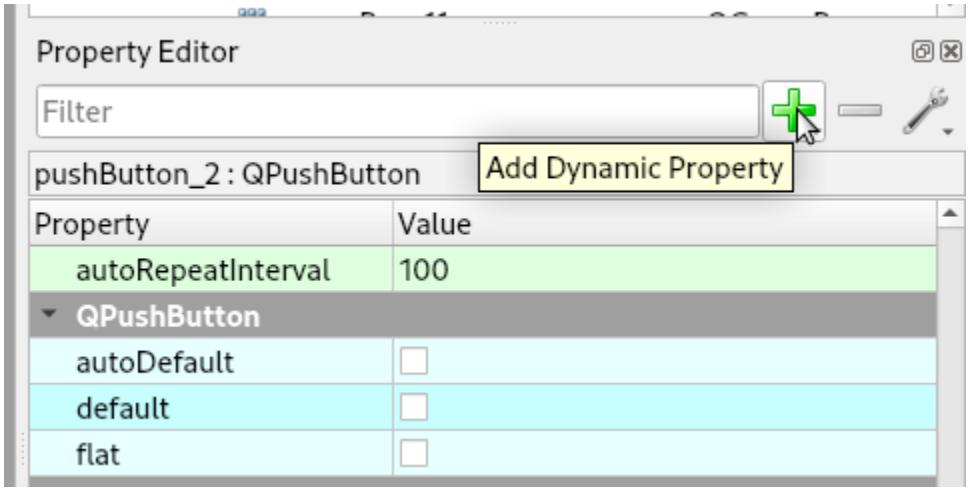
MDI buttons execute a MDI command when the button is pressed. These are created by adding two dynamic properties called *function* and *command* to a QPushButton.

Table 2: **MDI Command Button**

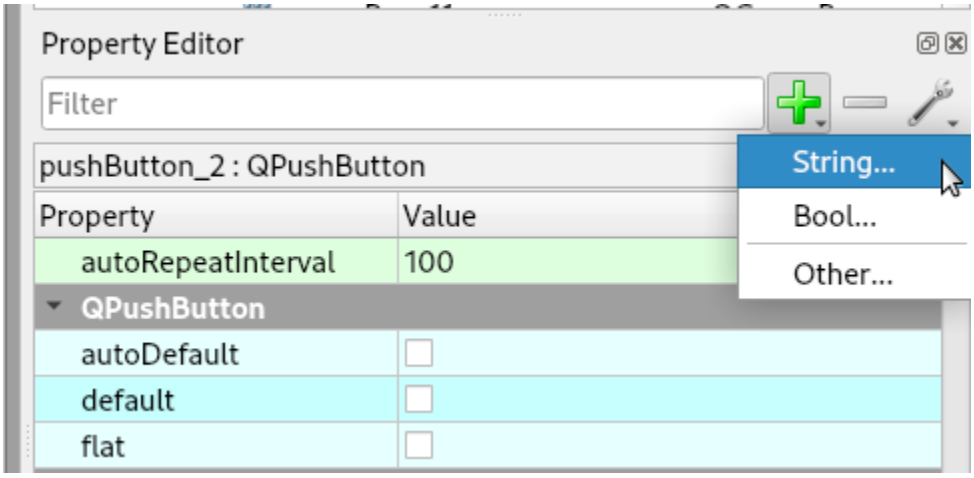
Property Type	Property Name	Pin Value
String	function	mdi
String	command	MDI command to execute

Note: If the *command* property is not found, the button will not be enabled!

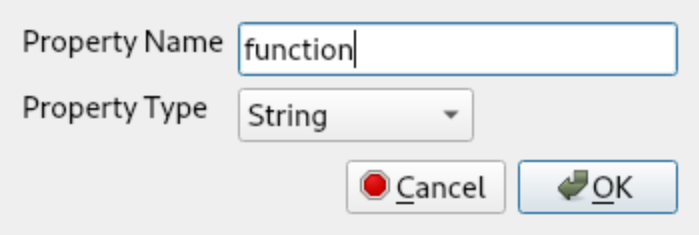
Select the button then create a dynamic property by pressing the green plus sign in the Property Editor



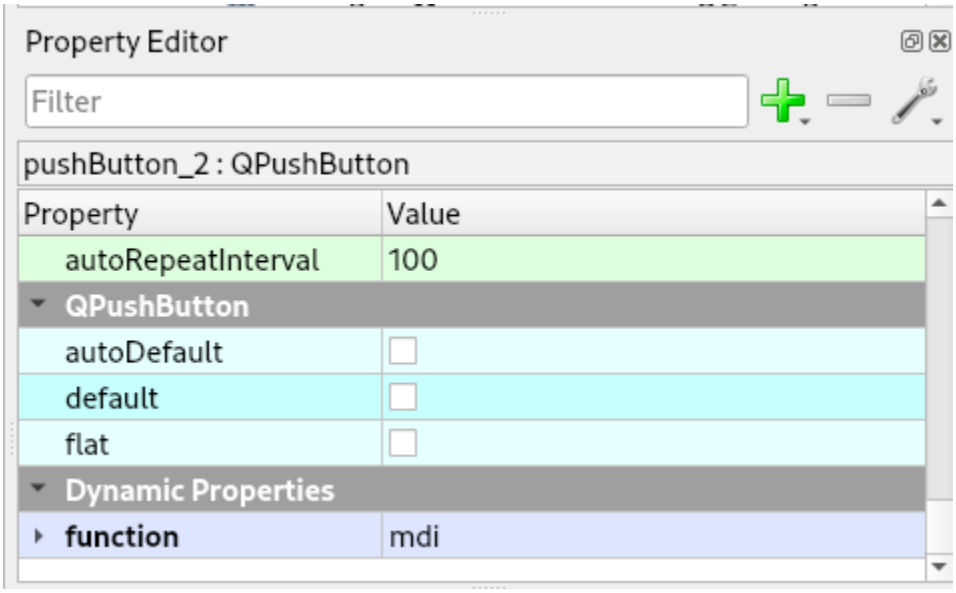
Then select *string*:



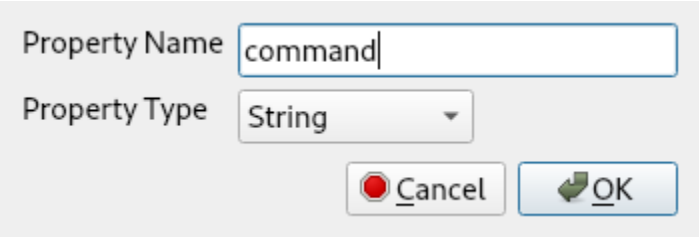
Name the property *function* and click OK



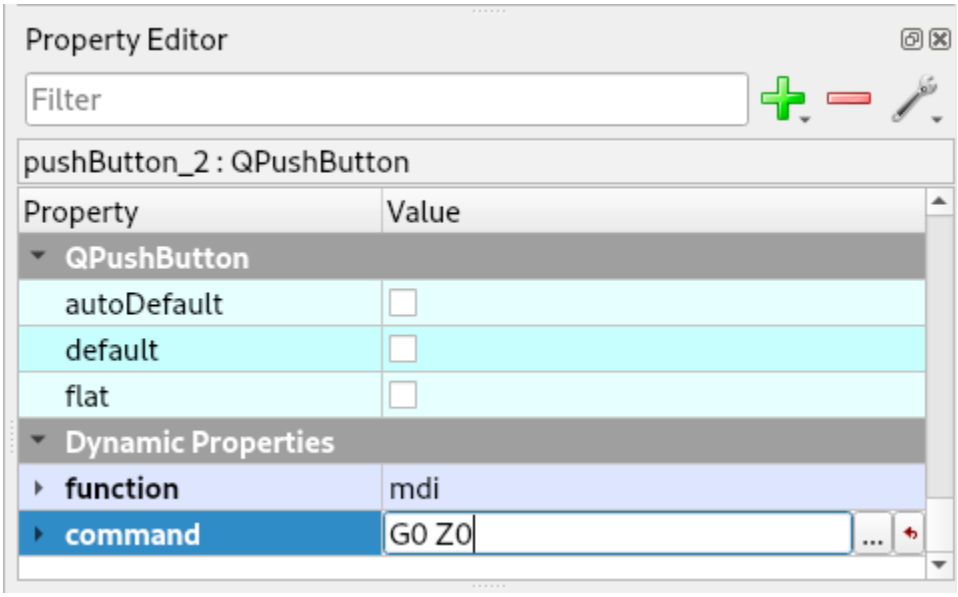
Set the value of the property to *mdi*



Add a property called *command*



Set the value of the property to your valid MDI command


Property Editor
Filter
pushButton_2 : QPushButton
Property
Value
QPushButton
autoDefault
default
flat
Dynamic Properties
function
mdi
command
G0 Z0
...

CHAPTER

FOURTEEN

SPINDLE

Video Tutorial

14.1 Multiple Spindles

Flex GUI can control and display status information for all 8 spindles in LinuxCNC. To configure more than one spindle you must add SPINDLES = 1 to 8 to the [TRAJ] section of the INI file.

In addition to the above you must add num_spindles=1 to 8 to the loadrt motmod line in your main hal file.

Optionally to control the spindle minimum RPM, maximum RPM, plus/minus increments and default RPM add a [SPINDLE_n] section to the ini file where *n* is the spindle number 0-7.

For more information see *Spindle INI Section*

See the LinuxCNC documents for more information about multiple spindles.

14.2 Spindle Status

Spindle status labels show the current status of the item. Linuxcnc can have up to 8 spindles. The spindles are numbered 0 through 7. If you have a single spindle it is spindle 0. Replace the *s* in the Object Name with the spindle number you want to monitor.

Table 1: **Spindle Status Labels**

Control Function	Object Type	Object Name
Spindle Brake	QLabel	spindle_brake_s_lb
Spindle Direction	QLabel	spindle_direction_s_lb
Spindle Enabled	QLabel	spindle_enabled_s_lb
Spindle Override Enabled	QLabel	spindle_override_enabled_s_lb
Spindle Speed	QLabel	spindle_speed_s_lb
Spindle Speed LCD	QLCDNumber	spindle_speed_s_lcd
Spindle Commanded Speed	QLabel	spindle_cmd_speed_s_lb
Spindle Override Percent	QLabel	spindle_override_s_lb
Spindle Orient State	QLabel	spindle_orient_state_s_lb
Spindle Orient Fault	QLabel	spindle_orient_fault_s_lb

Note: Replace the *n* with 0 - 7 to indicate which spindle to monitor

Note: Spindle Speed does not show override. Spindle Commanded Speed is commanded speed including override.

Note: The digitCount property of the LCD must be large enough to display the whole number.

Note: To get actual spindle speed you need to connect the spindle encoder to an encoder component and use the encoder.n.velocity-rpm float out connected to a HAL float label.

14.3 Spindle Controls

The following items control the spindle on/off direction and speed for a single spindle configuration. There are two ways to control the spindle speed, using a QSpinBox or a QSlider. You can have both the QSpinBox and the QSlider and changing one will update the other. The QSpinBox up/down arrow keys will change the value by the *INCREMENT* setting in the [SPINDLE_0] section down to the *MIN_RPM* and up to the *MAX_RPM*.

Table 2: **Single Spindle Controls**

Control Function	Object Type	Object Name
Spindle Forward	QPushButton	spindle_fwd_pb
Spindle Reverse	QPushButton	spindle_rev_pb
Spindle Stop	QPushButton	spindle_stop_pb
Spindle Faster	QPushButton	spindle_plus_pb
Spindle Slower	QPushButton	spindle_minus_pb
Spindle Speed	QSpinBox	spindle_speed_sb
Spindle Speed	QSlider	spindle_speed_sl

Note: The spindle can not be started with a spindle speed of zero.

The following items are manual spindle control each spindle up to 8 spindles. The spindles are numbered 0-7 with the default spindle being 0. Replace the *s* in the Object Name with the spindle number you want to control.

Table 3: **Multiple Spindle Controls**

Control Function	Object Type	Object Name
Spindle Forward	QPushButton	spindle_fwd_s_pb
Spindle Reverse	QPushButton	spindle_rev_s_pb
Spindle Stop	QPushButton	spindle_stop_s_pb
Spindle Faster	QPushButton	spindle_plus_s_pb
Spindle Slower	QPushButton	spindle_minus_s_pb
Spindle Speed	QSpinBox	spindle_speed_s_sb
Spindle Speed	QSlider	spindle_speed_s_sl

14.4 Set S Word

To set the *S* word to the current spindle RPM use the following button with the object name and the spindle number. Replace the *s* in the Object Name with the spindle number you want to set the *S* word for.

Table 4: **Set the S Word**

Control Function	Object Type	Object Name
Set S Word	QPushButton	set_speed_s_pb

Note: The Set S Word button uses MDI so all axes must be homed before it's enabled.

14.5 Spindle Overrides

If you just have one spindle you can use either the Multiple Spindle Override with n being 0 or the Single Spindle Override.

Table 5: **Single Spindle Override**

Control Function	Object Type	Object Name
Spindle Override	QSlider	spindle_override_sl

The spindle speed override is set using a QSlider. The Multiple Spindle Override is the preferred spindle override to use. The spindles are numbered 0-7 so replace the *s* in the Object Name with the number of the spindle you want to override.

Table 6: **Multiple Spindle Override**

Control Function	Object Type	Object Name
Spindle Override	QSlider	spindle_override_s_sl

See the Status Labels above for spindle override status labels.

14.6 Spindle Speed Presets

Each spindle can have QPushButtons configured to set a speed. Replace *s* in the Object Name with the spindle number and the *nnn* with the speed you want the spindle to change to when you press the button. The default spindle is *0*.

Table 7: **Spindle Speed Presets**

Function	Widget	Object Name
Spindle Speed Preset	QPushButton	spindle_set_s_nnn

14.7 Spindle Override Presets

A QPushButton can be used to set the spindle override to a set value. The value must be within the range of the override slider.

Table 8: **Single Spindle Override Presets**

Function	Widget	Object Name
Spindle Override Preset	QPushButton	spindle_percent_nnn

For multiple spindles you need to add the spindle number to the object name. Replace the *s* in the Object Name with the spindle number and the *nnn* with the override percentage. For example for spindle 2 a preset of 75% the QPushbutton name would be *spindle_percent_2_75*.

Table 9: **Multiple Spindle Override Presets**

Function	Widget	Object Name
Spindle Override Preset	QPushButton	spindle_percent_s_nnn

CHAPTER

FIFTEEN

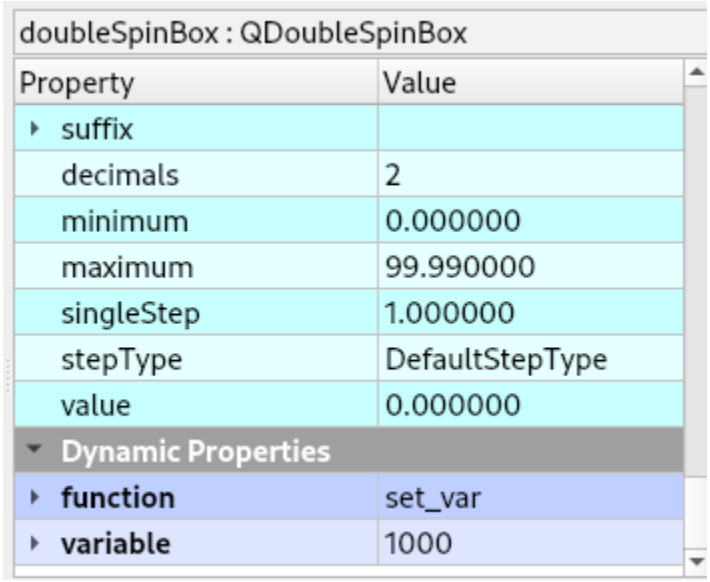
PARAMETERS

Parameter values in the var file can be set and watched for changes.

15.1 Setting Parameters

To set a user parameter value in the var file with a QDoubleSpinBox add a couple of string type Dynamic Properties. See *Dynamic Properties* The parameters 31 - 5000 are available for use in NC code programs. Replace *nnnn* with the variable number.

```
function set_var
variable `nnnn`
```



The user parameter must be in the var file. On startup Flex reads the var file and sets the value of the QDoubleSpinBox to that value. When you change the value of the QDoubleSpinBox the var file is updated with the new value. There is a 1 second timeout before the var file is updated to give time to type in a number.

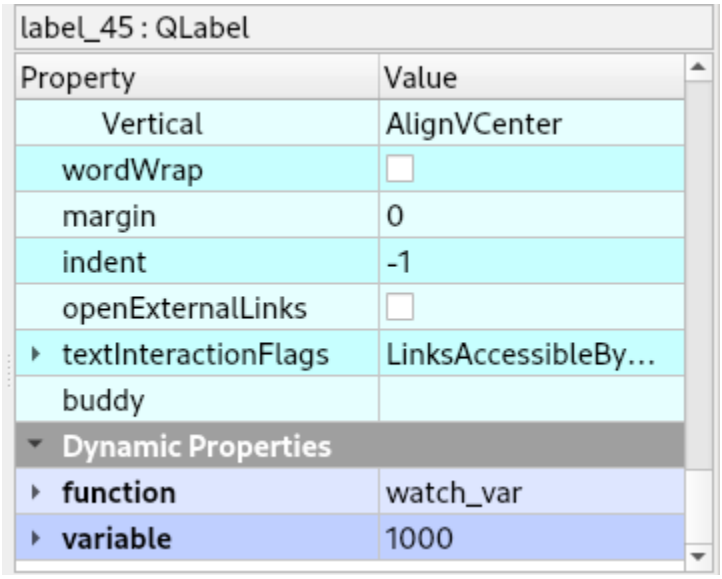
The configuration must be out of E-Stop, Power On and Homed before the QDoubleSpinBox is enabled.

Note: The Parameters are only updated when the mode is Manual

15.2 Watching Parameters

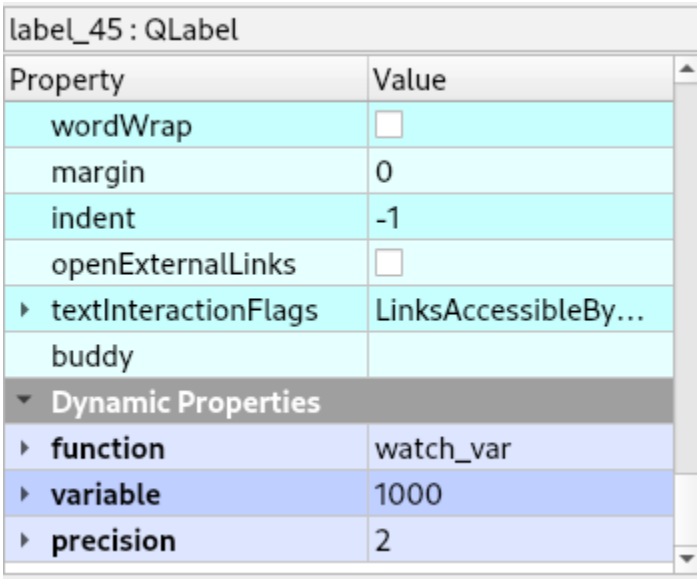
To watch the value of a user parameter a QLabel can be used with the following string type Dynamic Properties. See *Dynamic Properties*

```
function watch_var
variable `nnnn`
```



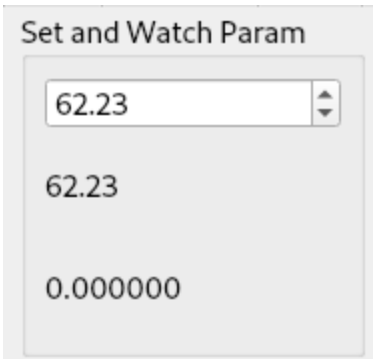
The value is updated at startup and any time the var file is updated.

The default precision for a parameter value is 6, to change the precision add a Dynamic Property called *precision* and set the value to the number of digits after the dot.



Note: The var file is not updated until the end of a NC file.

Example of setting and watching parameters


Set and Watch Param
62.23
62.23
0.000000

CHAPTER

SIXTEEN

PYTHON MODULE

Python Import Tutorial

To import a python module add the following to the INI [FLEXGUI] section using the name of the python file without the.py extension. The file name must be unique and can not be any python module name. You can have as many imports as you need to simplify your code.

```
[FLEXGUI]
IMPORT_PYTHON = testpy
```

Note: The module requires the .py extension to be able to import so the above module would be named testpy.py.

In each python file you import you must have a *startup* function where you make any connections from objects in the ui file to code in your module. The parent is passed to the startup function to give you access to all the objects in the GUI.

```
from functools import partial

def startup(parent):
        # connect a pushbutton without passing parent
        parent.my_test_pb.clicked.connect(test_1)
        parent.get_names_pb.clicked.connect(partial(get_names, parent))

        # connect a pushbutton and pass parent to the function
        parent.another_test_pb.clicked.connect(partial(test_2, parent))

def test_1():
        print('test 1')

def test_2(parent):
        # in this function you have access to all the objects in parent
        print(f'test 2 {parent.another_test_pb.text()}')

def get_names(parent):
        # get all the object names from the parent
        print(dir(parent))
```

16.1 Timer

A user timer is provided for use in the user python module.

```
from functools import partial

def startup(parent):
        parent.user_timer.timeout.connect(testit)
        parent.conn_pb.setEnabled(False) # prevent another connection
        parent.disc_pb.clicked.connect(partial(disc, parent))
        parent.conn_pb.clicked.connect(partial(conn, parent))
        parent.start_pb.clicked.connect(partial(start, parent))
        parent.stop_pb.clicked.connect(partial(stop, parent))

def testit():
        print('testing')

def disc(parent):
        parent.user_timer.timeout.disconnect(testit)
        parent.conn_pb.setEnabled(True) # allow a connection
        parent.disc_pb.setEnabled(False) # prevent trying to disconnect

def conn(parent):
        parent.user_timer.timeout.connect(testit)
        parent.conn_pb.setEnabled(False) # prevent trying to connect
        parent.disc_pb.setEnabled(True) # allow a disconnect

def start(parent):
        parent.user_timer.start(1000) # milliseconds

def stop(parent):
        parent.user_timer.stop()
```

CHAPTER

SEVENTEEN

PROBING

17.1 Probe Enable

Add a QPushButton named *probing_enable_pb* and if it is found it will be set as a toggle button. The button will only be enabled when the machine is homed and not running a program. The Probe Enable button will not function if a spindle is enabled.

Note: The *probing_enable_pb* requires at least one object that starts with *probe_* to be enabled.

17.2 Function

When the *probing_enable_pb* is toggled *OFF*, any widget with an object name that starts with *probe_* will be disabled.

When the *probing_enable_pb* is toggled *ON* the widgets that start with *probe_* will be enabled. In addition, spindle controls will disabled, spindle speed set to 0, run controls will be disabled, and MDI controls will be disabled.

QPushButtons with an objectName that start with *probe_* and configured as a *MDI Button* (to launch the probing subroutines) will be enabled when probing is enabled and disabled when probing is disabled.

You can create a *HAL Spin Box* to use in your probing subroutine. Set the objectName to start with *probe_* and it will be enabled and disabled with the probe buttons.

If you're using a touch-screen, the *QSpinBox* can add a Dynamic Property named *input* and set the value to *number*.

17.3 Stylesheet

The button is set to checkable in code so it can be styled with :checked and :enabled pseudo-states among others.

```
QPushButton#probing_enable_pb:enabled:checked {
        color: white;
        background-color: red;
}

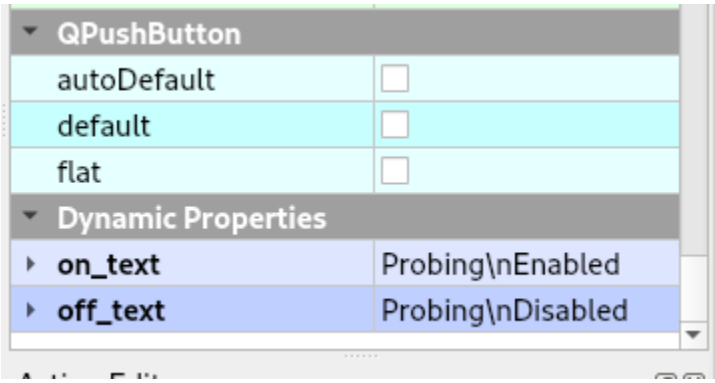
QPushButton#probing_enable_pb:enabled {
        color: green;
        background-color: yellow;
}
```

Note: A QPushButton with checkable set to true does not have an unchecked pseudo-state

For more style sheet options see the *StyleSheet* section

17.4 Button Text

The text on the Probe Enable button can be set by adding two Dynamic Properties named *on_text* and *off_text*. Both must be present or no change will take place as there is no default text for the Probe Enable button. See *Dynamic Properties*



This is what the button would look like with the above settings.

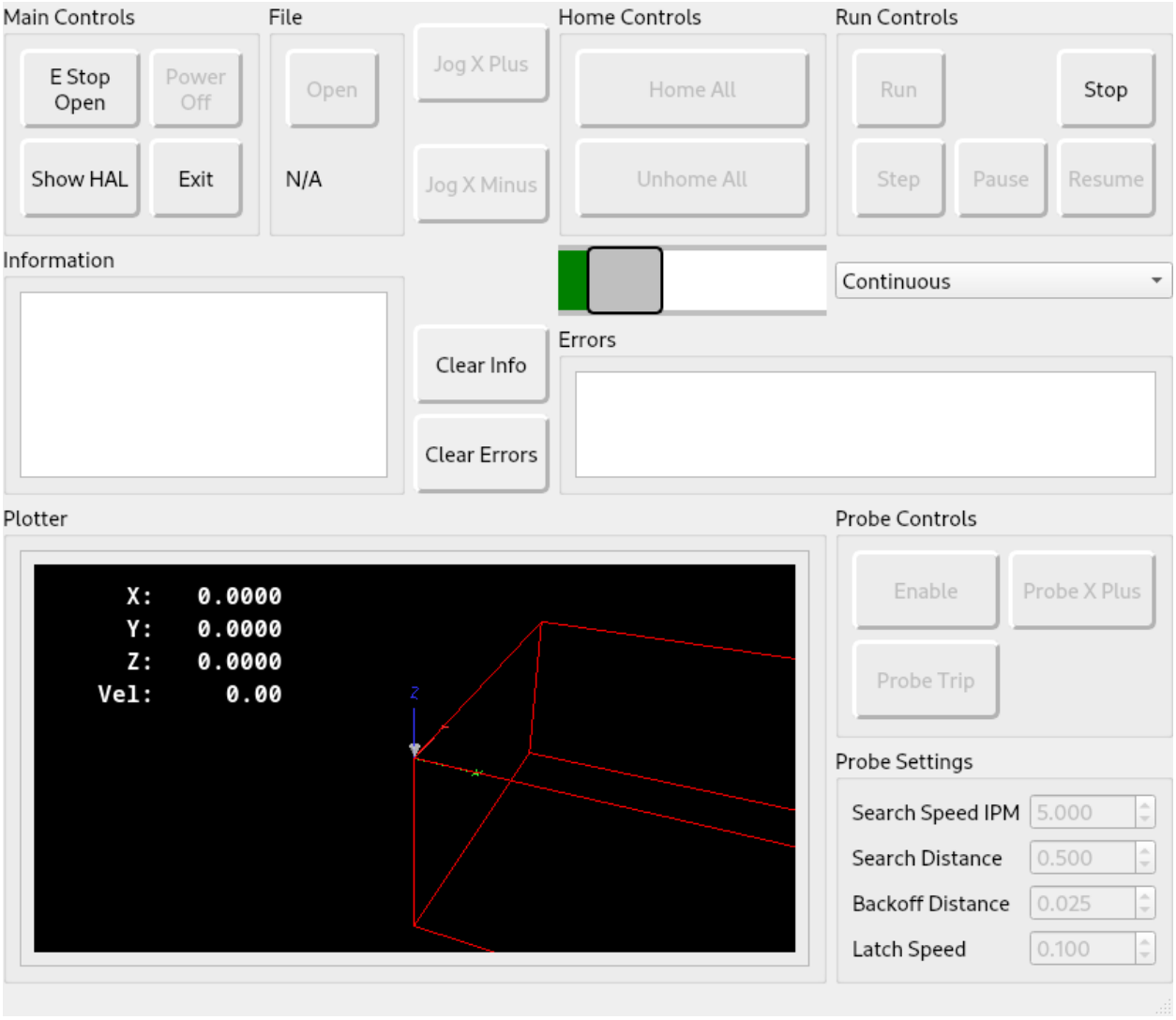


17.5 LED Indicator

The Probe Enable button can have a LED indicator. See *LED Buttons* for information on LED buttons

17.6 Example

A minimal example is in the Flex Examples in the Features directory



To run the example, close the E Stop, turn Power on, Home all, then toggle the Enable button. When probing is enabled many other controls are disabled including the spindle.

To test the probe routine press the Probe X Plus button and the X axis will start to move in the positive direction. If you do nothing when the Search Distance is reached you will get an error that the G38.2 move finished without making contact, which is expected.

Press the Probe X Plus button again and after it starts to move, press the Probe Trip button, the X axis will back off the Backoff Distance and start to move in the positive direction again. Pressing the Probe Trip button again will end the probe simulation. The debug information will show up in the Information window.

17.7 Subroutine

The probe subroutines can use the values from Probe Settings spin boxes or from user parameters. To use values from a spinbox you need to make the spin box a HAL pin. See the *HAL Spin Box* example in the HAL section.

The subroutine is located in a directory called *subroutines* that is in the configuration directory. The ini's [RS274NGC] SUBROUTINE_PATH sets the path that LinuxCNC looks for subroutines. Notice the leading ./ specifices that the path to the current directory is where the subroutine directory is.

```
SUBROUTINE_PATH = ./subroutines
```

The example files used are the following; notice that the xplus.ngc is in the ./subroutines directory

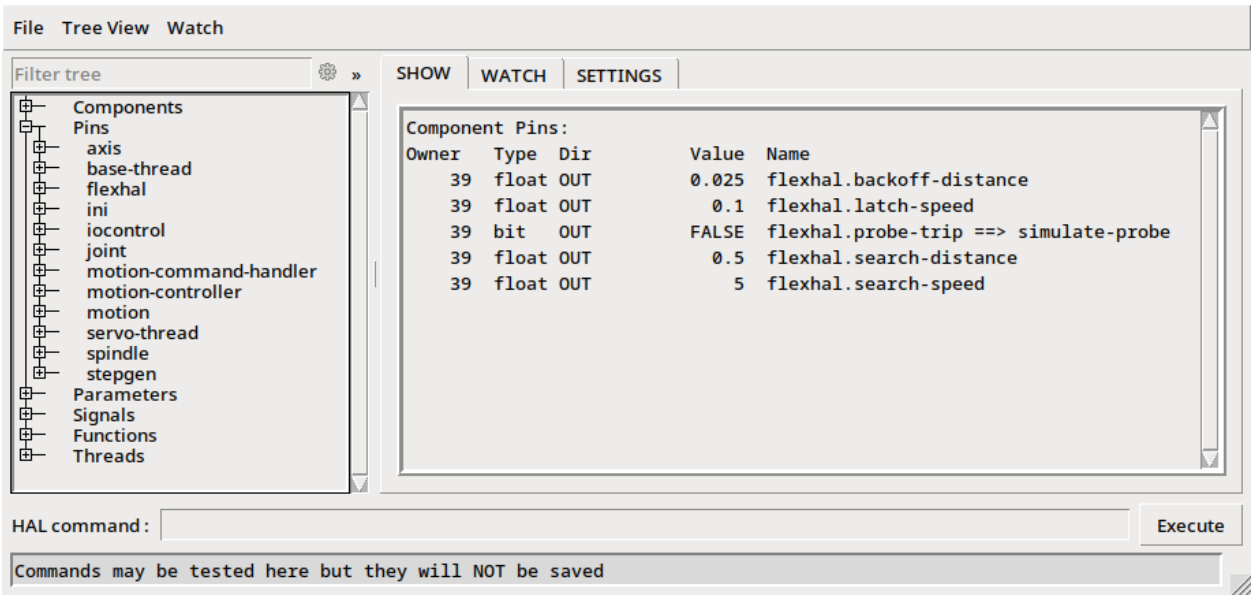
```
├── main.hal
├── parameters.var
├── parameters.var.bak
├── postgui.hal
├── probe.ini
├── probe.ui
├── README
├── sim_axis_probe.ini
├── subroutines
│   └── xplus.ngc
└── tool.tbl
```

In your subroutine you can use user parameters instead of using HAL pins. See the *Parameters* section.

The subroutine is a normal LinuxCNC subroutine. The magic is how you get the values from HAL pins with *#<_hal[pin_name]>* where pin_name is the actual pin name in HAL.

```
(filename xplus.ngc)
(HAL pins #<_hal[pin_name]>)
(G90 absolute distance mode G91 incremental distance mode)
o<xplus> sub
        (msg, xplus subroutine)
        G20
        ; initial search
        G91 G38.2 F#<_hal[flexhal.search-speed]> X#<_hal[flexhal.search-distance]>
        ;5061-5069 - Coordinates of a G38 probe result (X, Y, Z, A, B, C, U, V & W)
        (debug, Probe Contact at #5061)
        ; back off using #5061 to compensate for over travel on the probe
        G90 G0 X[#5061-#<_hal[flexhal.backoff-distance]>]
        ; final probe at latch speed
        G91 G38.2 F#<_hal[flexhal.latch-speed]> X[#<_hal[flexhal.backoff-distance]> + 0.
↪02]
        (debug, Probe Contact at #5061)
o<xplus> endsub
M2
```

Looking at the Halshow window which pops up when you press the Show HAL button, you can see the flexhal pin names for each spin box and for the Probe Trip button. Also notice that the Probe Trip button is connected to a signal which is connected to motion.probe-input in the postgui.hal file


File
Tree View
Watch
Filter tree
»
Components
Pins
axis
base-thread
flexhal
ini
iocontrol
joint
motion-command-handler
motion-controller
motion
servo-thread
spindle
stepgen
Parameters
Signals
Functions
Threads
SHOW
WATCH
SETTINGS
Component Pins:
Owner Type Dir Value Name
39 float OUT 0.025 flexhal.backoff-distance
39 float OUT 0.1 flexhal.latch-speed
39 bit OUT FALSE flexhal.probe-trip ==> simulate-probe
39 float OUT 0.5 flexhal.search-distance
39 float OUT 5 flexhal.search-speed
HAL command :
Execute
Commands may be tested here but they will NOT be saved

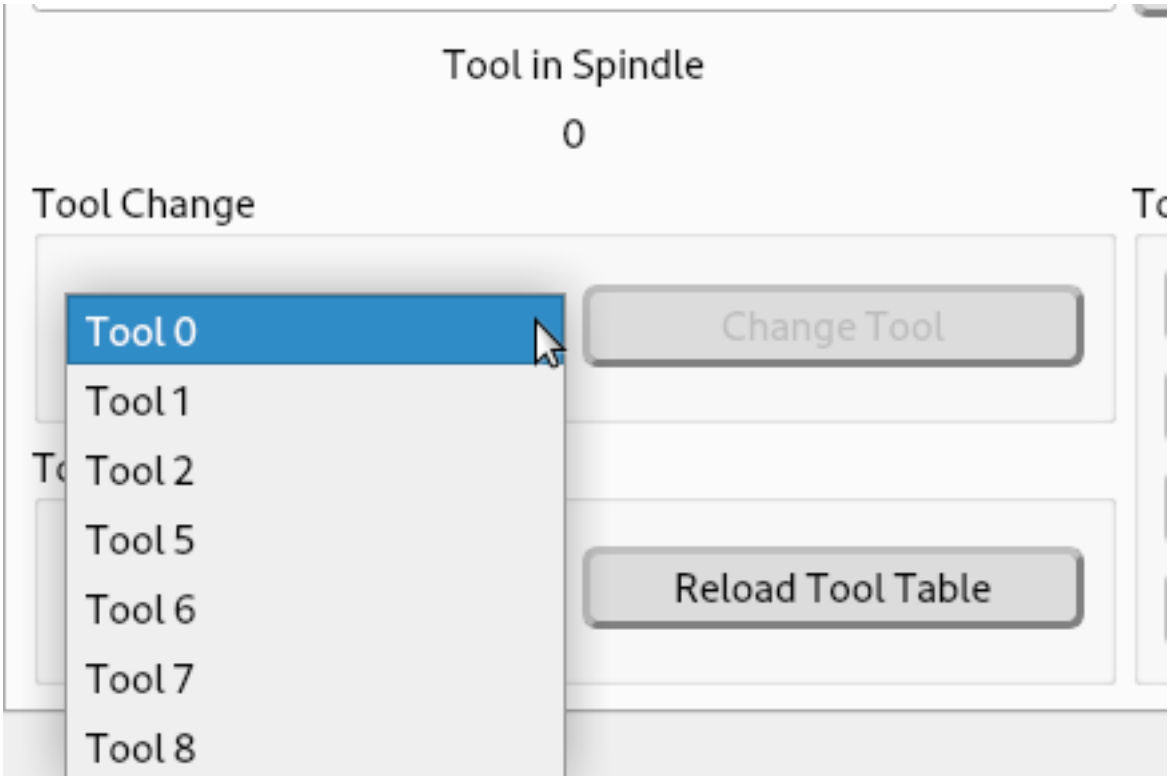
CHAPTER

EIGHTEEN

TOOLS

Tools Tutorial

18.1 Tool Change

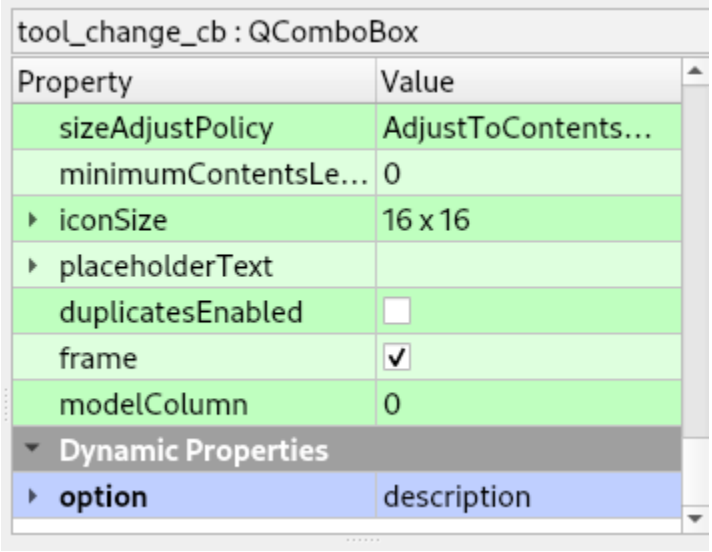


A tool change QPushButton, with a QComboBox to select the tool number to change to, is done with QPushButton named *tool_change_pb* and a QComboBox named *tool_change_cb*. The tool change combobox will automatically be populated with all the tools found in the tool table.

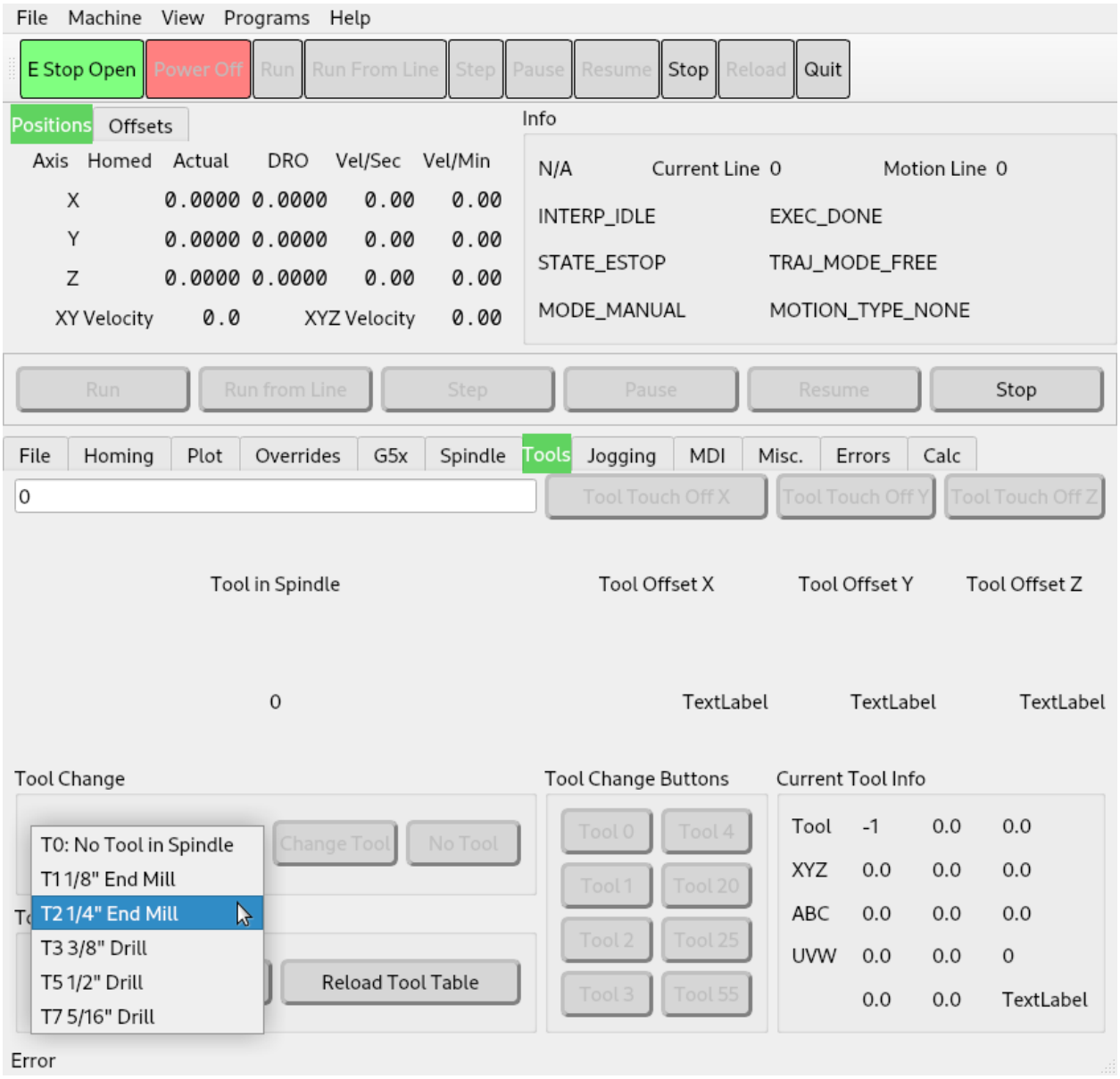
Table 1: **Tool Change Controls**

Control Function	Object Type	Object Name
Tool Change	QPushButton	tool_change_pb
Tool Selector	QComboBox	tool_change_cb

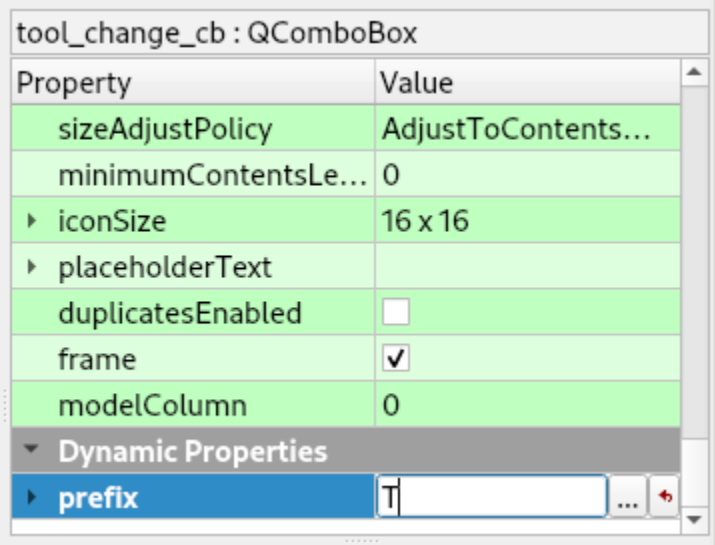
To add the description of the tools to the tool change combo box add a Dynamic Property named *option* and set the value to *description*. See *Dynamic Properties*



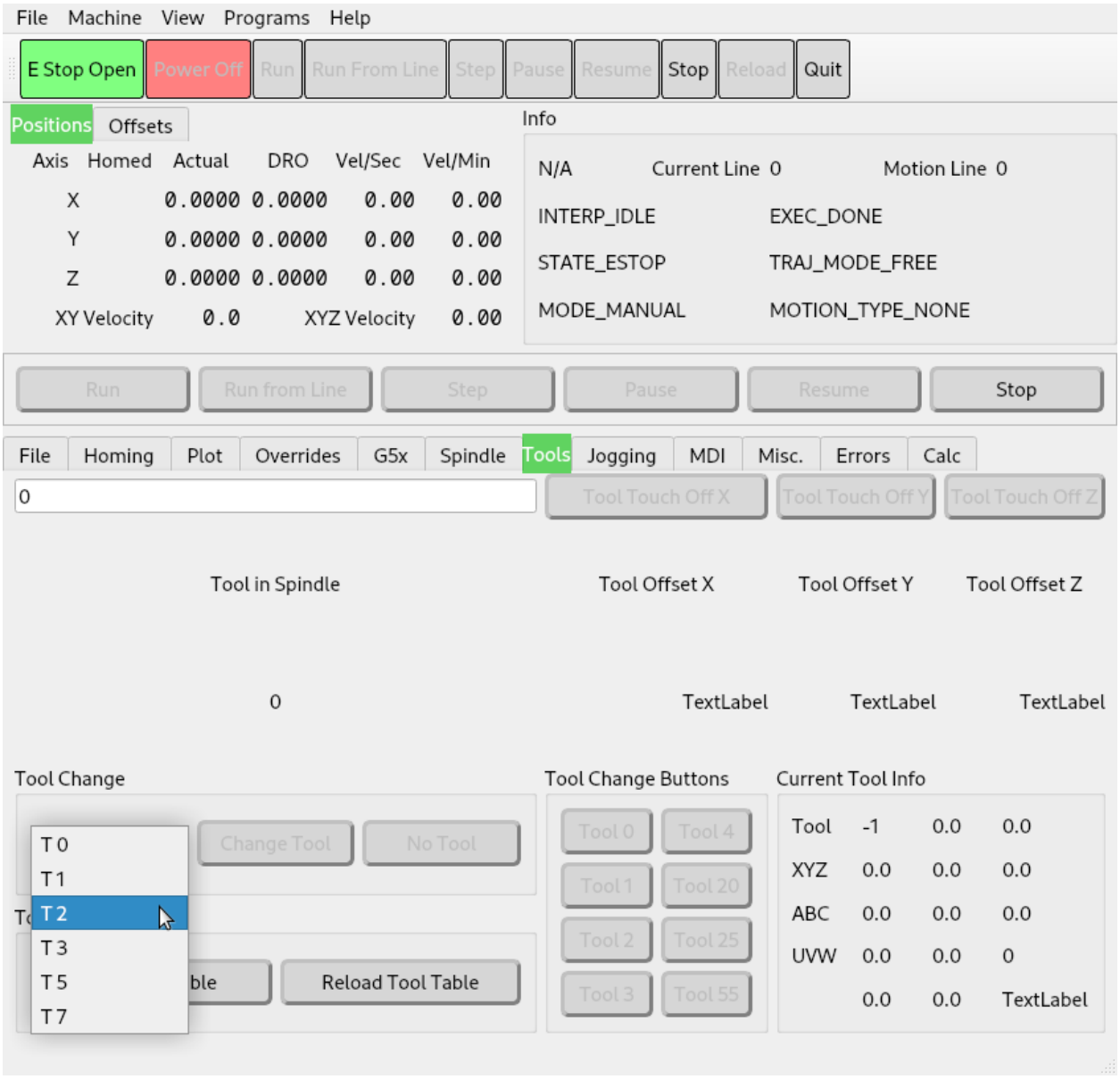
The description from the tool table will be appended to the tool number.



If you have limited space you can define the tool prefix by adding a Dynamic Property named *prefix* and set the value to the prefix you want.



The tool number will follow the prefix.



Note: Only one option can be used, if option is found it is used and prefix will be ignored.

18.2 Manual Tool Change

All that is needed to add a manual tool change is to add the following to the ini file in the [FLEXGUI] section.

```
[FLEXGUI]
MANUAL_TOOL_CHANGE = True
```

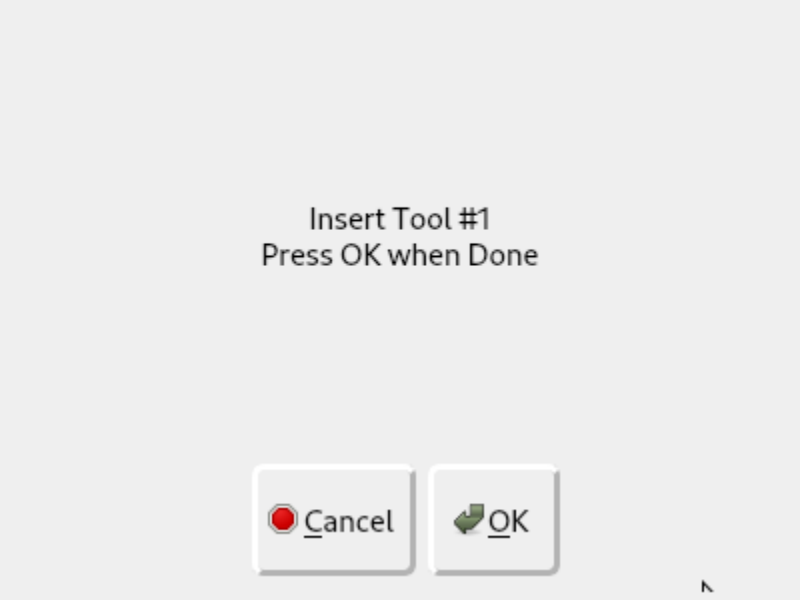


Fig. 1: This is without using a theme.



Fig. 2: This is the blue-touch theme.



Fig. 3: This is the dark-touch theme.

Warning: You can't use the hal_manualtoolchange at the same time as the built in Flex Manual Tool Change, you must comment out all the hal_manualtoolchange lines or remove them.

18.3 Input Accept

To use a HAL pin to accept the manual tool change connect your output pin to *flexhal.tc-ok* pin. The pin must be connected in the postgui.hal file. The manual tool change must be enabled in the INI file as shown above.

Note: The pin is only connected to the function during a manual tool change.

18.4 Manual Tool Change Error

If you get an error that hal_manualtoolchange component exists look in your hal files for the hal_manualtoolchange lines in the Manual Tool Change Option below.

If you're using a copy of one of the Axis sims the hal_manualtoolchange component can be hard to find. It's recommended that you start with a simple configuration like the Flex example simple-sim.

18.5 Manual Tool Change Option

The HAL Manual Tool Change requires at least the following HAL code in the main hal file if not using the builtin Flex Manual Tool Change above.

```
# manual tool change
loadusr -W hal_manualtoolchange
net tool-change iocontrol.0.tool-change => hal_manualtoolchange.change
net tool-changed iocontrol.0.tool-changed <= hal_manualtoolchange.changed
net tool-number iocontrol.0.tool-prep-number => hal_manualtoolchange.number
net tool-prepare-loopback iocontrol.0.tool-prepare => iocontrol.0.tool-prepared
```

18.6 Tool Change Button

Tool change QPushButtons can be used to change tools without a spinbox by adding up to 99 QPushButtons named *tool_change_pb_n*. With *n* being the number of the tool you wish to change to using that button

Table 2: **Tool Change Buttons**

Control Function	Object Type	Object Name
Tool Change Button	QPushButton	tool_change_pb_(n)

18.7 Tool Touchoff

To touch-off a tool to an axis, use a tool-touch-off QPushButton and a QLineEdit to enter the value of the touch off.

Table 3: **Tool Touchoff Controls**

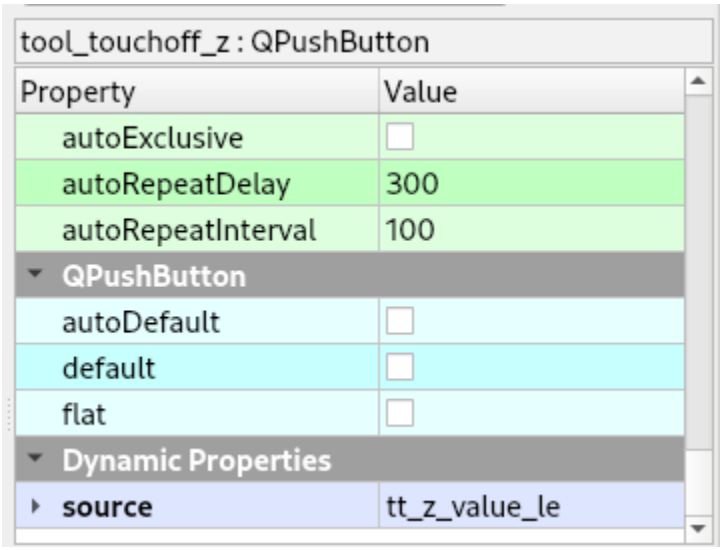
Control Function	Object Type	Object Name
Tool Touch Off Value	QLineEdit	tool_touchoff_le
Tool Touch Off	QPushButton	tool_touchoff_(axis letter lower case)

Optionally you can have a QLineEdit for each axis for tool touch off. Add a Dynamic Property named *source* to the tool touch off button and set the value to the name of the QLineEdit that is the source for that touch off button. See *Dynamic Properties*

Optionally, a QLineEdit can also have a dynamic property *return_button* with the objectName of a QPushButton that you want to receive a click event when the Enter Key is pressed, or when a value is saved using a touch popup.



Tool touch off QLineEdit for each axis.



18.8 Tool Touchoff Selected Axis

To have Axis style tool touch off add a QPushButton named *tool_touchoff_selected_pb*. You must have at least one QRadiobutton for an axis to select.

Table 4: **Tool Touchoff Selected Widgets**

Function	Widget	Name
Axis Select (0-8)	QRadioButton	axis_select_(0-8)
Tool Touchoff	QPushButton	tool_touchoff_selected_pb

18.9 Current Tool Status

Current Tool status of the tool loaded in the spindle. All the labels can have a Dynamic Property called *precision* with the number of digits you wish to show. The *tool_id_lb* and the *tool_orientation_lb* are integers.

Table 5: **Tool Table Status Labels**

tool_id_lb	tool_xoffset_lb	tool_yoffset_lb
tool_zoffset_lb	tool_aoffset_lb	tool_boffset_lb
tool_coffset_lb	tool_uoffset_lb	tool_voffset_lb
tool_woffset_lb	tool_diameter_lb	tool_frontangle_lb
tool_backangle_lb	tool_orientation_lb	

CHAPTER

NINETEEN

COORDINATE SYSTEMS

Coordinate System Tutorial

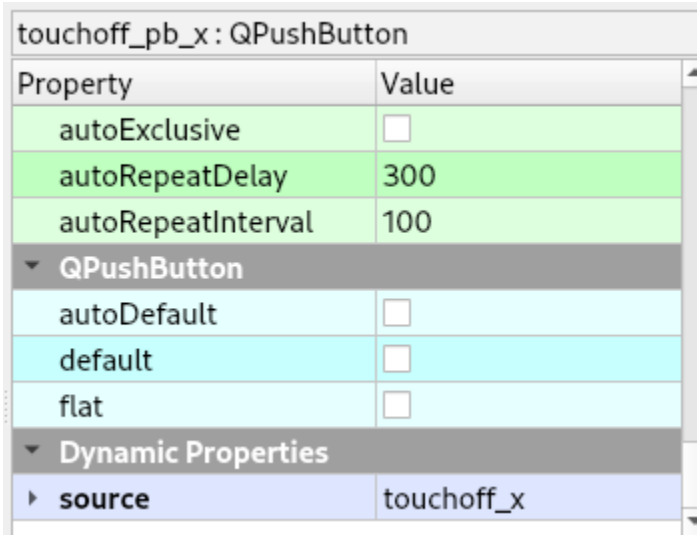
19.1 Coordinate System Touchoff

To touch-off an axis, use a QPushButton and QLineEdit to set the touch-off value. Optionally you can have a QComboBox to select the Coordinate System to touch off to.

Table 1: **Coordinate System Touch Off Controls**

Control Function	Object Type	Object Name
Touch Off Axis	QPushButton	touchoff_pb_(axis letter)
Touch Off Value	QLineEdit	touchoff_le
Optional		
Coordinate System	QComboBox	touchoff_system_cb

Optionally you can have a QLineEdit for any axis by adding a string type Dynamic Property named *source* to the QPushButton and the value contains the object name of the QLineEdit that you want to use. See *Dynamic Properties*



As you can see you can have a QLineEdit for each axis.



19.2 Change Coordinate System

To change the coordinate system via a button, use a change_cs_`n` QPushButton where *n* is 1-9 for G54 through G59.3

Table 2: **Coordinate System Change Buttons**

Control Function	Object Type	Object Name
Change Coordinate System	QPushButton	change_cs_(n)

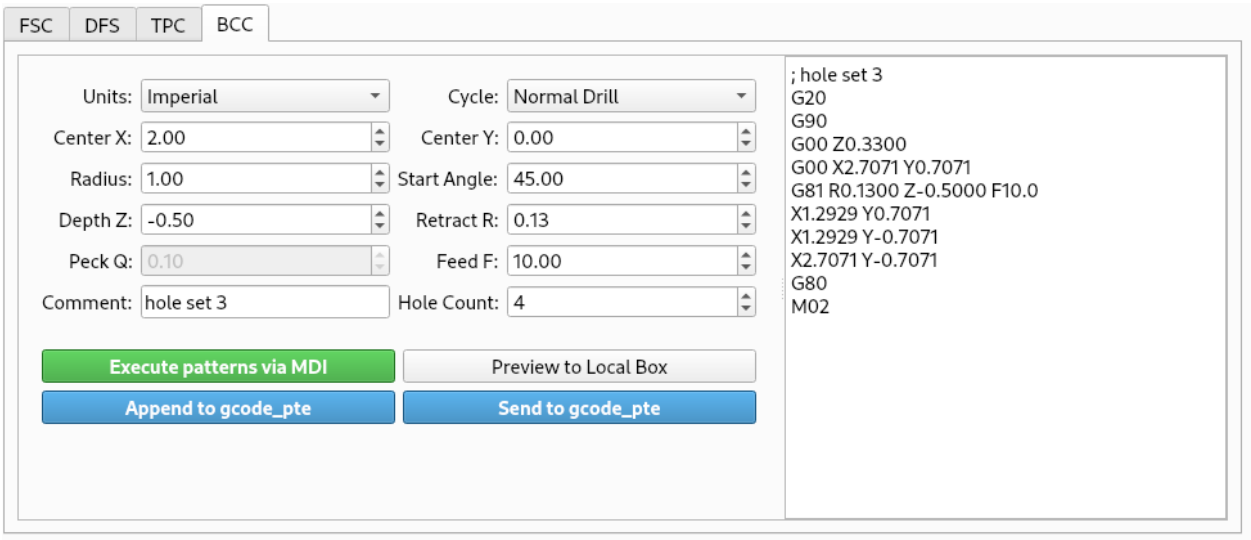
CHAPTER

TWENTY

NC CODE GENERATORS

20.1 Bolt Circle Generator

A Bolt Circle NC Code Generator can be added to your GUI by adding either a QWidget or QFrame and setting the object name to *bc_gen*.



- Units can be Metric or Imperial
- Drill Cycle can be Normal or Peck
- Start Angle 0 is to the right of center and increases CCW

CHAPTER

TWENTYONE

MISCELLANEOUS ITEMS

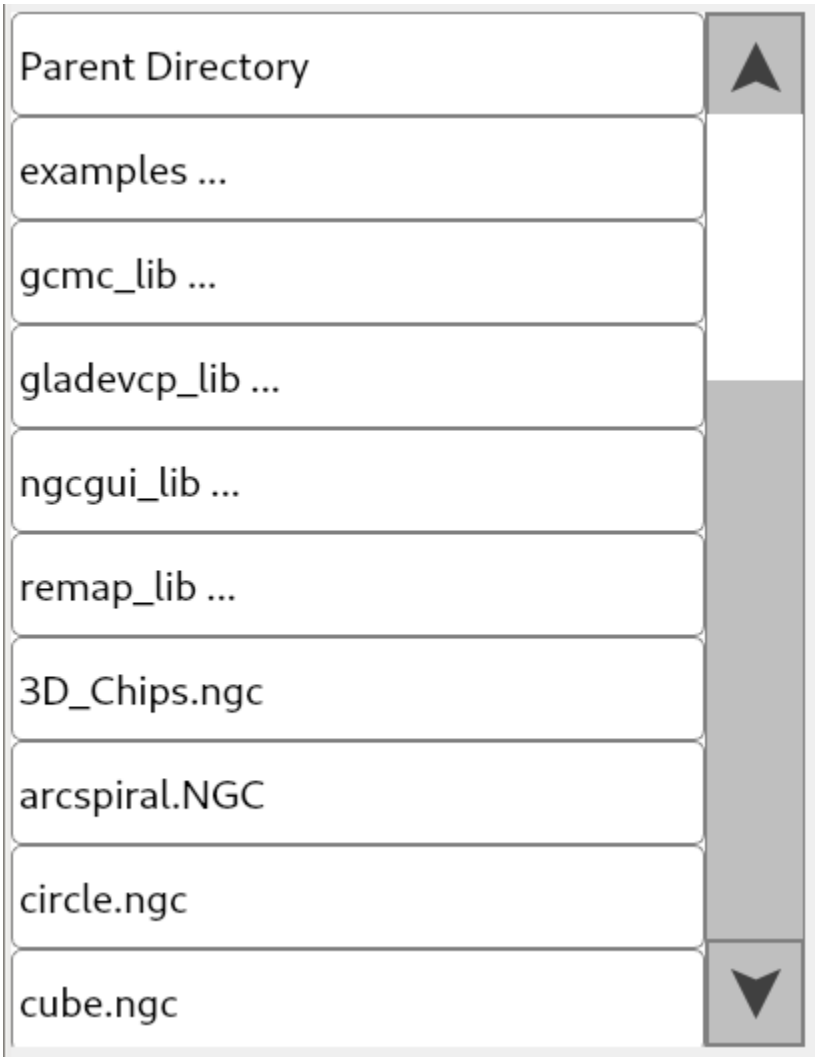
21.1 Line Edit

A QLineEdit can trigger a click event on a QPushButton by pressing the Enter key or if a touch screen popup keypad is used when Save is pressed. Add a String Type Dynamic Property called *return_button* with the value being the object name of the QPushButton.

21.2 File Selector

Add a QListWidget and name it *file_lw*, this can be used with a touch screen by specifying the touch input. A single left-click or touch is all that's needed to use the *File Selector*. A left-click or touch on a directory will change to that directory. A left-click or touch on the up or down arrow will move the list by one. A left-click or touch in between an arrow and the slider will move the list by one page. Touch-and-hold to move the slider.

If you use the touch input, the selector looks like this



Note: Make sure you use a QListWidget and not a QListView for the file selector.

File, Error and Information Tutorial

21.3 Code Viewer

To add a code viewer, add a *QPlainTextEdit* from Input Widgets and name it *gcode_pte*

```
( AXIS "splash G-code" Not intended for actual milling )
( To run this code anyway you might have to Touch Off the Z axis)
( depending on your setup. As if you had some material in your mill... )
( Hint jog the Z axis down a bit then touch off )
( Also press the Toggle Skip Lines with "/" to see that part )
( If the program is too big or small for your machine, change the scale below )
( LinuxCNC 19/1/2012 2:13:51 PM )
#<depth>=2.0
#<scale>=1.0
G21 G90 G64 G40
G0 Z3.0
( engraving )
```

21.4 Code Viewer Controls

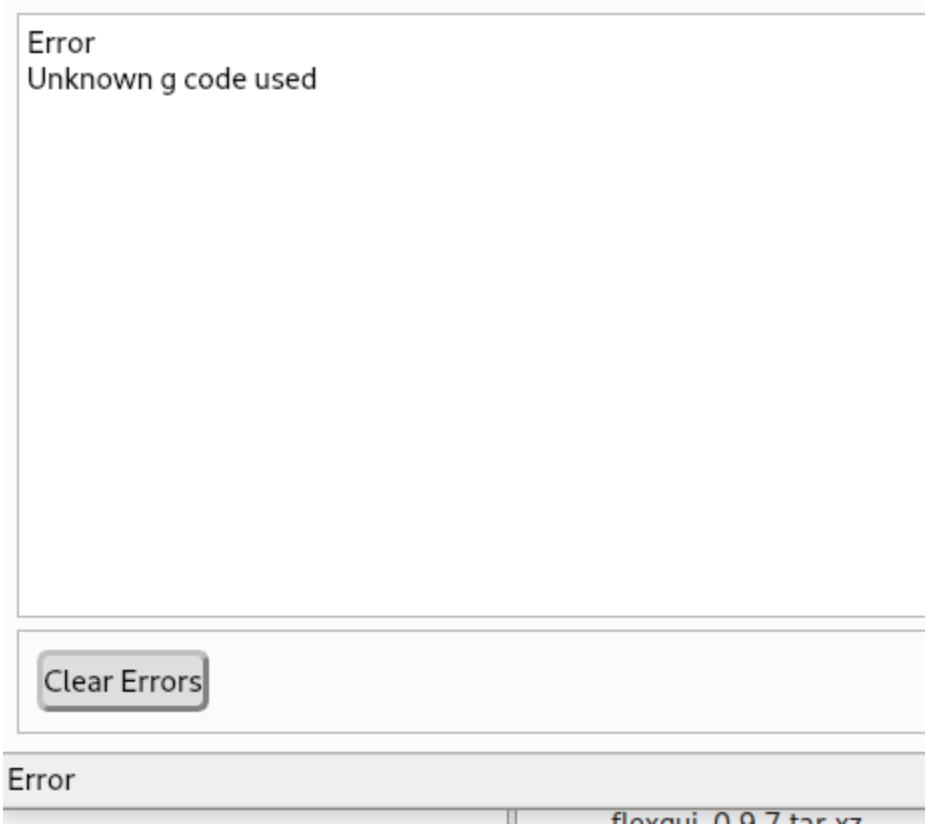
The Code Viewer allows you to edit the file in Flex GUI without using an external text editor. You can save the current code to the current file name, save the current code with a new file name and you can search the code.

Table 1: **Code Viewer Controls**

Function	Type	Object Name
Save	QPushButton	save_pb
Save As	QPushButton	save_as_pb
Search	QPushButton	search_pb

21.5 Error Viewer

To add an error viewer, add a *QPlainTextEdit* from Input Widgets and name it *errors_pte*



To clear the error history, add a QPushButton and set the objectName to *clear_errors_pb*.

To copy the errors to the clipboard, add a QPushButton and set the object name to *copy_errors_pb*.

Warning: The error viewer must be a QPlainTextEdit not a QTextEdit.

21.6 Information Viewer

To add an information viewer, add a *QPlainTextEdit* from Input Widgets and name it *info_pte*. Information messages from MSG, DEBUG and PRINT will show up in the Information Viewer if it exists.

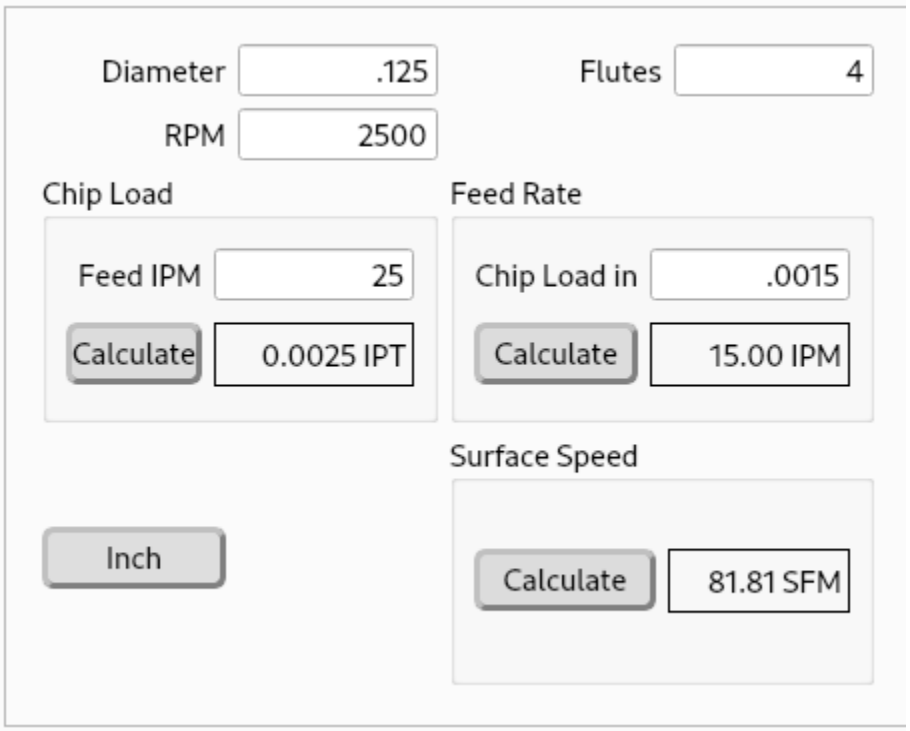
If *info_pte* is not found and the *errors_pte* is found, then information messages will show up in the Error Viewer.

To clear the information viewer, add a QPushButton and name it *clear_info_pb*.

Warning: The information viewer must be a QPlainTextEdit not a QTextEdit.

21.7 Speed & Feed Calculators

To add a milling Speeds and Feeds Calculator, add a *QFrame* or *QWidget* and set the Object Name to *fsc_container*

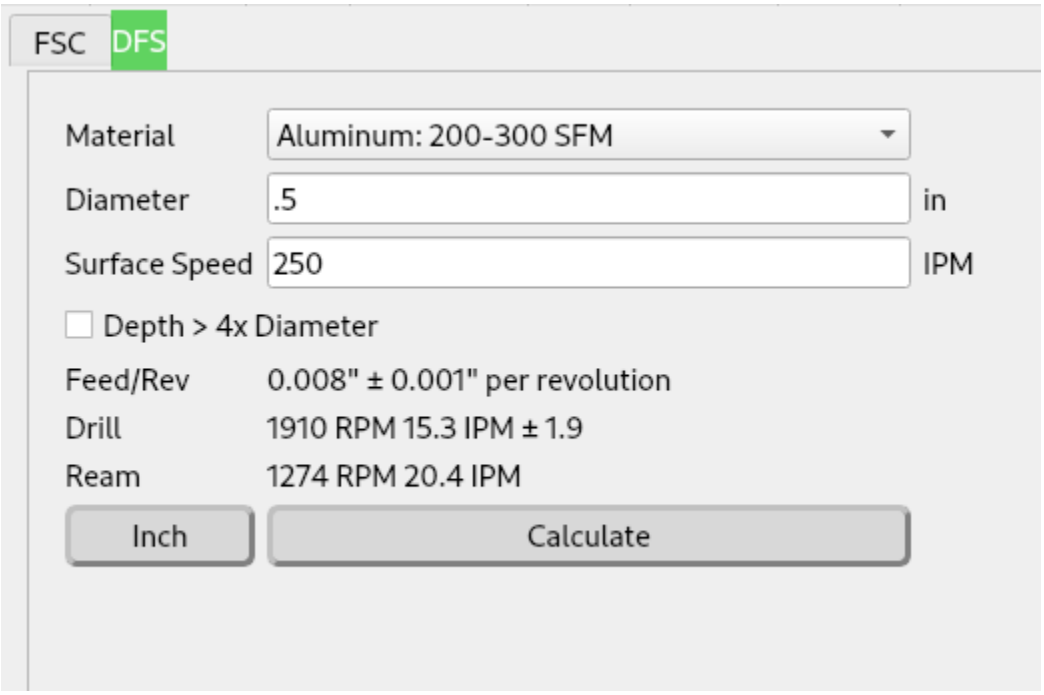


To make the entry boxes touch-screen aware, add a Dynamic Property called *mode* and set the value to *touch*. Then when you touch an entry field, a numeric popup will show up to allow you to enter the value without a keyboard. See *Dynamic Properties*



To add a Drill Feed and Speed calculator, add a *QFrame* or *QWidget* and set the Object Name to *dsf_container*.

To make the entry boxes touch-screen aware, add a Dynamic Property called *mode* and set the value to *touch*. Then when you touch it, a numeric popup will appear, allowing you to enter the numbers



21.8 Help System

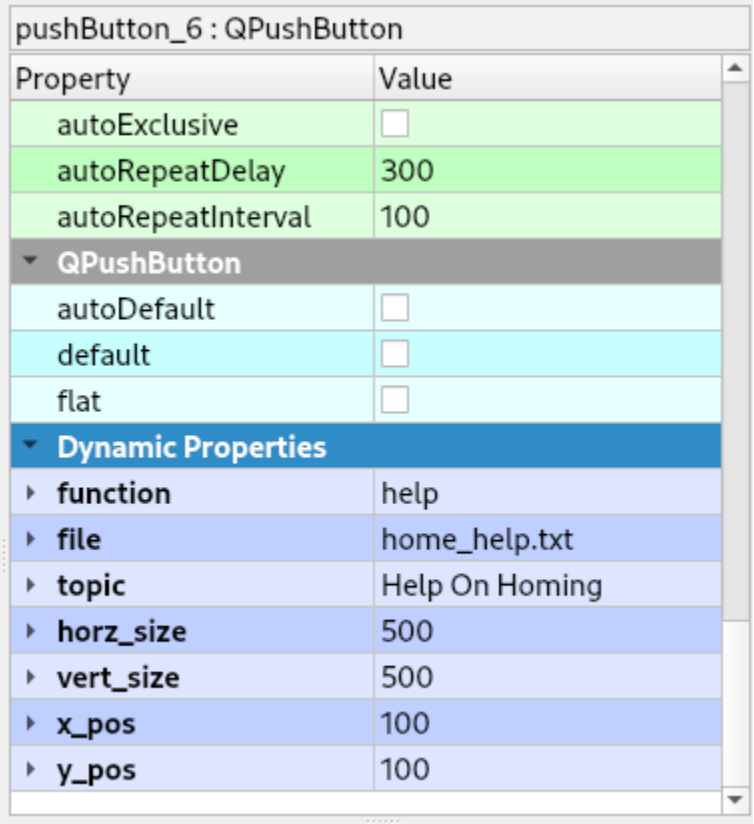
A QPushButton can be setup to launch a Help dialog which contains text from a file in the configuration directory. A help button can be placed on multiple places with different file names. Only one Help dialog can be open at a time. The Help dialog will be centered over the GUI.

Table 2: **Help Button Dynamic Properties**

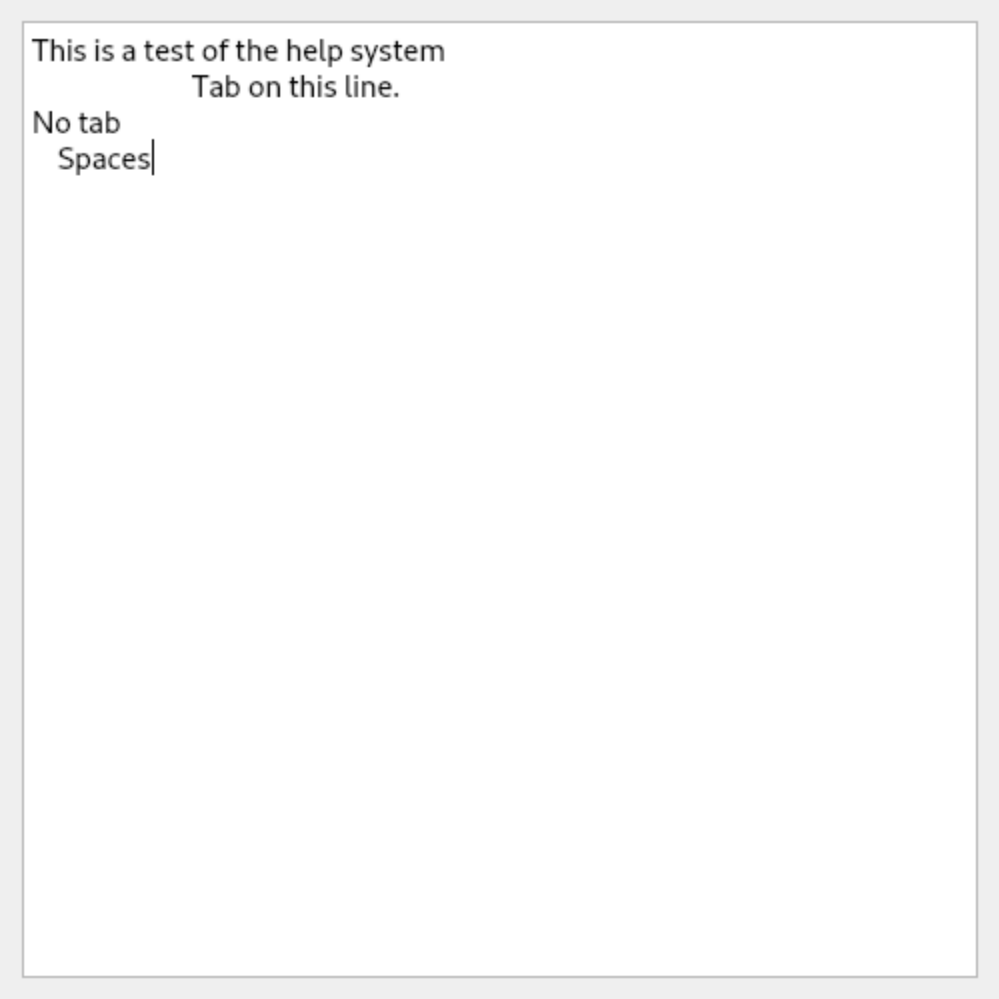
Property Type	Property Name	Value
String	function	help
String	file	file name
String	topic	title of topic
String	horz_size	width
String	vert_size	height

Note: The x_pos is from the left edge of the screen and the y_pos is from the top of the screen.

Dynamic Properties


pushButton_6 : QPushButton
Property
Value
autoExclusive
autoRepeatDelay
300
autoRepeatInterval
100
QPushButton
autoDefault
default
flat
Dynamic Properties
function
help
file
home_help.txt
topic
Help On Homing
horz_size
500
vert_size
500
x_pos
100
y_pos
100

Help Dialog


This is a test of the help system
Tab on this line.
No tab
Spaces

CHAPTER

TWENTYTWO

HAL PINS

HAL Tutorial

Creating widgets that connect to HAL (Hardware Abstract Layer) is as simple as adding a few Dynamic Properties to a widget. See *Dynamic Properties* for step by step instructions to add a Dynamic Property.

Typically the Dynamic Properties are String type with some exceptions being Bool and Color. Connections from Flex HAL objects to other HAL objects must be done in the file assigned to the POSTGUI_HALFILE variable in the [HAL] section typically named *postgui.hal*.

```
[HAL]
HALFILE = main.hal
POSTGUI_HALFILE = postgui.hal
```

The property *pin_name* defines the HAL pin name that is prefixed with *flexhal*. A pin_name of my-button would be *flexhal.my-button* in HAL.

Note: Dynamic Property names are case sensitive and must be all lower case. Hal types and directions are case sensitive and must be all caps. The function value must be lower case.

Note: Hal pin names can containe a-z, A-Z, 0-9, underscore _, or dash -. Use a dash to not conflict with program variables of the same name which use an underscore.

22.1 HAL Types

Starting with version 2.10 Linuxcnc added two 64 bit HAL types. You must be using LinuxCNC 2.10 or newer to use *HAL_S64* or *HAL_U64*.

HAL Pin Types

```
HAL_BIT
HAL_FLOAT
HAL_S32
HAL_U32
HAL_S64
HAL_U64
```

HAL Pin Directions

```
HAL_IN
HAL_OUT
HAL_IO
```

22.1.1 Options

The default behavior for HAL objects that are a HAL_OUT or HAL_IO is to be enabled when the E Stop is released. To change this behaviour add a Dynamic Property to control when the HAL object is enabled. The *always_on* will not disable the control for any state. The *state_on* property will enable the control when the power is on. The *all_homed* property will enable the control when all joints are homed. They can be combined to make a HAL object only enabled when power is on and all joints are homed.

HAL objects that are HAL_IN are always enabled.

Table 1: **HAL Options**

Property Type	Property Name	Pin Value
Bool	always_on	True
Bool	state_on	True
Bool	all_homed	True

22.1.2 HAL Buttons

22.2 HAL Button

A QPushButton, QCheckBox or QRadioButton can be assigned to a HAL *bit* pin by adding two string type Dynamic Properties. A pin_name of my-button would be *flexhal.my-button* in HAL. HAL pins must be connected in the post-gui.hal file.

The HAL direction is OUT and the HAL type is bit for a QPushButton, QCheckBox or QRadioButton.

The option always_on will not disable the HAL Button for any state.

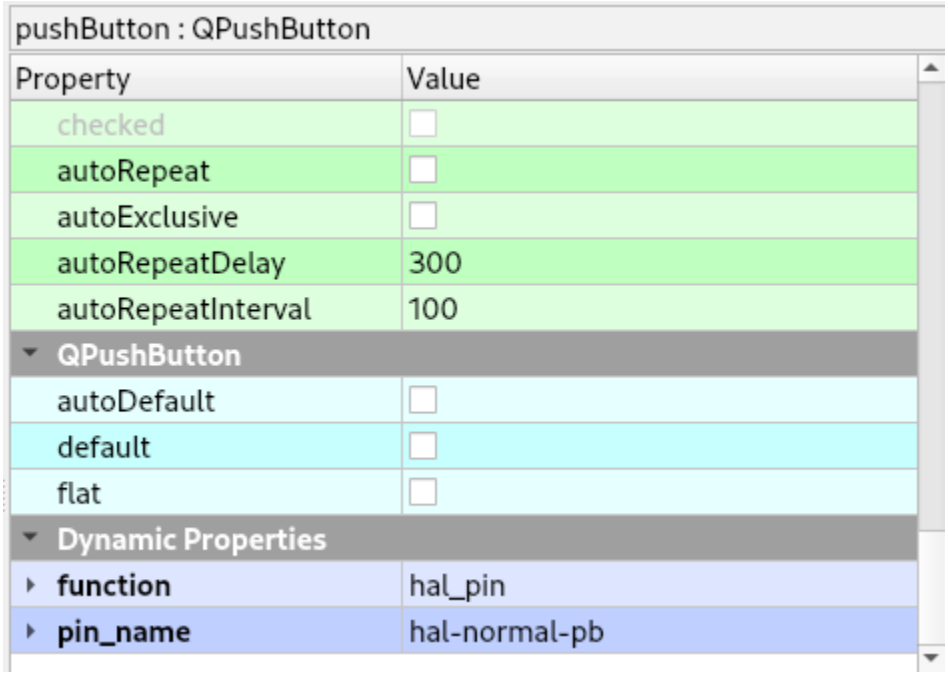
The option state_on will disable the HAL Button until the Power is on.

The option all_homed will disable the HAL Button until all joints are homed.

The option confirm can be used with a checkable QPushButton or QCheckBox. When toggled a popup will ask for confirmation of the action. If Cancel is selected the checked state will be set back to the previous state and the HAL pin will not change.

Table 2: **HAL Push Button**

Property Type	Property Name	Pin Value
String	function	hal_pin
String	pin_name	any unique name
Optional		
Bool	always_on	True
Bool	state_on	True
Bool	all_homed	True
Bool	confirm	True



Warning: By default, no QRadioButtons are checked unless you set one checked in the Designer. Starting up with none checked could be a problem if you expect one to be selected at startup.

22.3 HAL LED Button

A QPushButton can be a HAL LED button by adding two dynamic properties. The *pin_name* property will be the HAL pin name you use to connect the button state in the postgui.hal file. The button can be momentary or checkable. The default colors are Red when Off and Green when On. The round LED will be in the upper right corner.

The HAL direction is OUT and the HAL type is bit for a HAL LED QPushButton.

The option always_on will not disable the HAL LED Button for any state.

The option state_on will disable the HAL LED Button until the Power is on.

The option all_homed will disable the HAL LED Button until all joints are homed.

Table 3: **HAL LED Button Dynamic Properties**

Property Type	**Property Name**	**Pin Value**
String	function	hal_led_button
String	pin_name	any unique name
Optional		
Bool	always_on	True
Bool	state_on	True
Bool	all_homed	True

22.3.1 HAL Spinboxs

22.4 HAL Spinbox

A QSpinBox can be a HAL *number* pin by adding three string type Dynamic Properties. The pin_name used will create a HAL pin prefixed with *flexhal.* A pin_name of my-spinbox would be in HAL *flexhal.my-spinbox*. The spinbox is an Out type that will set the value of the HAL pin to match the value of the spinbox.

The HAL direction is OUT for a HAL Spinbox.

Table 4: **HAL Spin Box**

Property Type	**Property Name**	**Pin Value**
String	function	hal_pin
String	pin_name	any unique name
String	hal_type	HAL_S32 or HAL_U32
LinuxCNC 2.10 or greater		
String	hal_type	HAL_S64 or HAL_U64
Optional		
Bool	state_on	True
Bool	all_homed	True

22.5 HAL Double Spinbox

A QDoubleSpinBox can be a HAL *number* pin by adding two string type Dynamic Properties. The pin_name used will create a HAL pin prefixed with *flexhal.* A pin_name of my-spinbox would be in HAL *flexhal.my-spinbox*. The spinbox is an Out type that will set the value of the HAL pin to match the value of the spinbox.

The HAL direction is OUT and the HAL type is float for a HAL Double Spinbox.

Table 5: **HAL Spin Box**

Property Type	**Property Name**	**Pin Value**
String	function	hal_pin
String	pin_name	any unique name
Optional		
Bool	state_on	True
Bool	all_homed	True

22.6 HAL Slider

A QSlider can be a HAL pin by adding these three string type Dynamic Properties. The pin_name used will create a HAL pin prefixed with *flexhal.* A pin_name of my-slider would be in HAL *flexhal.my-slider*. A QSlider supports only integers so to connect it to a float HAL pin use conv_s32_float or conv_u32_float.

The HAL direction is OUT for a HAL Slider.

See *Dynamic Properties* for step by step instructions to add a Dynamic Property

Table 6: **HAL Slider**

Property Type	Property Name	Pin Value
String	function	hal_pin
String	pin_name	any unique name
String	hal_type	HAL_S32 or HAL_U32
LinuxCNC 2.10 or greater		
String	hal_type	HAL_S64 or HAL_U64
Optional		
Bool	state_on	True
Bool	all_homed	True

22.7 HAL I/O

A HAL I/O object has an input and output on the same pin. The pin can set an input pin of another HAL object and the pin can be set by another HAL object output pin. The HAL I/O will stay synchronized with the pin it's connected to.

Note: The connected pins must be of the same HAL type.

A QPushButton (set to checkable), QCheckBox, QRadioButton, QSpinBox, QDoubleSpinBox or a QSlider can be a HAL I/O object.

The HAL direction is I/O and the HAL type is bit for QPushButton, QCheckBox or QRadioButton.

The HAL direction is I/O for a HAL I/O QSpinBox.

The HAL direction is I/O and the HAL type is float for a HAL I/O QDoubleSpinBox.

Note: The hal_type for QSpinBox and QSlider must be specified.

Table 7: **HAL I/O**

Property Type	Property Name	Pin Value
String	function	hal_io
String	pin_name	any unique name
For a QSpinBox or QSlider		
String	hal_type	HAL_S32 or HAL_U32
LinuxCNC 2.10 or greater		
String	hal_type	HAL_S64 or HAL_U64
Optional		
Bool	state_on	True
Bool	all_homed	True

Note: The hal_type is required for QSpinBox, QSlider or QDoubleSpinBox

22.7.1 HAL Labels

22.8 HAL Label

A QLabel can be used to monitor HAL pins. HAL connections must be made in the post gui HAL file. The pin_name used will create a HAL pin prefixed with *flexhal.* A pin_name of my-reader would be *flexhal.my-reader* in HAL.

The HAL direction is IN for a HAL Label.

Table 8: **HAL Label**

Property Type	Property Name	Pin Value
String	function	hal_pin
String	pin_name	any unique name
String	hal_type	HAL_FLOAT or HAL_S32 or HAL_U32
LinuxCNC 2.10 or greater		
String	hal_type	HAL_S64 or HAL_U64
Optional		
String	precision	Number of digits for HAL_FLOAT
String	zero_padding	Left padded zeros for integer types

22.9 HAL Bool Label

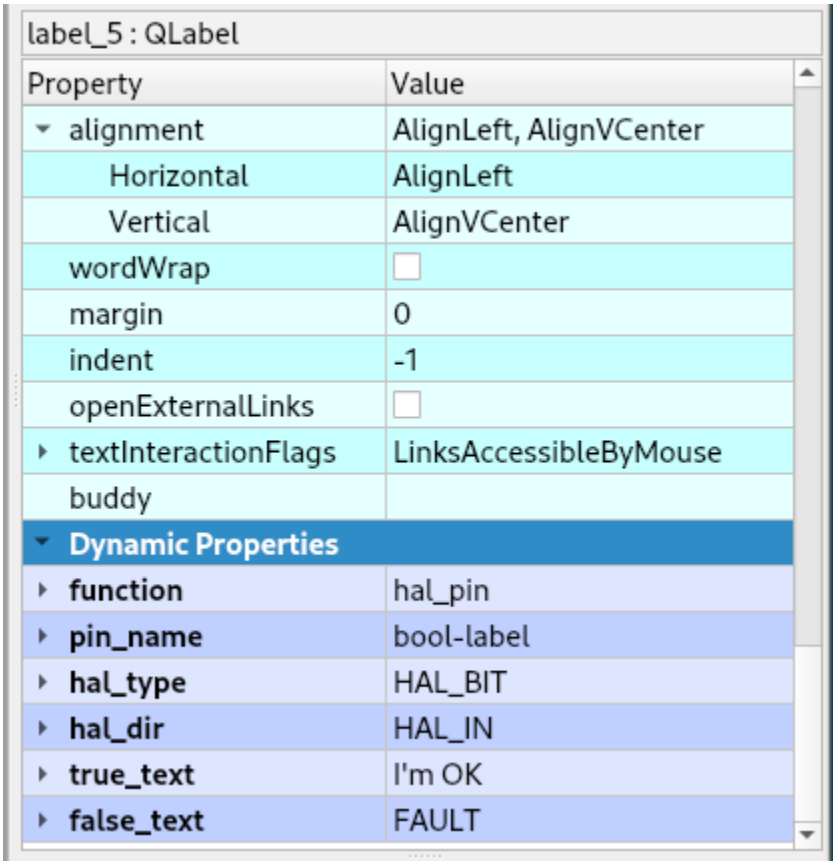
A QLabel can have optional True and False text by adding two additional Dynamic Properties. Default is *True* and *False*.

The HAL direction is IN and the hal_type is bit for a HAL Bool Label

See *Dynamic Properties* for step by step instructions to add a Dynamic Property

Table 9: **HAL Bool Label**

Property Type	Property Name	Pin Value
String	function	hal_pin
String	pin_name	any unique name
String	hal_type	HAL_BIT
Optional		
String	true_text	text to display when True
String	false_text	text to display when False



22.10 HAL Average Float Label

A QLabel can be used to monitor HAL float number pins and display an average of the number of samples. The sample stack is LIFO so a new value pushes the oldest value out of the stack. This could be useful to display RPM from a spindle encoder or any numeric value that changes.

The *samples* option can change the number of samples that are averaged out to help smooth the display output.

The *rounding* option can be used to round the number. Use 1 to round to the nearest tenth and 2 to round to the nearest hundred, etc.

The *precision* option is used to set the number of decimal places. Set to 0 to not have any decimal places.

HAL connections must be made in the post gui HAL file. The pin_name used will create a HAL pin prefixed with *flexhal.* A pin_name of my-reader would be *flexhal.my-reader* in HAL.

The HAL direction is IN and the hal_type is float for a HAL Average Float Label.

Table 10: **HAL Average Float Label**

Property Type	Property Name	Pin Value
String	function	hal_avr_f
String	pin_name	any unique name
Optional		
Int	samples	The number of samples to use default is 10
Int	rounding	number of digits to the left of the decimal to round
Int	precision	Number of decimal digits

22.11 HAL Average Integer Label

A QLabel can be used to monitor HAL integer pins and display an average of the number of samples. The sample stack is LIFO so a new value pushes the oldest value out of the stack. This could be useful to display RPM from a spindle encoder or any numeric value that changes.

The *samples* option can change the number of samples that are averaged out to help smooth the display output.

HAL connections must be made in the post gui HAL file. The pin_name used will create a HAL pin prefixed with *flexhal.* A pin_name of my-reader would be *flexhal.my-reader* in HAL.

The HAL direction is IN.

Table 11: **HAL Average Integer Label**

Property Type	**Property Name**	**Pin Value**
String	function	hal_avr_i
String	pin_name	any unique name
String	hal_type	HAL_S32 or HAL_U32
LinuxCNC 2.10 or greater		
String	hal_type	HAL_S64 or HAL_U64
Optional		
Int	samples	The number of samples to use default is 10

22.12 HAL Multi-State Label

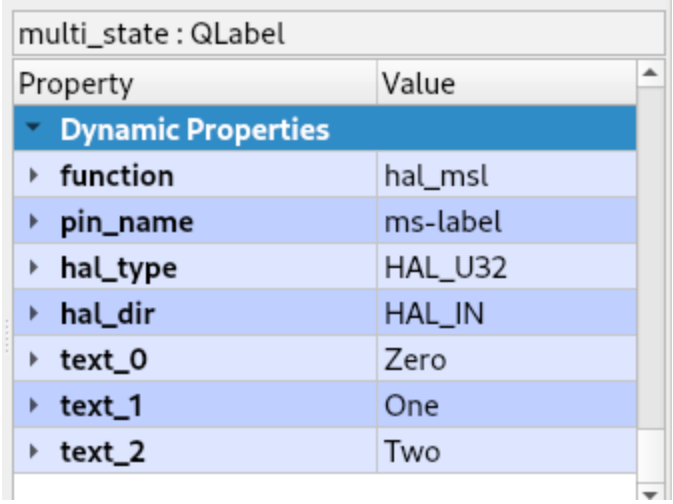
A QLabel can have multiple text by adding as many Dynamic Properties as needed for each text. The *text_n* property the *n* is the integer value for that text. Setting the pin name value will change the label text

The HAL direction is IN and the hal_type is U32 for a HAL Multi-State Label.

Table 12: **HAL Multi-State Label**

Property Type	**Property Name**	**Pin Value**
String	function	hal_msl
String	pin_name	any unique name
String	text_n	text to display when value is equal to n

Note: Setting the HAL pin to a value that is not a text_n will show an error with the HAL pin value.



22.13 HAL LCD

A QLCDNumber can be used to monitor HAL pins. HAL connections must be made in the post gui HAL file. The pin_name used will create a HAL pin prefixed with *flexhal.* A pin_name of my-reader would be in HAL *flexhal.my-reader*.

The HAL direction is IN for a HAL LCD

Table 13: **HAL LCD**

Property Type	**Property Name**	**Pin Value**
String	function	hal_pin
String	pin_name	any unique name
String	hal_type	HAL_FLOAT or HAL_S32 or HAL_U32
LinuxCNC 2.10 or greater		
String	hal_type	HAL_S64 or HAL_U64
Optional		
String	integer_digits	Number of left pad zeros for HAL_S32 or HAL_U32

Note: A HAL_FLOAT QLCDNumber can have a string Dynamic Property called *precision* with a value of the number of decimal digits.

22.13.1 HAL LED's

22.14 HAL LED

A QLabel can be used as a HAL LED indicator by adding the following properties to a blank label. The default colors are Red when Off and Green when On. The pin_name is the hal name the LED will have. The default shape is round.

The HAL LED needs to be connected in the postgui.hal file and can only be connected to a HAL pin of type bit with a HAL direction of OUT or a signal that is connected to a HAL pin of type bit with a HAL direction of OUT. Only one OUT direction can be connected to a signal while multiple IN directions can be connected to a signal.

The HAL direction is IN and the hal_type is bit for a HAL LED

Table 14: **HAL LED**

Property Type	Property Name	Pin Value
String	function	hal_led
String	pin_name	any unique name
Optional		
Color	on_color	color of your choice
Color	off_color	color of your choice
Int	edge_margin	space between circle and edge of the label
String	led_shape	square

Choosing 'square' for LED shape will fill the entire label.

Note: Select Other to get the list and select Color. You can copy and paste the hex color value into the color picker.

22.15 HAL LED Text Label

Similar to the HAL LED except it has a border and the on and off text can be set as well as other options. In Qt Designer it's recommended to set the Horizontal Alignment to *AlignHCenter*.

Table 15: **HAL Text LED**

Property Type	Property Name	Pin Value
String	function	hal_led_text
String	pin_name	any unique name
Optional		
String	on_text	text to display when HAL pin is True
String	off_text	text to display when HAL pin is False
Color	on_bg_color	color of background when HAL pin is True
Color	off_bg_color	color of background when HAL pin is False
Color	on_text_color	color of text when HAL pin is True
Color	off_text_color	color of text when HAL pin is False
Font	font_family	Font and size of text to use
Bool	font_bold	True for bold font

Default values for the options are:

- on_text, True
- off_text, False
- on_bg_color, Green
- off_bg_color, Red
- on_text_color, Black
- off_text_color, White
- font_family, Courier 16
- font_bold, False

22.16 HAL LED Label

Similar to the HAL LED except the LED is in the upper right corner so the label can have text. The default colors are Red when Off and Green when On. The default shape is round.

The HAL direction is IN and the hal_type is bit for a HAL LED Label.

Table 16: **HAL LED Label**

Property Type	Property Name	Pin Value
String	function	hal_led_label
String	pin_name	any unique name
Optional		
Color	led_on_color	color of your choice
Color	led_off_color	color of your choice
Int	led_diameter	diameter of led
Int	led_right_offset	offset from right edge
Int	led_top_offset	offset from top edge
String	led_shape	square

22.16.1 HAL Progress Bar

A QProgressBar can be used to monitor HAL pins. HAL connections must be made in the post gui HAL file. The pin_name used will create a HAL pin prefixed with *flexhal.* A pin_name of my-bar would be in HAL *flexhal.my-bar*. Typically the minimum value is 0 and the maximum value is 100. If the HAL value exceeds the maximum value 0 is displayed.

The HAL direction is IN and the hal_type is U32 for a HAL Progress Bar.

Table 17: **HAL Progressbar**

Property Type	Property Name	Pin Value
String	function	hal_pin
String	pin_name	any unique name

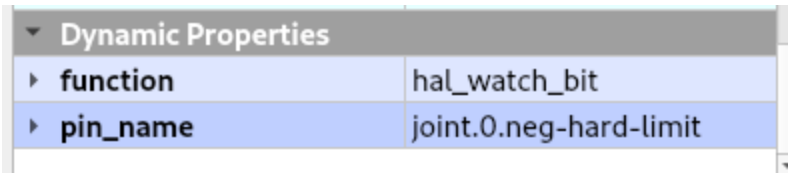
22.16.2 HAL Watch Labels

You can watch the status of a HAL pin in your GUI by adding a QLabel with the following Dynamic Properties. You can watch Bit (True/False) pins, S32/U32 integer pins, Float pins.

22.17 HAL Watch Bool

Table 18: **HAL Watch Bit**

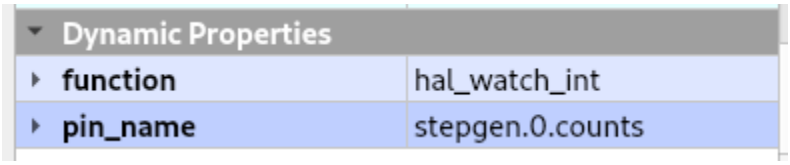
Property Type	Property Name	Pin Value
String	function	hal_watch_bit
String	pin_name	any HAL pin that is a bool



22.18 HAL Watch Integer

Table 19: **HAL Watch Integer**

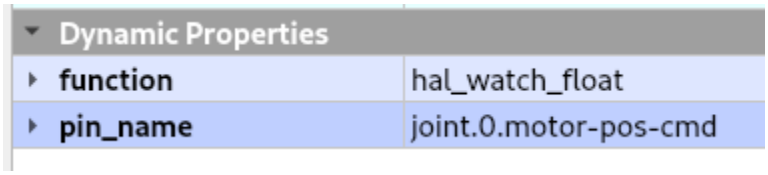
Property Type	Property Name	Pin Value
String	function	hal_watch_int
String	pin_name	any HAL pin that is a integer



22.19 HAL Watch Float

Table 20: **HAL Watch Float**

Property Type	Property Name	Pin Value
String	function	hal_watch_float
String	pin_name	any HAL pin that is a float



22.20 HAL Watch Time

To have a label monitor the HAL time component add the following Dynamic Properties to a QLabel. You can have as many Time Watch labels as you want. There are two types of HAL Watch Time labels, the *hal_watch_time_hms* displays the hours, minutes and seconds and the *hal_watch_time_hm* displays the hours and minutes. You need to load the *time* component on your hal file and use the *start* pin to start and stop the *time* component.

Table 21: **HAL Watch Hours, Minutes, Seconds**

Property Type	Property Name	Pin Value
String	function	hal_watch_time_hms
String	hours	time hours pin
String	minutes	time minutes pin
String	seconds	time seconds pin



Table 22: **HAL Watch Hours, Minutes**

Property Type	Property Name	Pin Value
String	function	hal_watch_time_hm
String	hours	time hours pin
String	minutes	time minutes pin



22.20.1 Step by Step

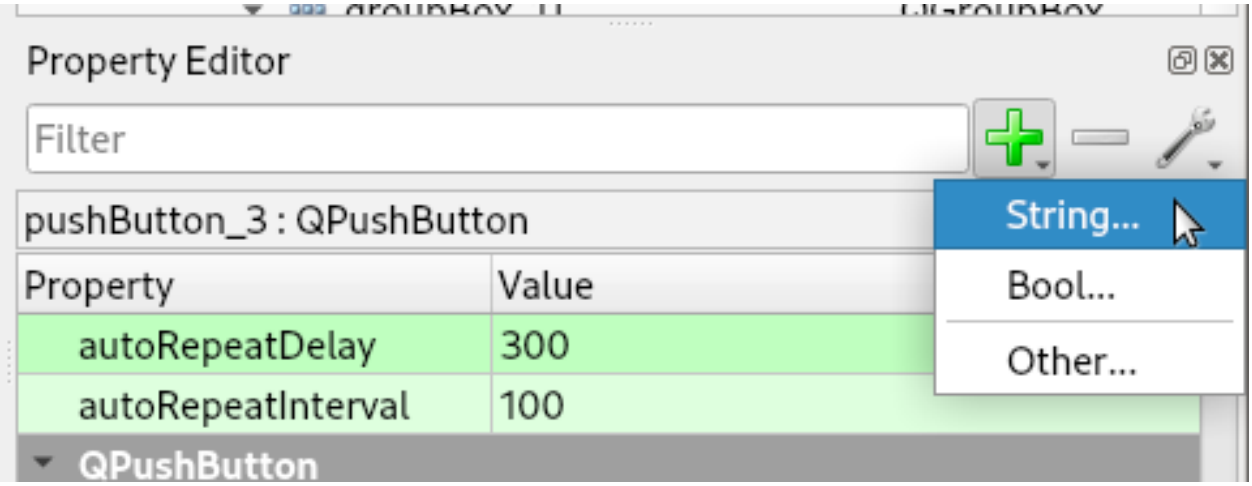
Note: This example is for a QPushButton

You can use a QPushButton as a momentary output, or with *checkable* selected for a toggle type output, or QCheckBox or QRadioButton for a HAL output control.

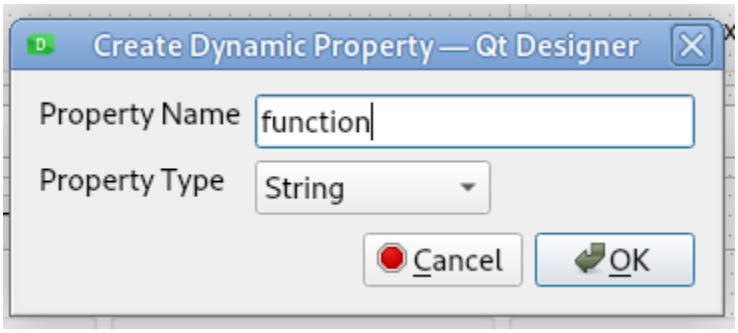
Drag the widget into the GUI and the widget can have any name you like; names are not used by HAL controls in Flex GUI - it is the following that matters.

Click on the widget to select it then click on the green plus sign in the Property Editor for that widget to add a Dynamic Property and select String.

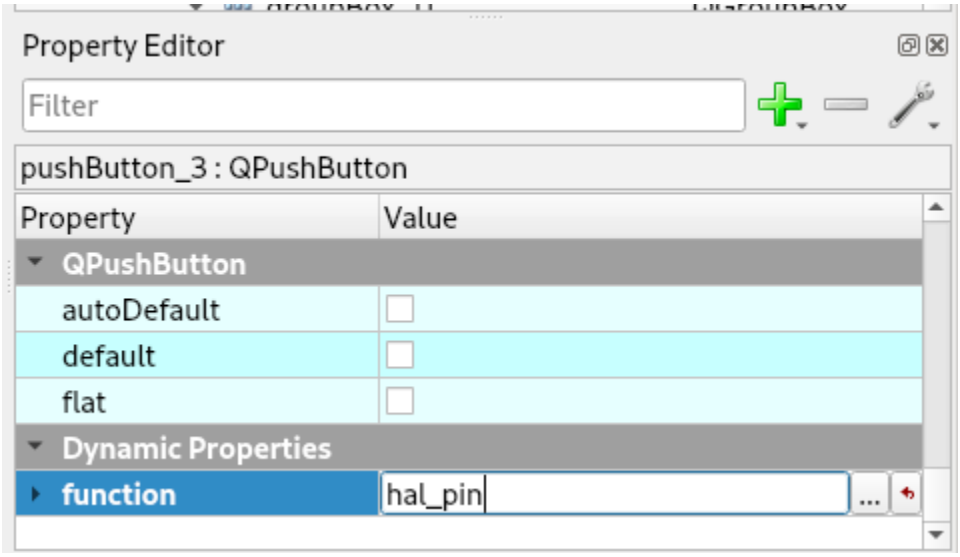
See *Dynamic Properties* for step by step instructions to add a Dynamic Property



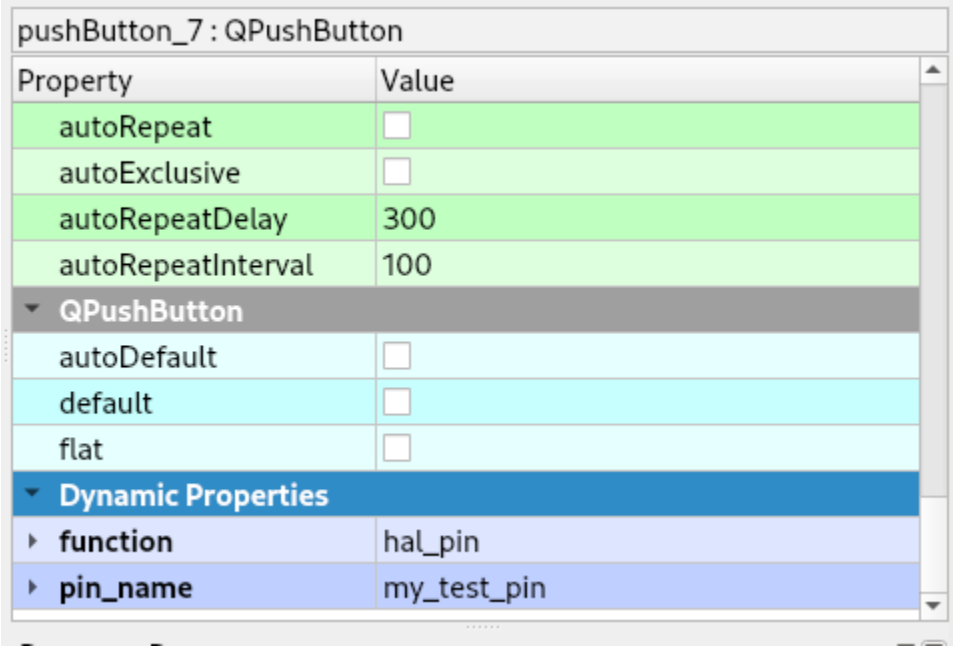
Set the Property Name to *function* and click Ok



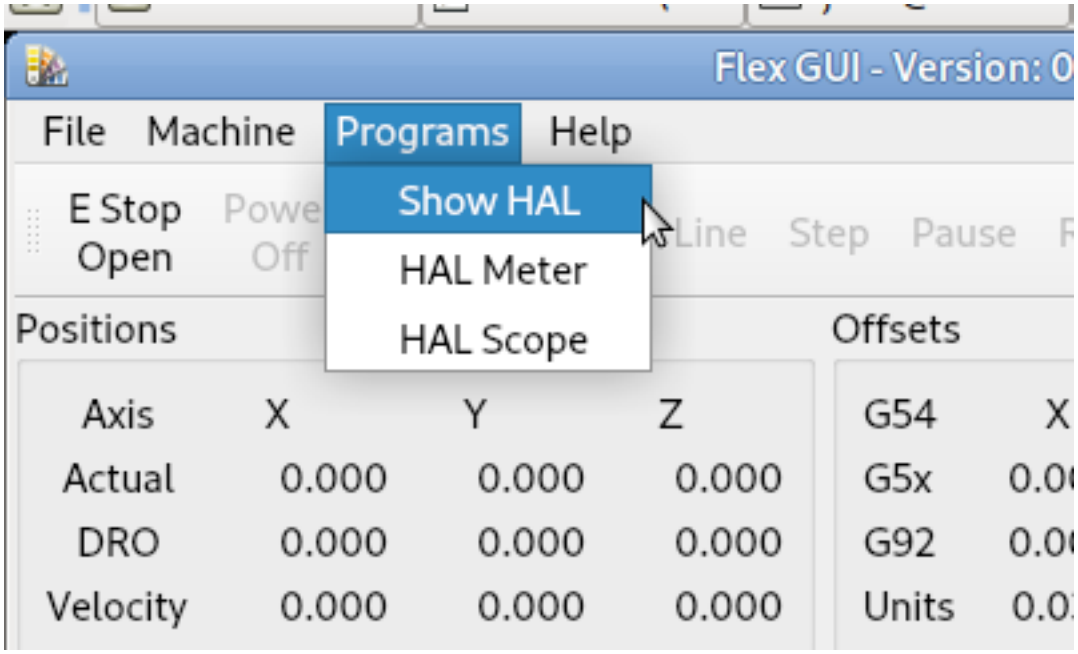
Set the Value to *hal_pin*; this tells Flex GUI that this widget is going to be for a HAL pin



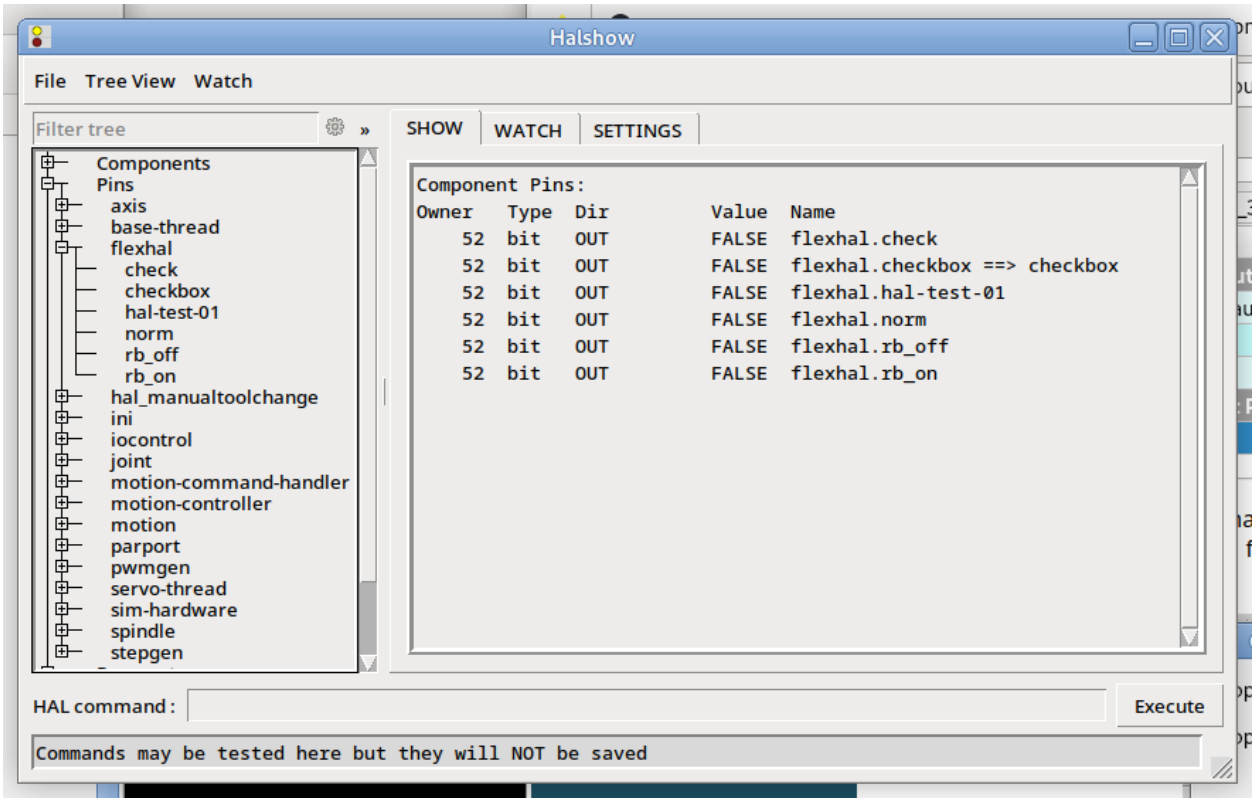
Add another string Dynamic Property named *pin_name* and set the value to any unique name



If you added Show HAL to your menu, you can open up the *Halshow* program and view the pin names



The pin names will all start with *flexhal* plus the unique name you gave them



Now you can connect the Flex HAL pin in the postgui.hal file like normal

```
net some-signal-name flexhal.hal-test-01 => some-other-pin-in
```

You can also preset a numeric HAL value by using *setp* in this example the HAL object is a QSpinBox which uses integer values.

```
setp flexhal.launch-velocity 10
```

After installing Flex GUI, from the CNC menu, you can copy the Flex GUI examples and look at the hal-btn example.

CHAPTER

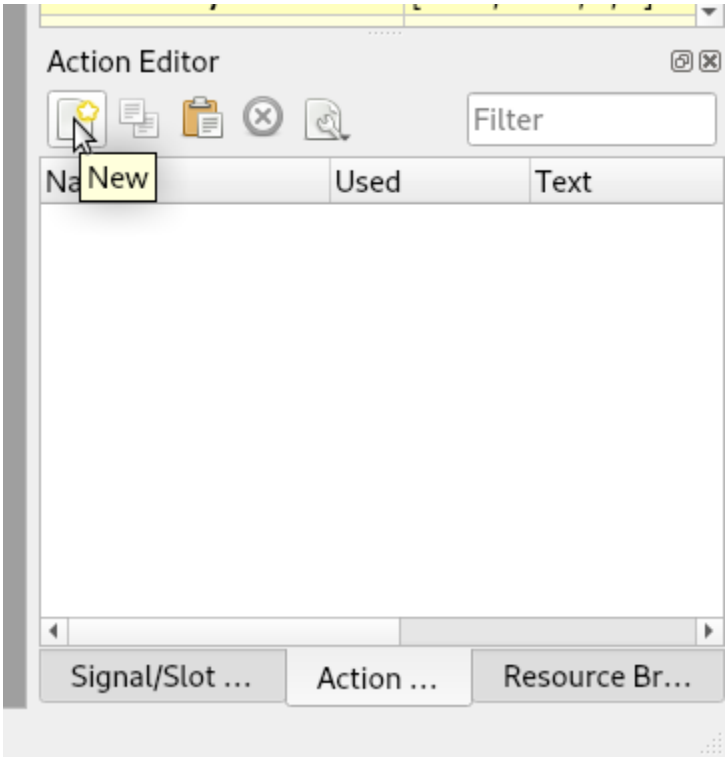
TWENTYTHREE

TOUCH SCREENS

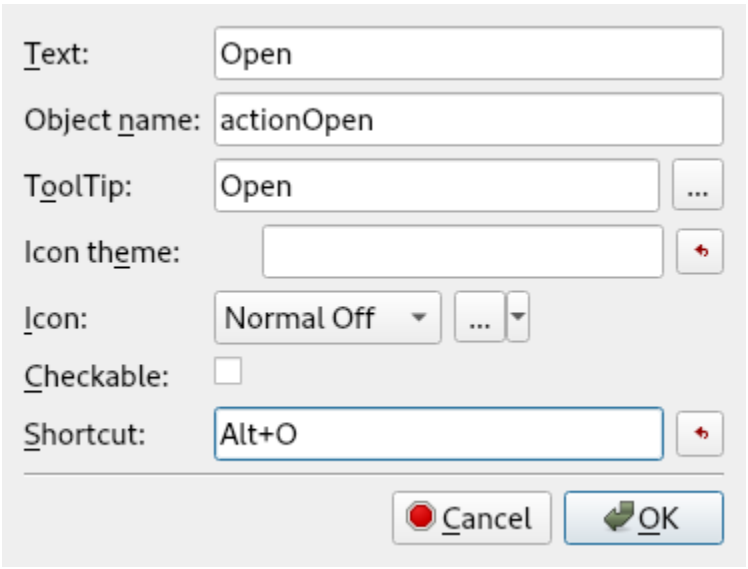
Some entry widgets like MDI and Touch-Off have a touch-screen popup available to make it easier for those users to enter the data.

23.1 Tool Bar

To add a button to the tool bar without having a menu item that creates an action, you just have to create the action yourself

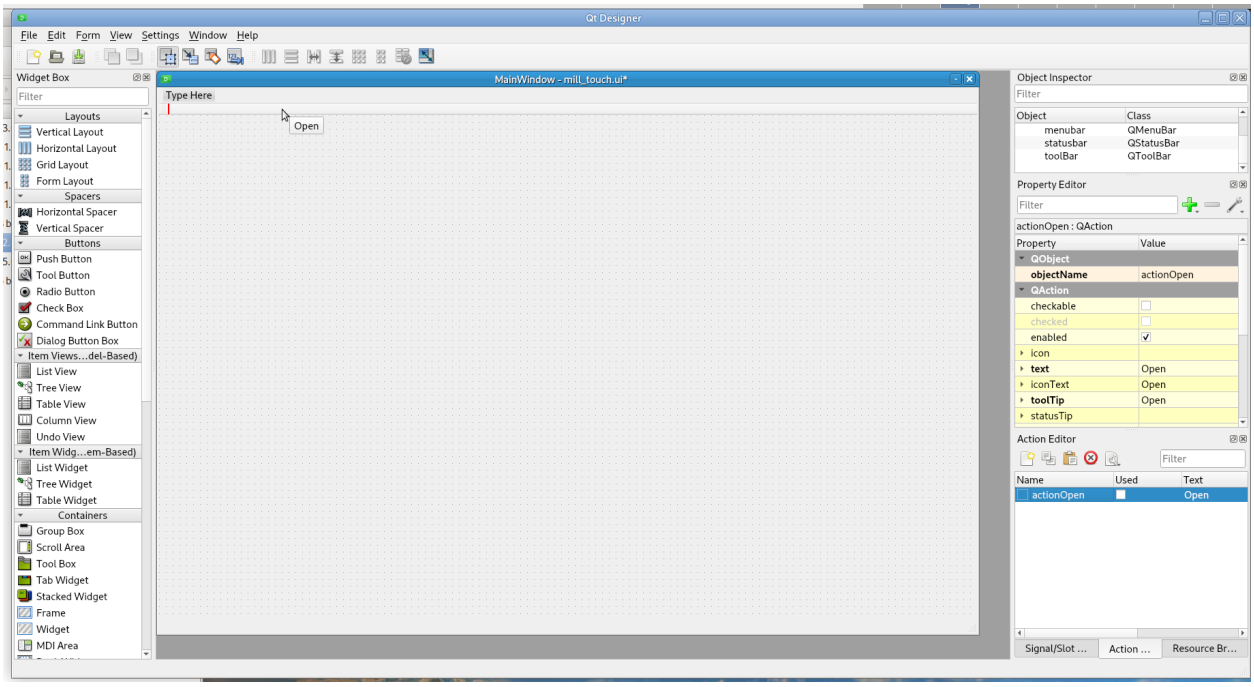


The action creating window, when you type in the Text name, the Object name is created for you



Warning: Make sure the Object name matches the Action Name created when you create a menu item - see the *Menu* section for the full list of Action Names.

Now you just drag the action into the tool bar to create a new tool bar button



Another option is to just use QPushButtons in a QFrame, as every menu action has a QPushButton as well that executes the same function.

23.2 Popup Keypads

If you drag a popup keypad to a different location, next time you use the keypad for that object the popup keypad will remember the last location and move to that location.

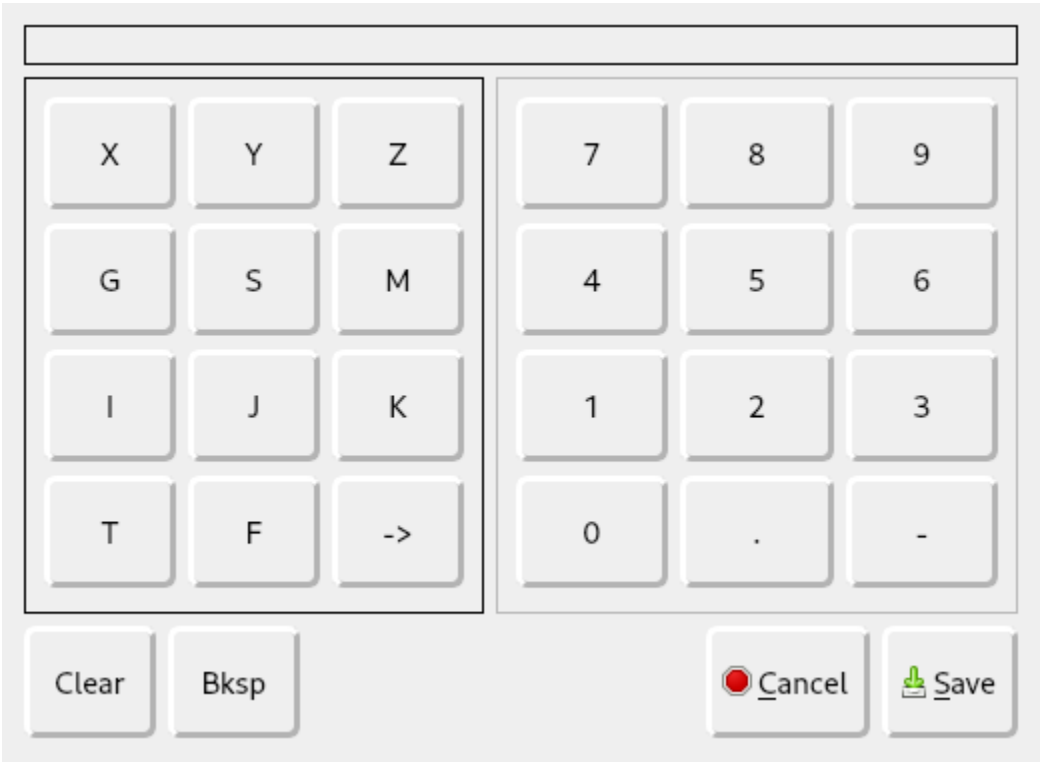
If you use a custom stylesheet for the popup keypads the letters and numbers start with key so they can be targeted with the following QSS style for example to change the font size and weight. See *Touch Section* of the ini settings document.

```
QPushButton[objectName*="key"] {
        font-size: 24px;
        font-weight: 700;
}
```

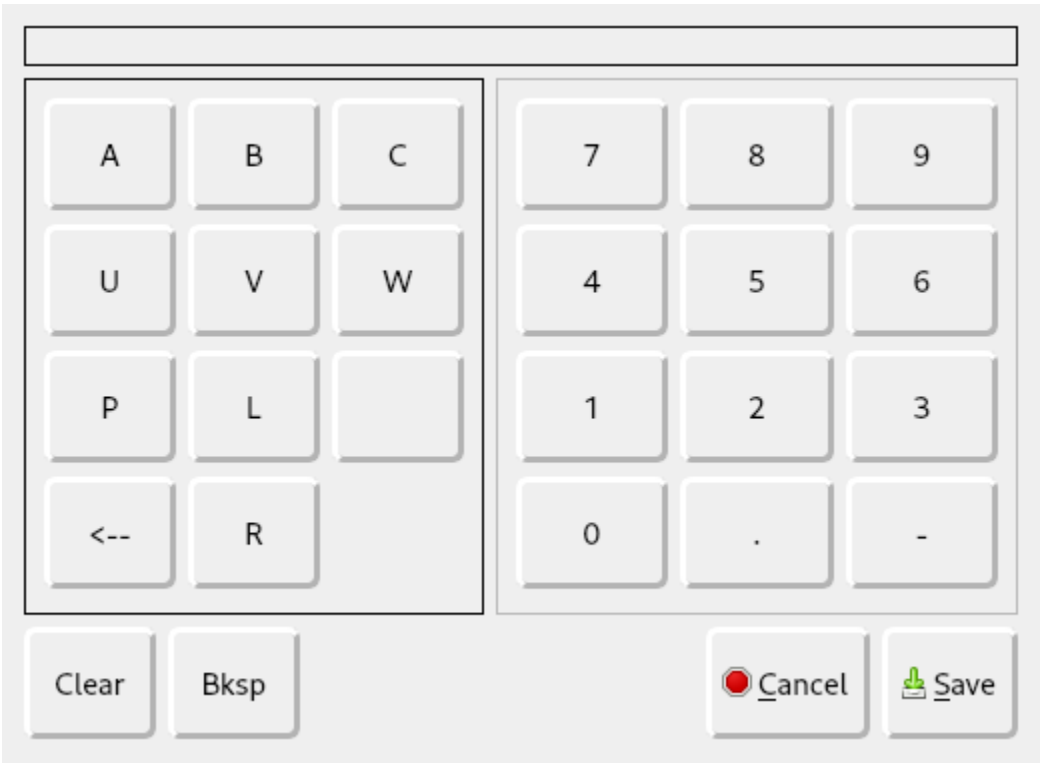
23.3 MDI

To enable the popup entry dialogs for the MDI entry, the QLineEdit object name must be *mdi_command_le* and the Dynamic Property *input* must be *nccode* for the NC codes popup or *keyboard* for a full keyboard popup.

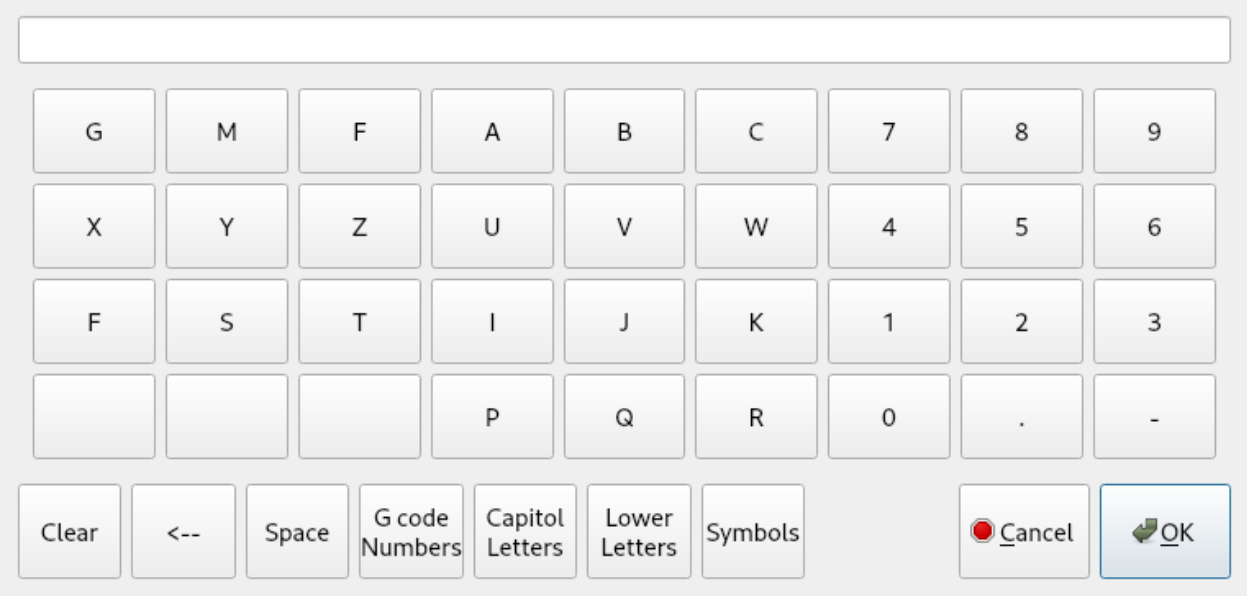
The G codes dialog will appear when you touch the MDI entry box



The arrow buttons change the letters section to different letters



The full keyboard

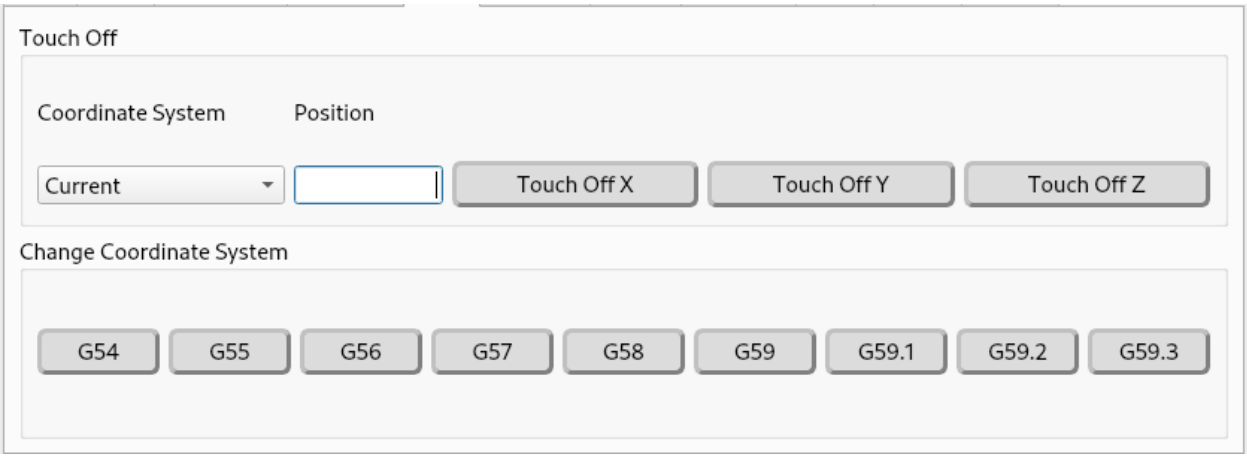


23.4 Touch Off

The Coordinate System Touch-Off offset is a QLineEdit named *touchoff_le*. To enable the number pad popup for the offset entry, add a Dynamic Property named *input* and set the value to *number*



Touch-Off:



23.5 Tool Touch-Off

The Tool Touch-Off offset is a QLineEdit named *tool_touchoff_le*. To enable the number pad popup for the offset entry, add a Dynamic Property named *input* and set the value to *number*.

23.6 Spin Boxes

QDoubleSpinBox and QSpinBox can use the popup numbers keypad by adding a Dynamic Property named *input* and setting the value to *number*. If you enter a float value for a QSpinBox the value will get converted to an integer

23.7 Line Edits

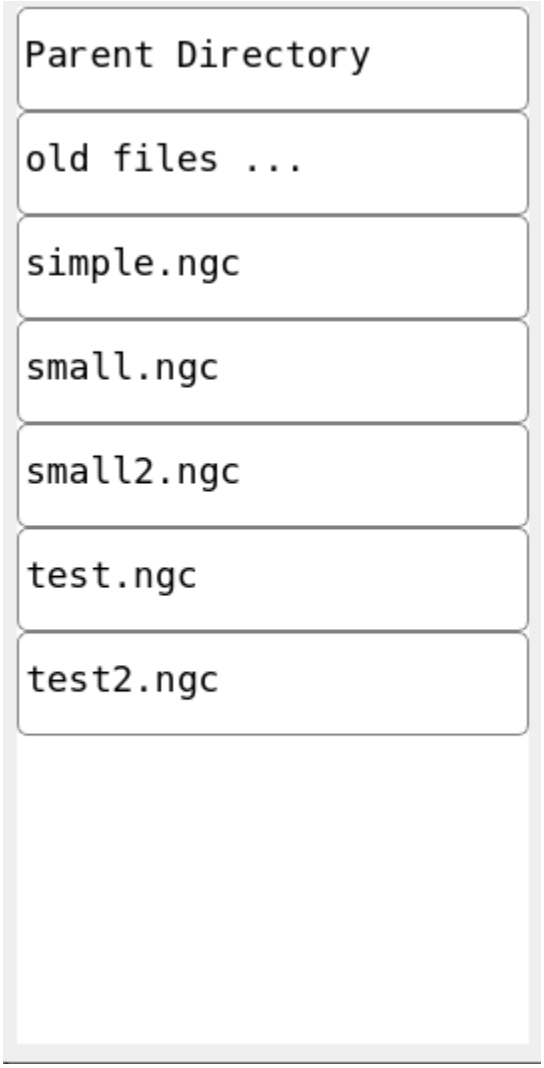
A QLineEdit can have a popup entry for numbers, G codes, or a full keyboard. Add a Dynamic Property named *input* and set the value to one of these *number*, *nccode*, or *keyboard*.

Optionally, a QLineEdit can have a String type Dynamic Property *return_button* with the objectName of a QPushButton that you want to receive a click event when the Save button is pressed using a touch popup.

23.8 File Navigator

If a QListWidget with an objectName of *file_lw* is found, a touch-friendly file selector is added. A Parent Directory and possibly a directory name with an ellipsis can be used to change directories. Touch a file name and it is loaded into the GUI.

If PROGRAM_PREFIX is specified, that will be the starting directory:

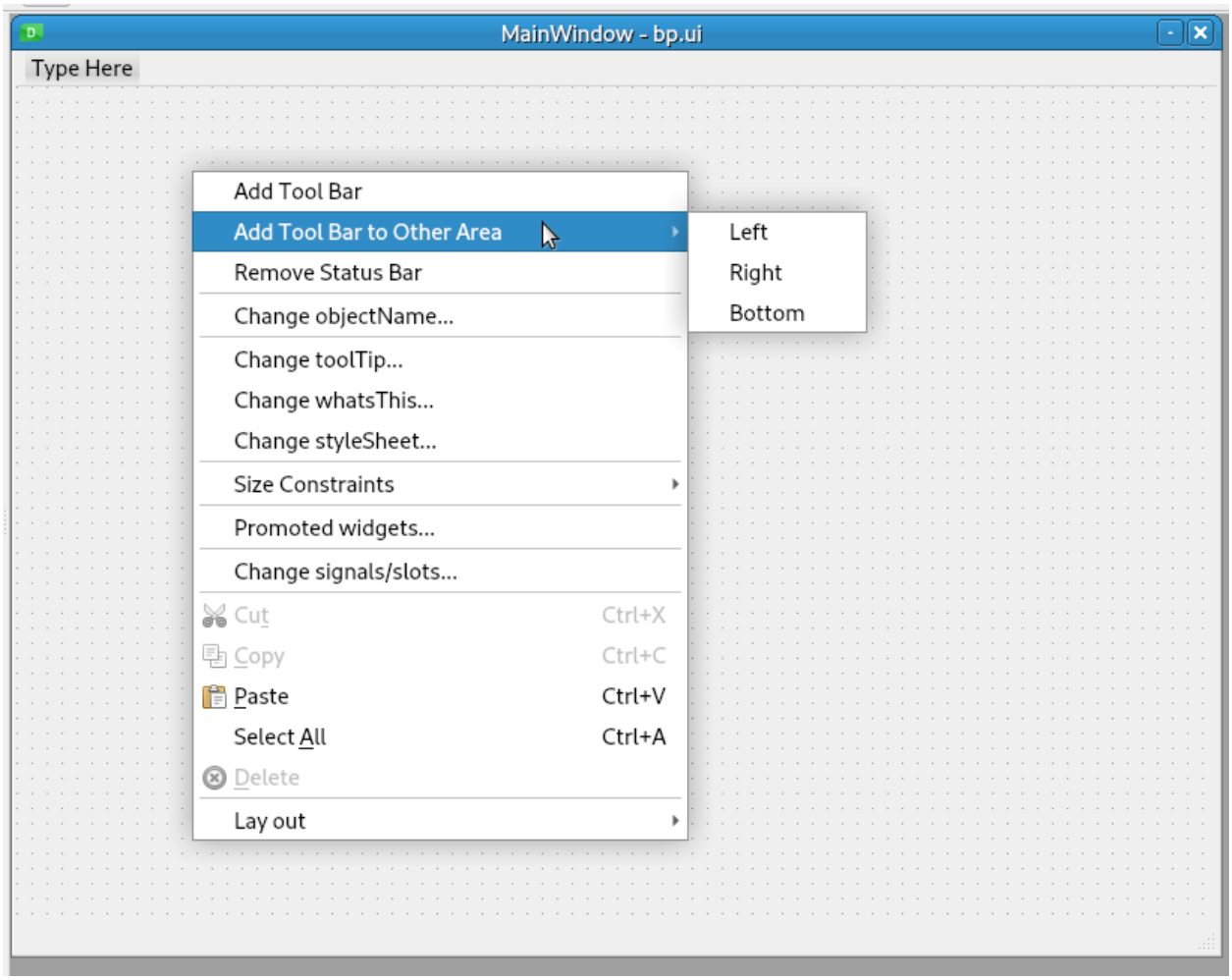

Parent Directory
old files ...
simple.ngc
small.ngc
small2.ngc
test.ngc
test2.ngc

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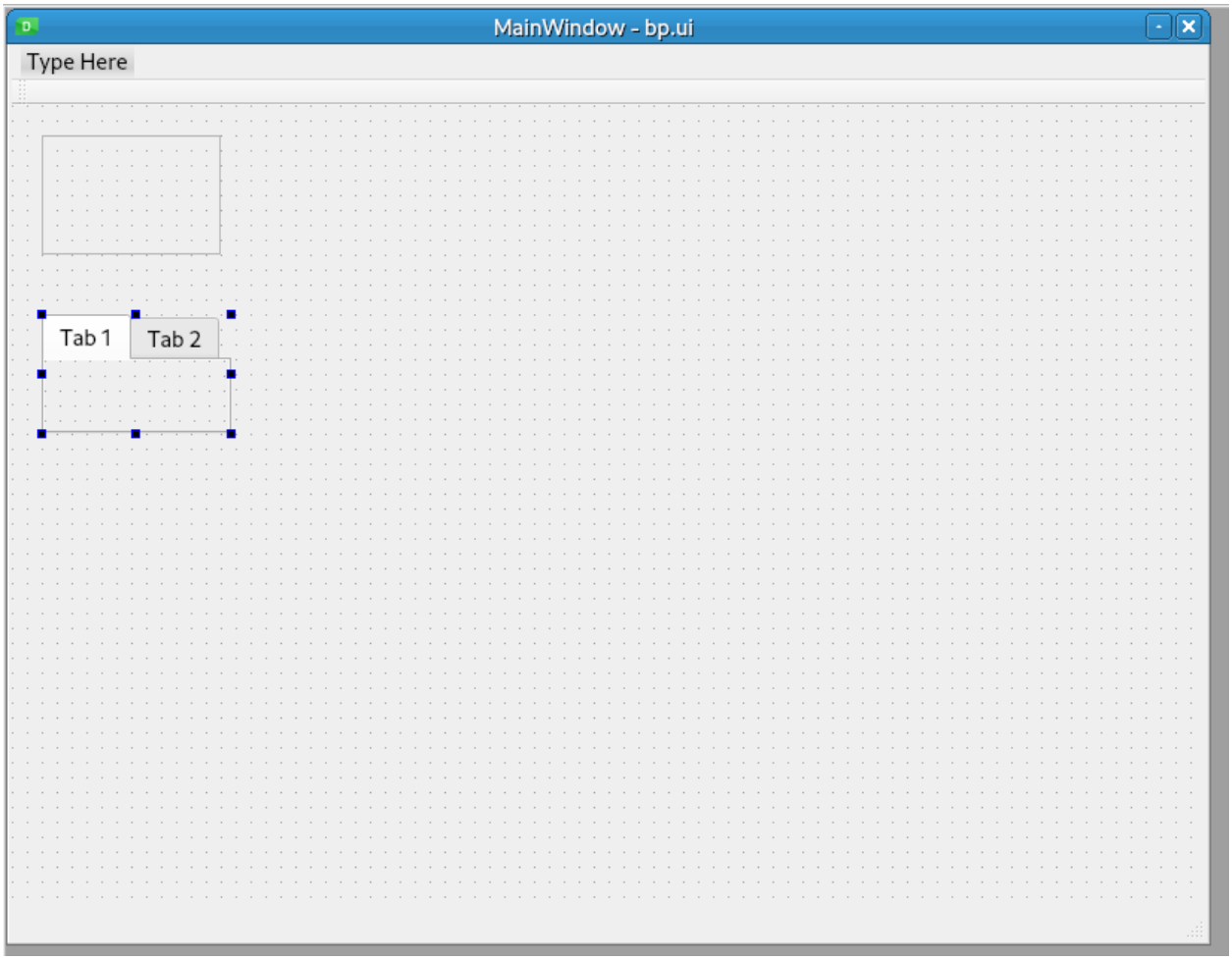
TWENTYFOUR

MASTER LAYOUT

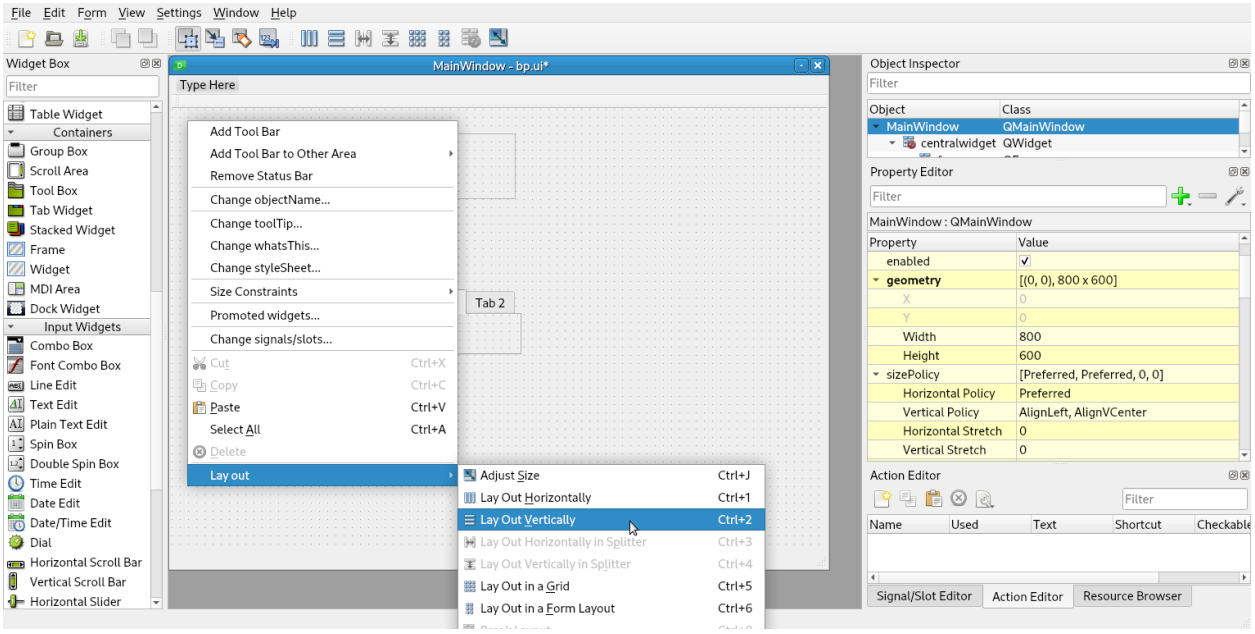
Starting with an empty Main Window, if you right-click in it you can add a tool bar or remove the status bar



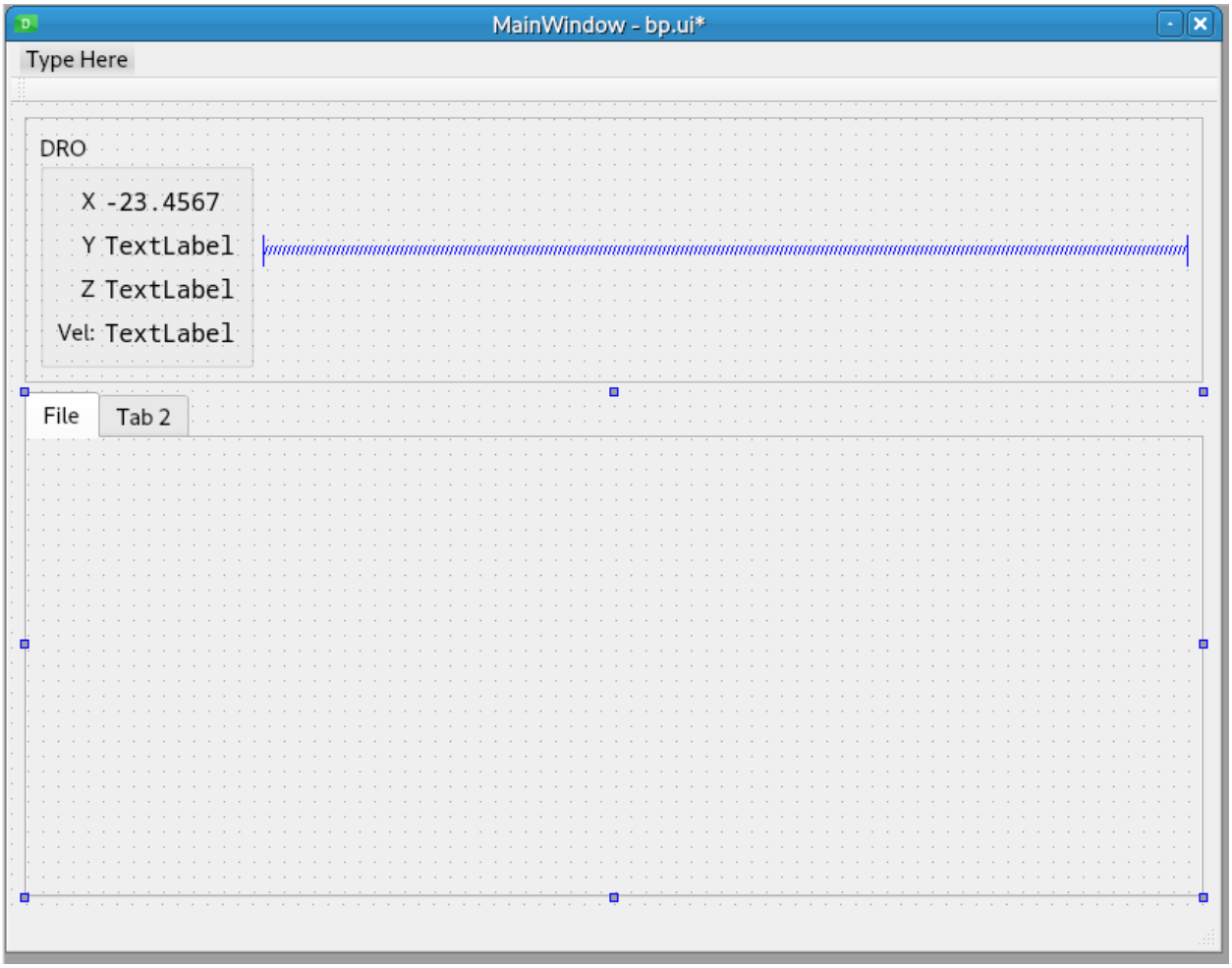
Next, if you have some items you want visible all the time you can add a QFrame or QWidget then below that, add a QTabWidget



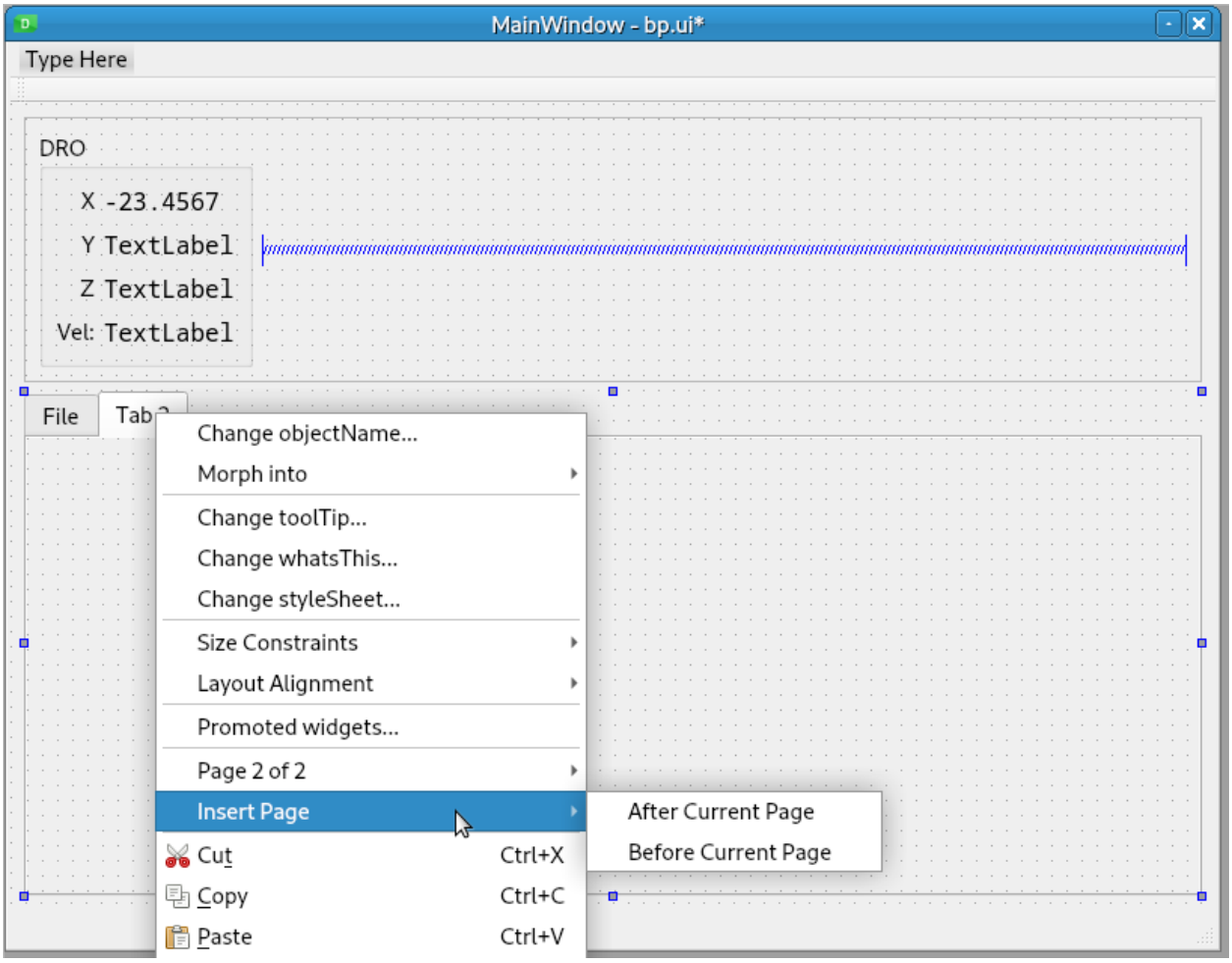
Now that you have at least one widget in the main window, you can right-click and select the layout you want to use



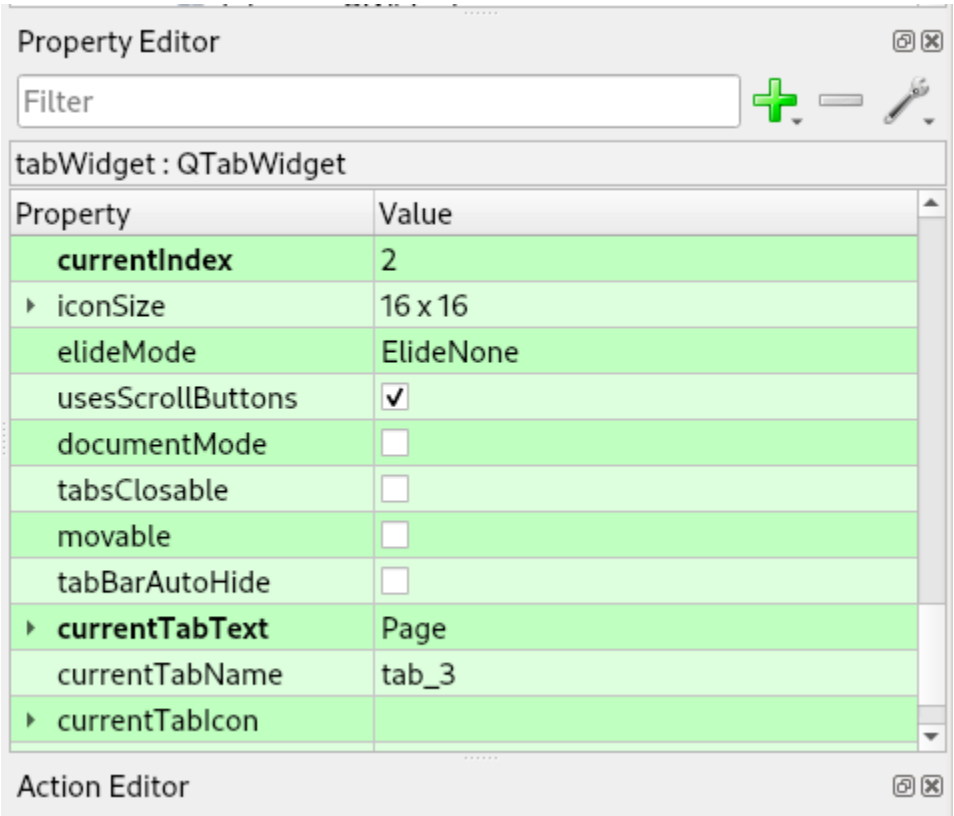
Example layout:



To add more tabs to a tab widget, right click on the tab then select Insert Page and where you want it to be inserted



To change the tab name, in the Property Editor QTabWidget section, change the currenTabText value to the new desired name.


Property Editor
Filter
tabWidget : QTabWidget
Property
Value
currentIndex
2
iconSize
16 x 16
elideMode
ElideNone
usesScrollButtons
documentMode
tabsClosable
movable
tabBarAutoHide
currentTabText
Page
currentTabName
tab_3
currentTabIcon
Action Editor

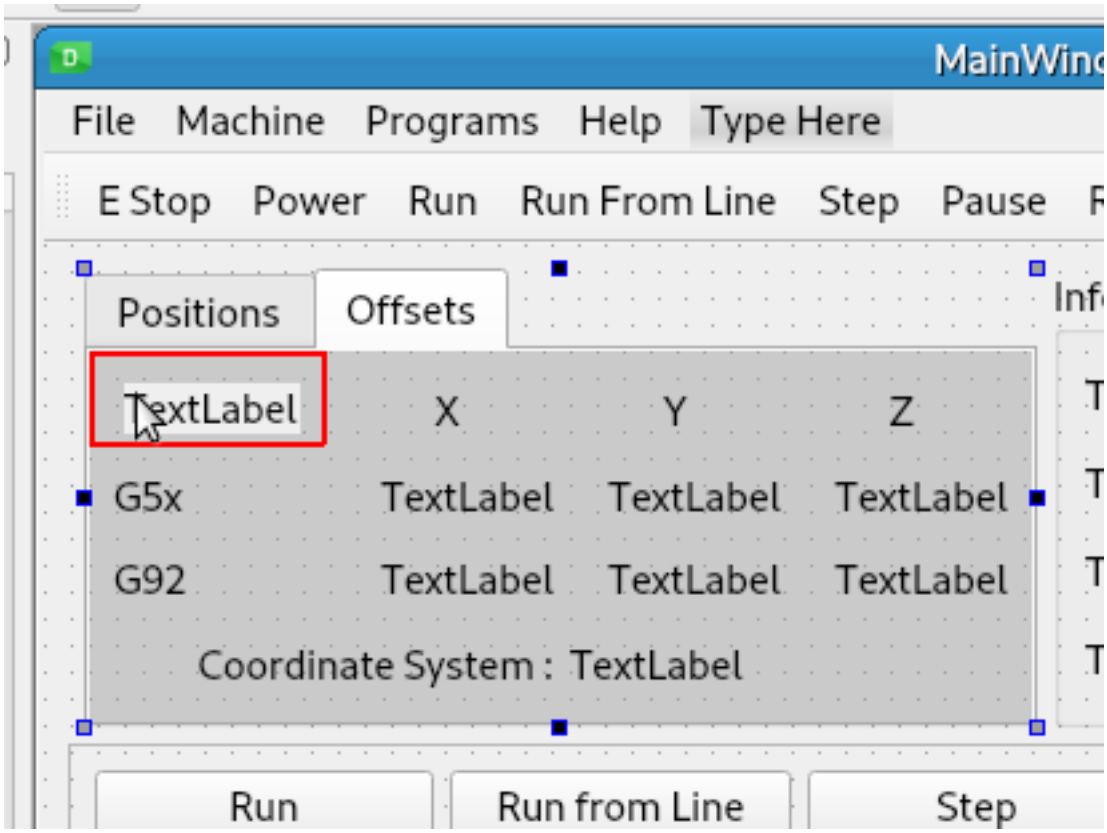
CHAPTER

TWENTYFIVE

GUI TIPS

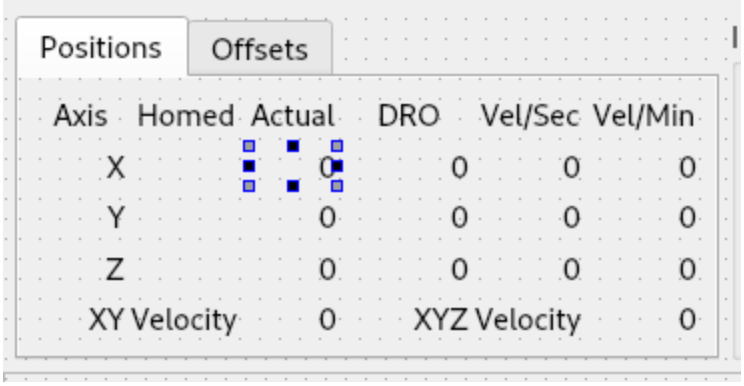
To group items together, use a container like a QFrame, QGroupBox or a QTabWidget.

If the contents of a container are rows and columns after adding at least one widget, right click and select *Layout* then *Lay Out in a Grid*. Now you can drag and drop widgets into the container. The blue line or red box indicate where it will be placed in the grid.

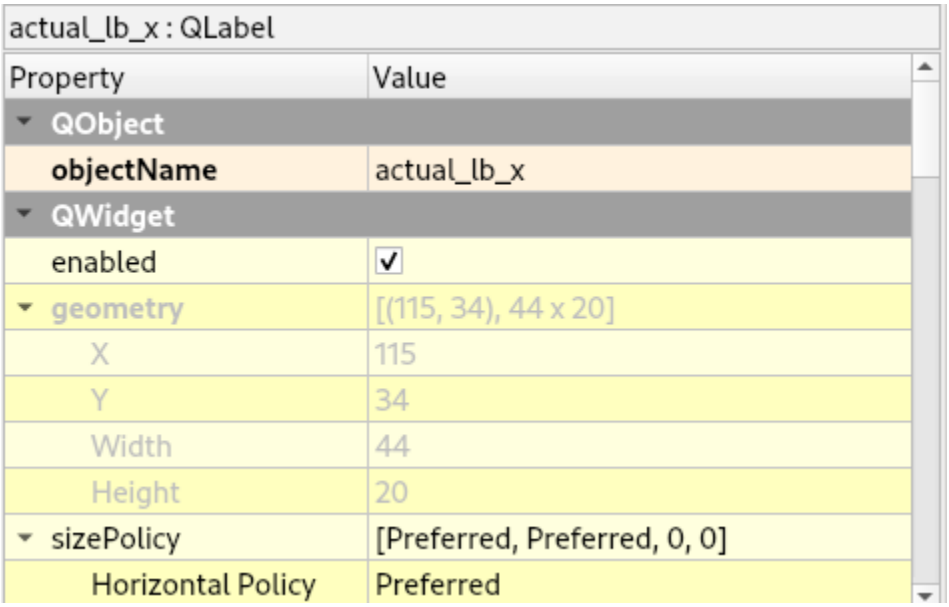


When using a grid layout for items that may change like Dro labels, change the text to represent the longest number including a minus sign. Next, in the property editor look at the Width and select (I usually use the column title) a widget and set the minimum width a tad bigger than the widest widget in that column. This will prevent the column from resizing as the values change.

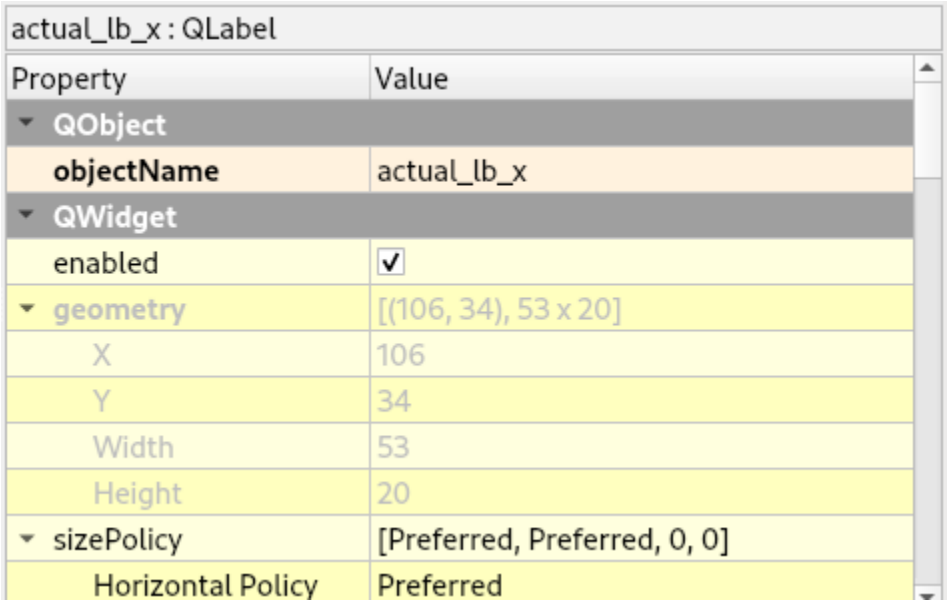
For example, the numbers in the Actual column can contain up to 8 characters like -23.4567. In the next image no minimum width has been set



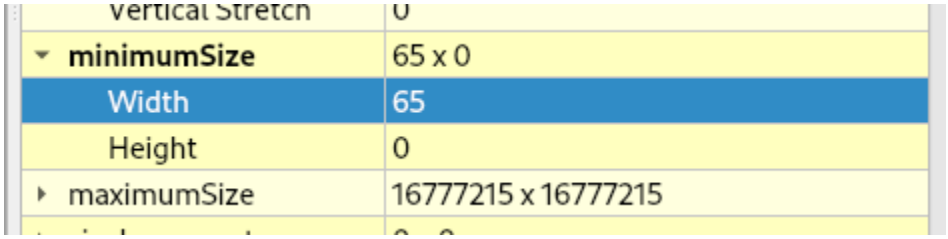
All the cells in the column will have the same width - here you can see it has a width of 44



If we double-click in the label and add -23.4567 the width changes to 61



I usually set the title of a column width to be a bit wider than the widest widget in the column



If you drag a container into another container that has a layout and it's real short, just set the minimum height to make it larger and easier to drag and drop into.

Ctrl + left click to select several widgets at once to change all their properties.

The Monospace font is good for numbers that need a fixed width like DRO values.

CHAPTER

TWENTYSIX

STYLESHEET

You can use your own .qss style sheet by creating a valid .qss file in the configuration directory and setting it in the *INI Settings*.

```
[DISPLAY]
QSS = name_of_file.qss
```

Note: If a THEME is found in the ini file the QSS entry is ignored

The Qt Style Sheets Reference and the Style Sheet Syntax and the Style Sheet Examples are good references to use when creating your own stylesheets.

Note: If there is an error in the stylesheet syntax, no warning is issued, it is just ignored. So don't forget the ; at the end of each setting. And do not accidentally use any backslashes it will break the whole file.

Warning: If you only set a background-color on a QPushButton, the background may not appear unless you set the border property to some value, even if border is set to none.

26.1 Rules

When multiple rules apply, QSS follows specificity rules similar to CSS. More specific selectors (e.g., those with pseudo-states or object names) take precedence. If specificity is equal, the last rule defined in the stylesheet takes precedence. In this example if a QPushButton state is hover or pressed or disabled the background-color will change.

```
QPushButton {
        background-color: lightgray;
        color: black;
        border: 1px solid gray;
        padding: 5px;
}
QPushButton:hover {
        background-color: lightblue;
        color: white;
}
QPushButton:pressed {
```

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```
        background-color: darkblue;
        border-style: inset;
}
QPushButton:disabled {
        background-color: #cccccc;
        color: #666666;
}
```

26.2 Colors

Most colors can be specified using Hex, RGB or RGBA color model. RGB is Red, Green, Blue and A means Alpha or transparency. The alpha parameter is a number between 0.0 (fully transparent) and 1.0 (not transparent at all). Hex is red, green blue in hexadecimal number pairs from 00 to ff.

```
#0000ff
rgb(0, 0, 255) Blue
rgba(0, 0, 255, 25%) Light Blue
```

26.3 Controls

The style can be set in the qss stylesheet for an individual QPushButton by using the QPushButton object name. For example to target the E Stop button use *#estop_pb* to target the Power button use *#power_pb*.

```
QPushButton#estop_pb {
        font-size: 24px;
        font-weight: 700;
        background-color: yellow;
        border-style: outset;
        border-width: 5;
        border-color: red;
        border-radius: 10;
}
QPushButton#estop_pb:checked {
        color: white;
        background-color: red;
        border-style: inset;
        border-color: yellow;
}

QPushButton#power_pb {
        font-size: 24px;
        font-weight: 800;
        background-color: red;
        border-style: outset;
        border-width: 5;
        border-color: green;
        border-radius: 10;
}
```

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```
QPushButton#power_pb:checked {
        color: white;
        background-color: green;
        border-style: inset;
        border-color: black;
}
```

To make a font bold use the font-weight, 400 is normal and 700 is bold.

Warning: Any errors like forgetting a ; will make the rest of the stylesheet not apply.

26.4 Flashing

Checkable buttons, like estop_pb, power_pb, or hal_buttons that are checkable can flash when either checked or not checked. To add flashing add the following Dynamic Property to the QPushButton.

The default flash time is 1000ms it can be changed in the INI file. See *Flash Timer* for more information on setting the flash time.

Table 1: **Flashing Push Button**

Property Type	Property Name	Pin Value
String	flash_state	checked or unchecked

When the button checked state matches the *flash_state* value, the button will flash between the normal background-color and the checked or !checked (not checked) background-color. The *[flashing="True"]* must be added to the state that matches the *flash_state* value.

To make the E Stop QPushButton flash when not checked the flash_state Dynamic Property must be set to *unchecked* and following example could be used.

```
QPushButton#estop_pb {
        font-size: 12px;
        font-weight: 700;
        background-color: yellow;
        border-style: outset;
        border-width: 2;
        border-color: red;
        border-radius: 5;
}

QPushButton#estop_pb:checked {
        color: white;
        background-color: red;
        border-style: inset;
        border-color: yellow;
}

QPushButton#estop_pb:!checked[flashing="True"] {
        background-color: red;
}
```

In the above example the E Stop push button will flash if not checked.

To make the Power QPushButton flash if not checked and enabledthe flash_state Dynamic Property must be set to *unchecked* and following example could be used.

```
QPushButton#power_pb {
        font-size: 12px;
        font-weight: 700;
        background-color: yellow;
        border-style: outset;
        border-width: 2;
        border-color: red;
        border-radius: 5;
}

QPushButton#power_pb:checked {
        color: white;
        background-color: red;
        border-style: inset;
        border-color: yellow;
}

QPushButton#power_pb:!checked:enabled[flashing="True"] {
        background-color: red;
}
```

26.5 Examples

```
/* Set the font size and weight 400 is normal and 700 is bold*/
QLabel {
        font-size: 24px;
        font-weight: 800;
}

/* Set the background color for all QPushButtons, border is required * /
QPushButton {
        background-color: rgba(224, 224, 224, 50%);
        border: 1px;
}

/* Set the background color and style for all QPushButtons when Pressed * /
QPushButton:pressed {
        background-color: rgba(192, 192, 192, 100%);
        border-style: inset;
}

/* Set settings for a QPushButton named exit_pb * /
QPushButton#exit_pb {
        border: none;
        background-color: rgba(0, 0, 0, 0);
}
```

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```
/* Using sub controls * /
QAbstractSpinBox::up-button {
        min-width: 30px;
}

/* Combining sub controls and state * /
QTabBar::tab:selected {
        background: lightgray;
}

/* Target by Object Name starts with something common*/
QLabel[objectName*="dro"] {
        font-family: Courier;
        font-size: 14pt;
        font-weight: 700;
}
```

26.6 Tool Bar Buttons

A tool bar button created from a menu action can be styled by using the *QToolButton* selector:

```
QToolButton:hover {
        background-color: rgba(255, 0, 0, 75%);
}
```

To set the style of a single tool bar button, you need to use the widget name for that action. The tool bar button must exist in the tool bar.

Table 2: **Tool Button Names**

Menu Item	Action Name	Widget Name
Open	actionOpen	flex_Open
Edit	actionEdit	flex_Edit
Reload	actionReload	flex_Reload
Save As	actionSave_As	flex_Save_As
Edit Tool Table	actionEdit_Tool_Table	flex_Edit_Tool_Table
Reload Tool Table	actionReload_Tool_Table	flex_Reload_Tool_Table
Ladder Editor	actionLadder_Editor	flex_Ladder_Editor
Quit	actionQuit	flex_Quit
E Stop	actionE_Stop	flex_E_Stop
Power	action_Power	flex_Power
Run	actionRun	flex_Run
Run From Line	actionRun_From_Line	flex_Run_From_Line
Step	actionStep	flex_Step
Pause	actionPause	flex_Pause
Resume	actionResume	flex_Resume
Stop	actionStop	flex_Stop
Clear MDI History	actionClear_MDI_History	flex_Clear_MDI_History
Copy MDI History	actionCopy_MDI_History	flex_Copy_MDI_History
Show HAL	actionShow_HAL	flex_Show_HAL
HAL Meter	actionHAL_Meter	flex_HAL_Meter
HAL Scope	actionHAL_Scope	flex_HAL_Scope
About	actionAbout	flex_About
Quick Reference	actionQuick_Reference	flex_Quick_Reference

The syntax to select a tool bar button by name (here the flex_Quit button) is:

```
QToolButton#flex_Quit:hover {
        background-color: rgba(255, 0, 0, 75%);
}
```

CHAPTER

TWENTYSEVEN

RESOURCES

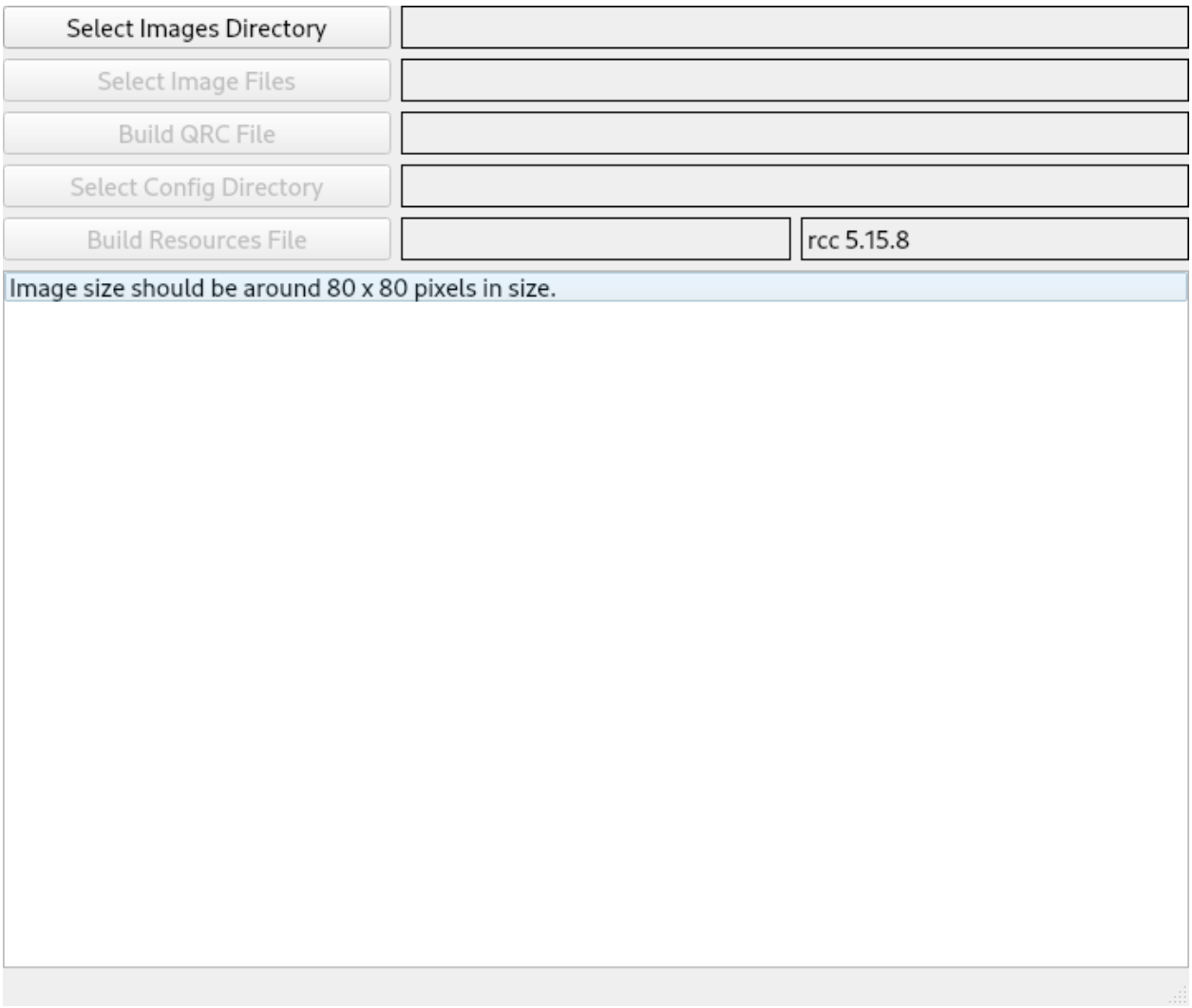
To create a resources.py file with images to use with the .qss stylesheet, start by placing all the images in a different directory than the configuration files. A subdirectory in the configuration directory is fine

```
└── configs
   └── my_mill
       └── images
```

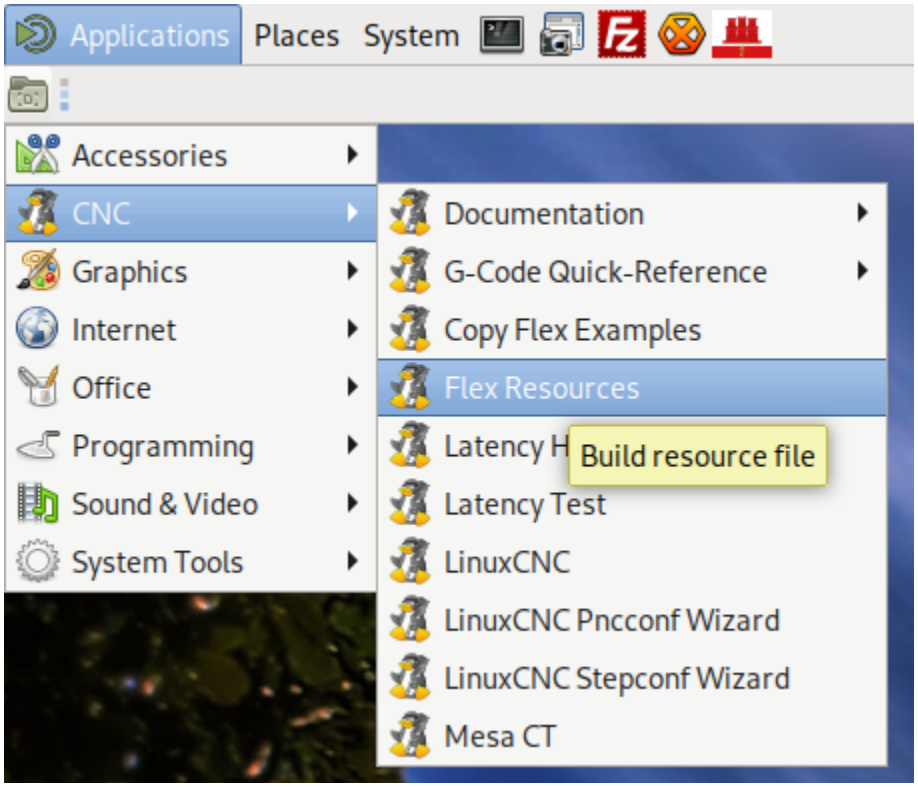
Add the following library if not installed

```
sudo apt install qtbase5-dev-tools
```

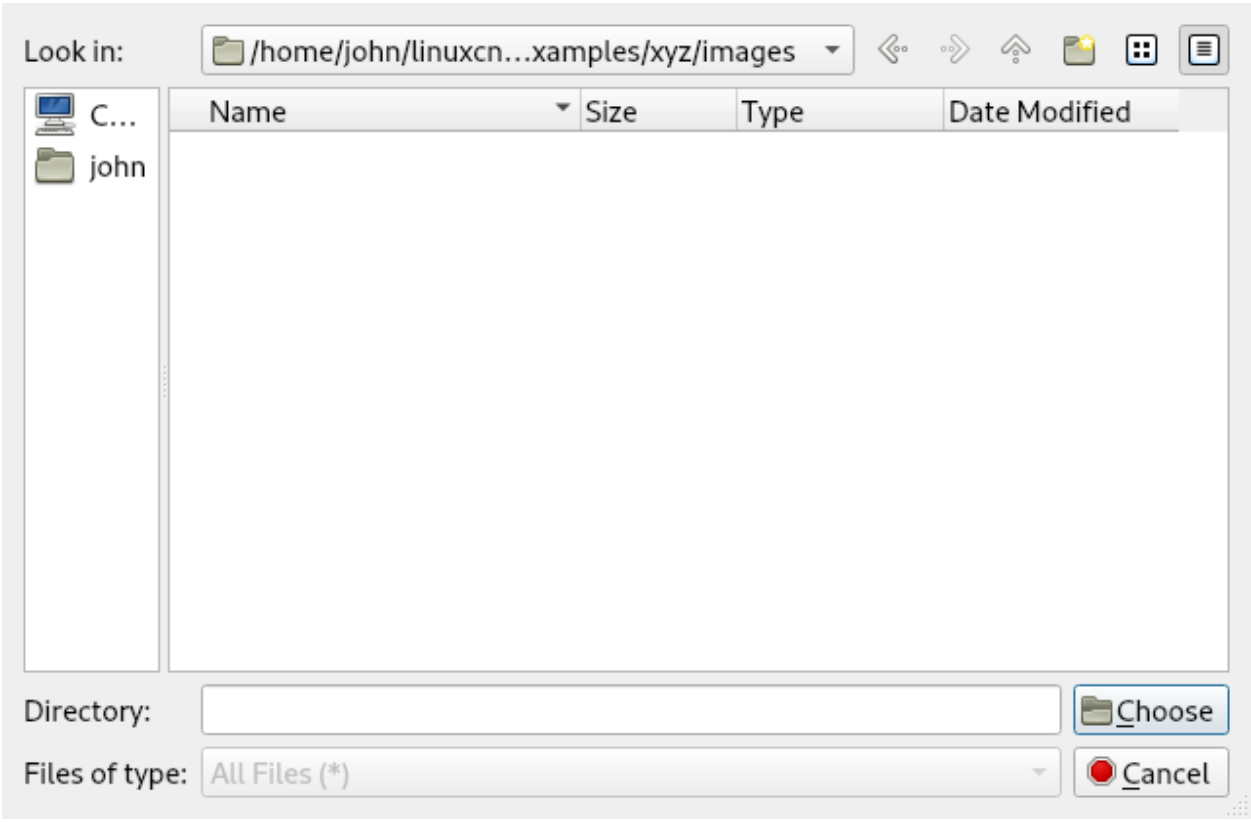
After installing Flex GUI on the CNC menu run *Flex Resources*


Select Images Directory
Select Image Files
Build QRC File
Select Config Directory
Build Resources File
rcc 5.15.8
Image size should be around 80 x 80 pixels in size.

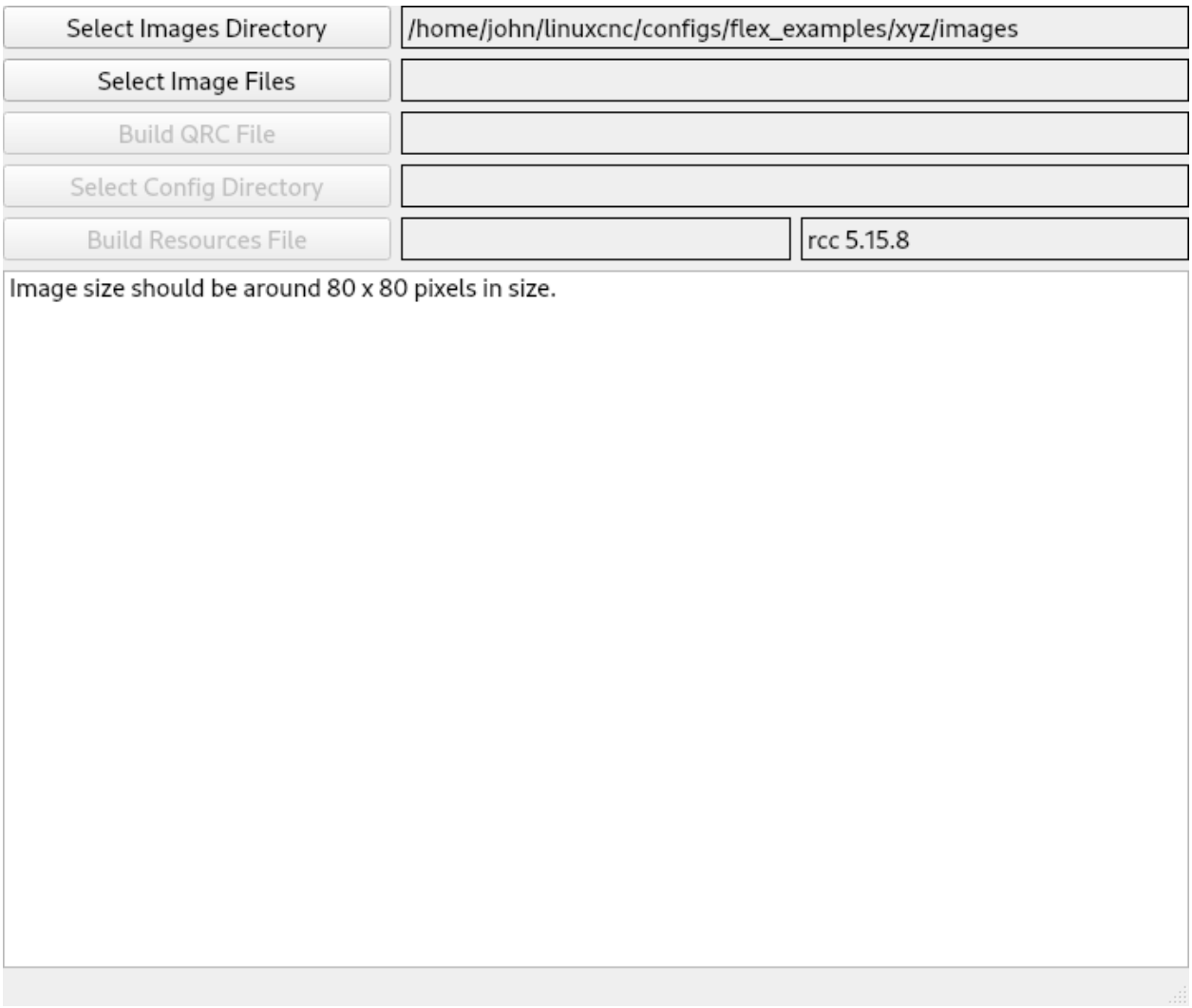
Startup



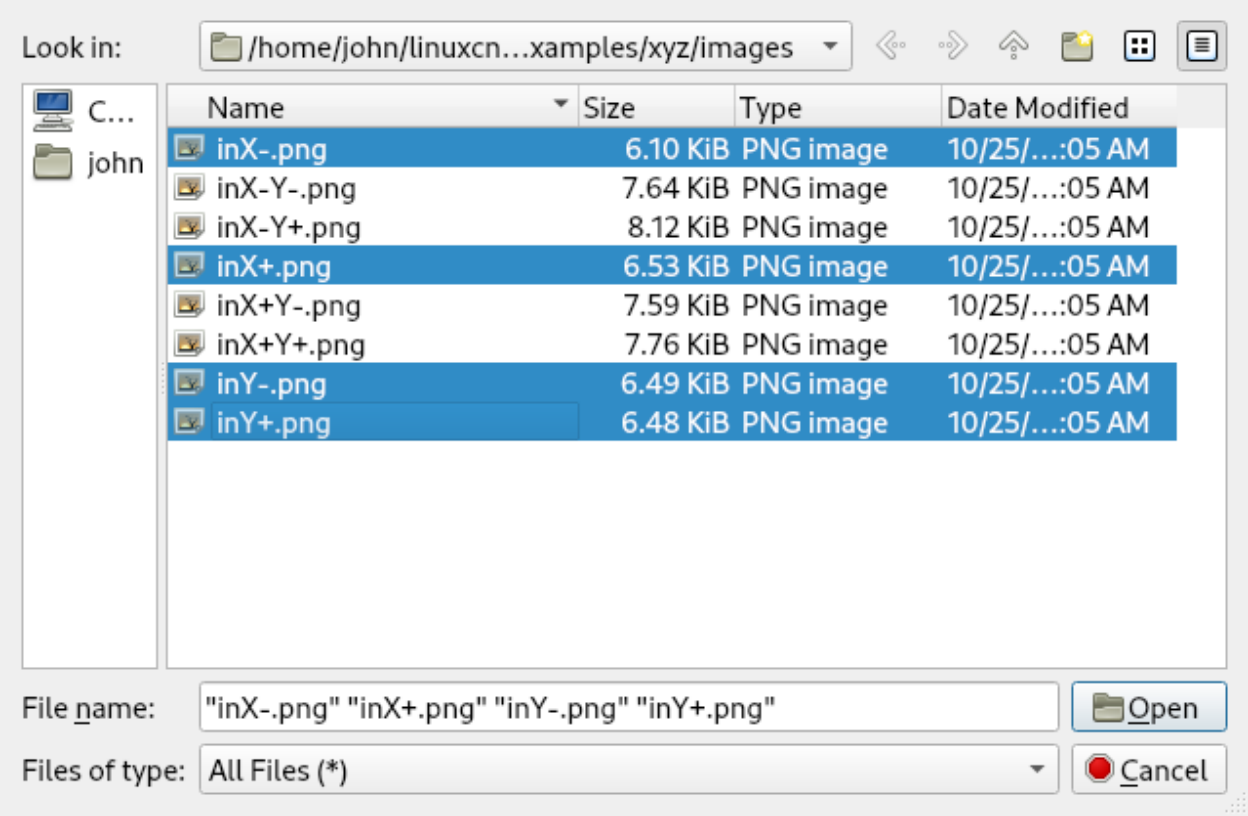
Next Select Images Directory



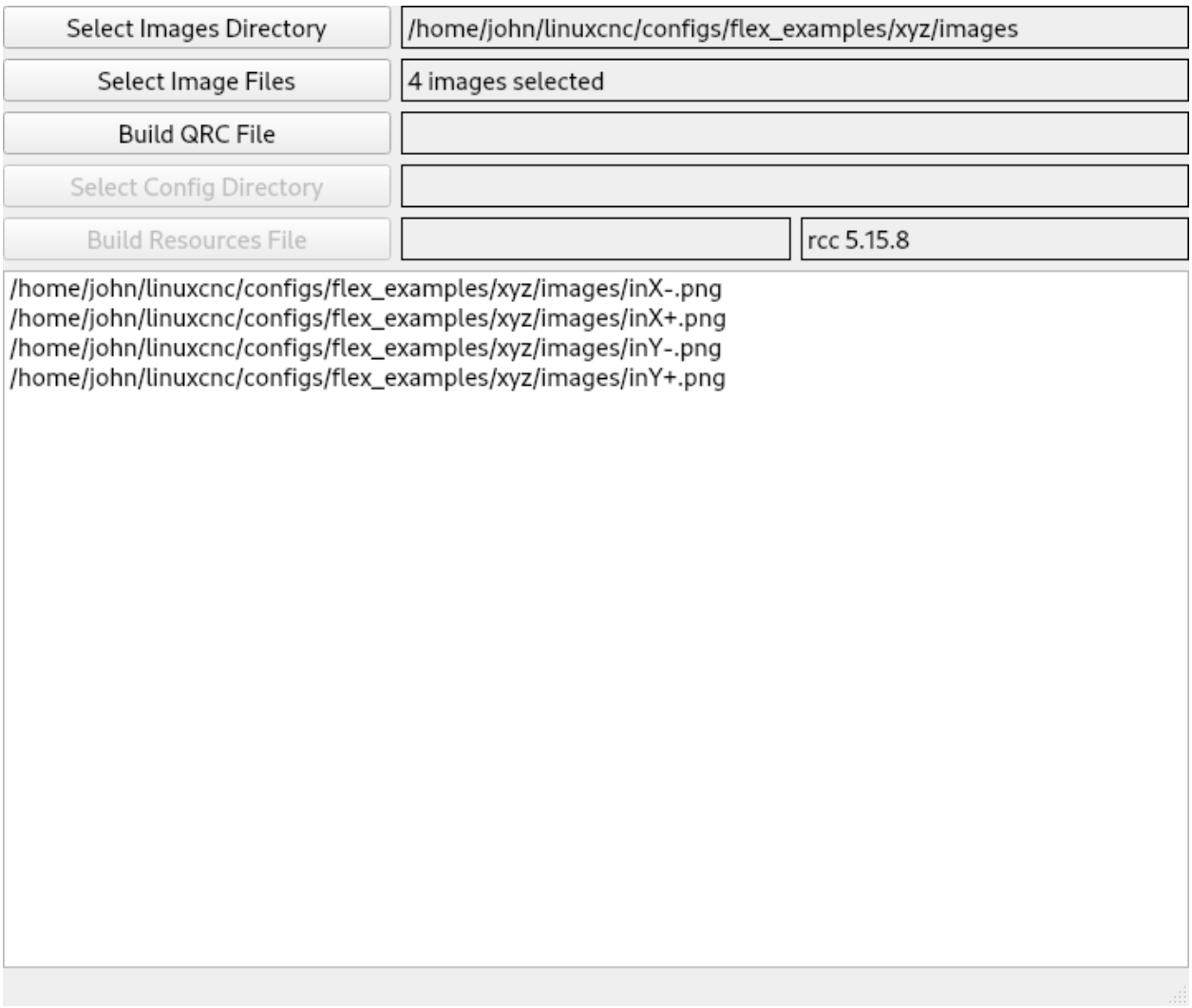
The selected directory is shown in the label



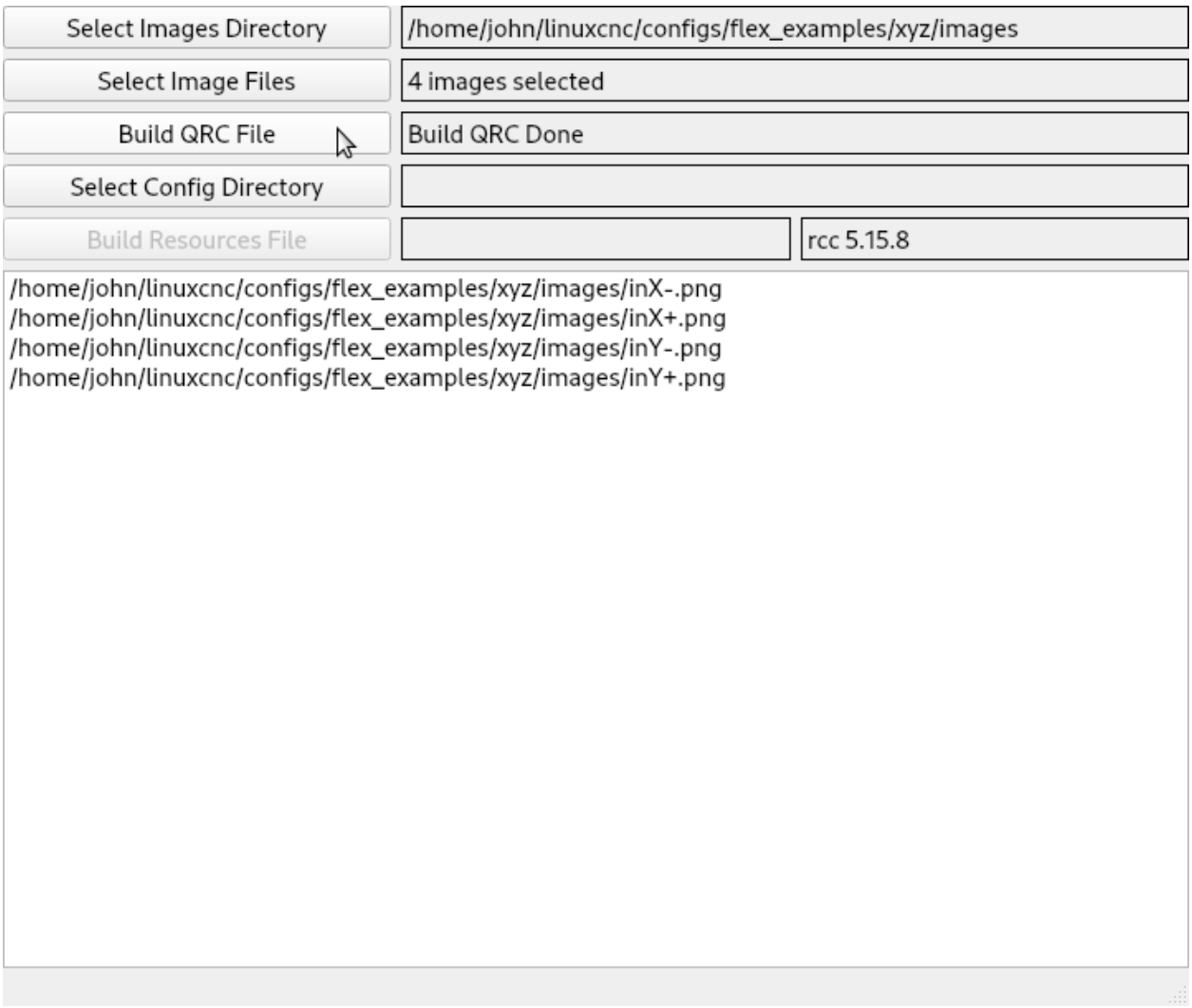
Next Select Image Files. To select all the images left click on the first one and hold down the shift key and left click on the last one. To pick several images but not all hold down the ctrl key while you left click on each one.



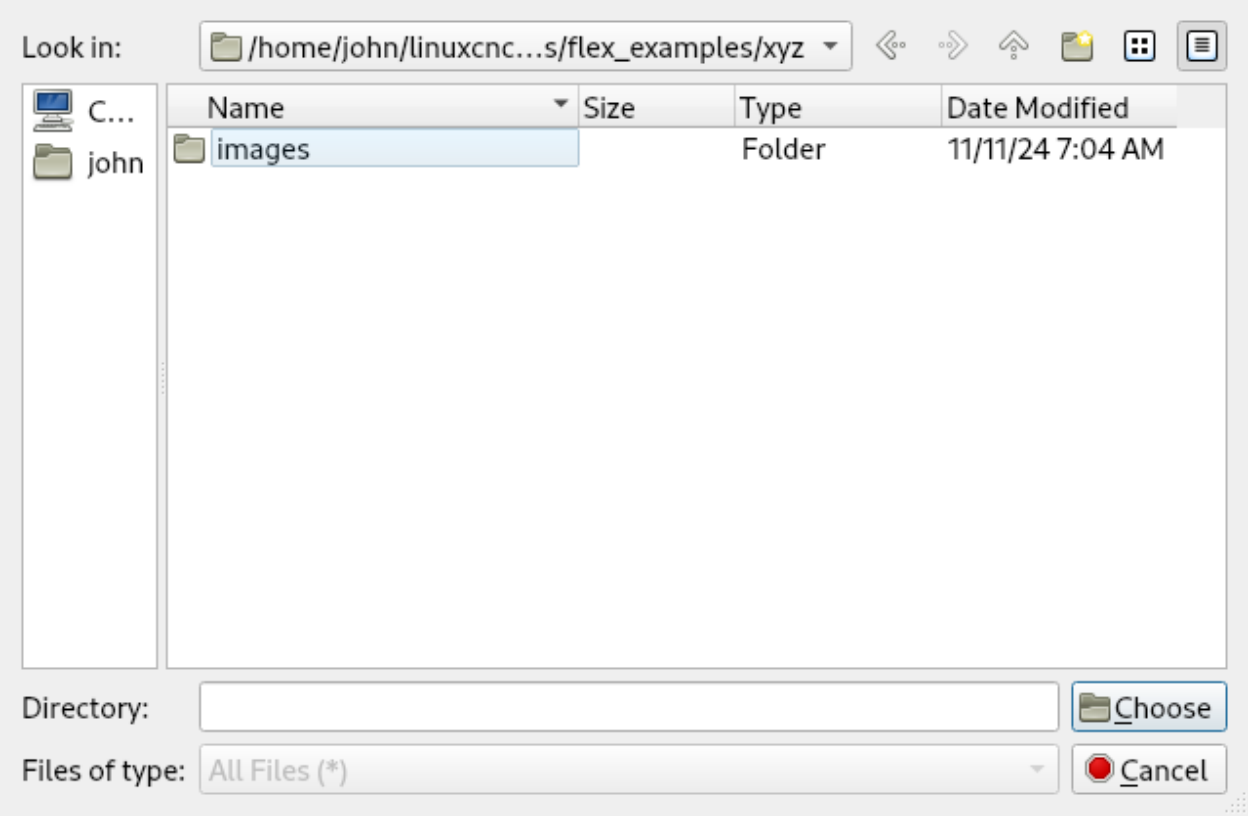
The images selected are shown below



Next Build QRC File

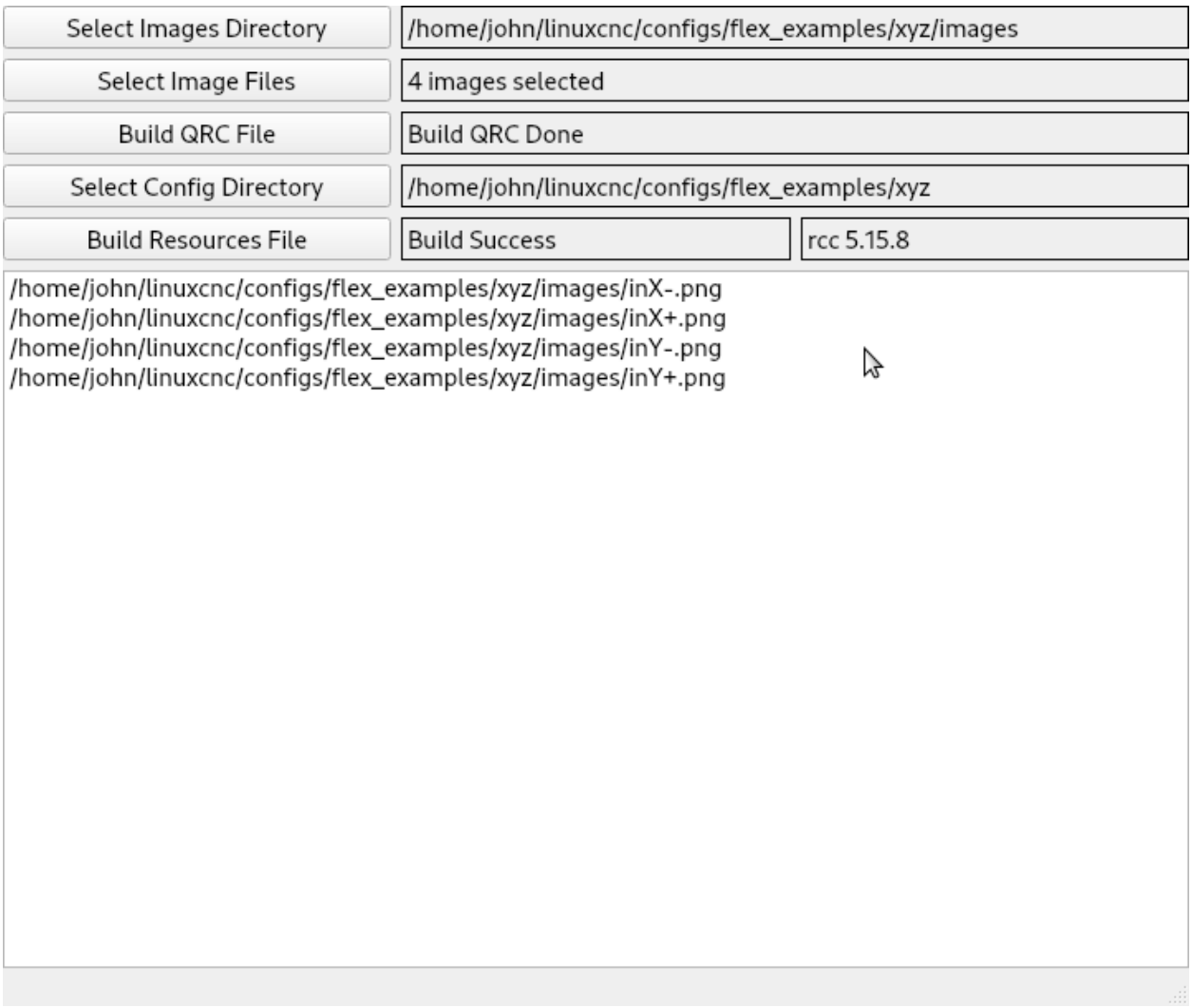


Next Select Config Directory



Note: The Image directory and the configuration directory must be different

Next Build Resources File



The Flex Resource Builder can be closed now. In the configuration directory you will have a resources.py file that contains the images used by the stylesheet.

Next edit the ini file and in the [DISPLAY] section add the following line

```
RESOURCES = resources.py
```

In the [DISPLAY] section add the style sheet

```
QSS = xyz.qss
```

To add an image named my-image.png to a QPushButton with an object name of my_pb add the following to the qss file

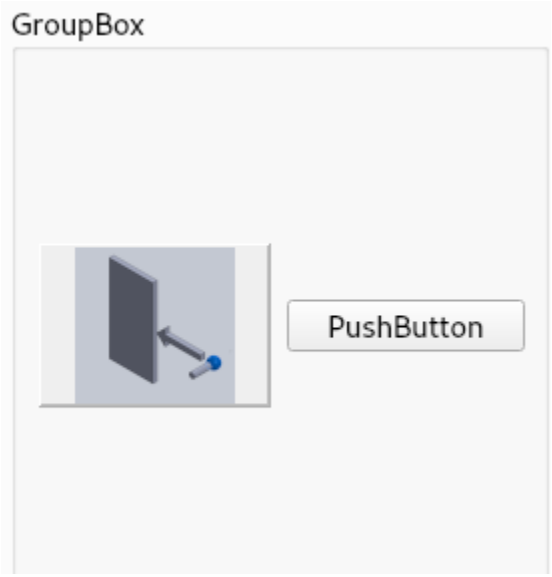
```
QPushButton#my_pb {
        min-height: 80px;
        min-width: 80px;
        margin: 2px;
        background-position: center;
        background-origin: content;
        background-clip: padding;
```

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```
        background-repeat: no-repeat;
        background-image: url(:my-image.png);
}
```

Now when you run the configuration the image will be on the QPushButton



Note: Delete any text in the QPushButton or it will be on top of the image

CHAPTER

TWENTYEIGHT

DIAGNOSTICS AND TOOLS

28.1 Raspberry Pi 5

To add the SMP PREEMPT_RT kernel it's easier than ever starting with Linux kernel version 6.12, the PREEMPT_RT patch is fully integrated into the mainline code for ARM64 architecture, making it significantly easier to set up.

Back up any critical data on your SD card or SSD before running experimental updates.

Open your terminal and run the update command explicitly requesting the 64-bit RT branch:

```
sudo WANT_64BIT_RT=1 rpi-update rpi-6.12.y
```

Reboot your Raspberry Pi 5

To activate the PREEMPT_RT kernel you just downloaded, follow these steps to update your boot configuration:

Open your system's boot configuration file:

```
sudo nano /boot/firmware/config.txt
```

Scroll to the end of the file or look under the [all] section and add the following lines:

```
kernel=kernel8_rt.img
auto_initramfs=0
```

Press Ctrl + O then Enter to save, then Ctrl + X to exit the editor.

Note: Disabling auto_initramfs ensures the system loads the raw RT image directly without trying to wrap a standard non-RT ramdisk around it

---

Restart your Raspberry Pi to apply the configuration

```
sudo reboot
```

Once the system comes back up, run the verification command

```
uname -a
```

You should now see the line include *SMP PREEMPT_RT*, confirming that the real-time kernel is active!